

Where Optimal Networks Stop Existing: Certified Spectral Limits of Cyclic Architectures for Quantum Controllability and Bottleneck-Free Routing

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September 23, 2026

Abstract

This paper studies one family of graphs – the generalized Petersen graphs $P(n, k)$, and cyclic covers more generally – with one engine, and answers two questions about it. Both questions are about the spectrum of a matrix attached to the graph; they differ in whether the matrix is fixed or ranges over a class. The engine is one object, the *symbol* $M(\zeta)$ of the cyclic cover: a 2×2 matrix of Laurent polynomials in ζ , which evaluated at the n -th roots of unity gives the n Fourier blocks of any rotation-invariant matrix on $P(n, k)$. Part I asks where the eigenvalues of $M(\zeta)$ lie as ζ runs over the grid, for the one matrix with adjacency weights; Part II asks how often $\det M(\zeta)$ can vanish on the same grid when the weights are free, and then what survives when the weights are allowed to vary around the ring. Same grid, same symbol, two questions.

Part I: which of these graphs satisfy the Riemann Hypothesis. Attached to a finite graph is its Ihara zeta function, which has an Euler product, a functional equation, and a prime number theorem for closed geodesics. For a $(q + 1)$ -regular graph one can ask the Riemann Hypothesis for it, and by the Ihara determinant formula the answer is equivalent to a finite spectral condition: the graph satisfies the RH if and only if it is *Ramanujan*, every eigenvalue with $|\lambda| \neq q + 1$ obeying $|\lambda| \leq 2\sqrt{q}$ [8, Ex. 9]. The analogous question has been settled for other families: Droll classified Ramanujan unitary Cayley graphs, and Le and Sander [12] classified Ramanujan integral circulant graphs of prime power order, in both cases finding only finitely many members qualify. Which $P(n, k)$ satisfy the Riemann Hypothesis appears not to have been asked, although the spectrum has been completely known since [11] — a paper containing no occurrence of *Ramanujan*, *expander* or *spectral gap*. We answer it completely, for every n and every k .

Clearing both radicals in the eigenvalue comparison — twice legitimately, since $|A| \leq 4 < 4\sqrt{2}$ and $7 + P \geq 3$ make each squaring an equivalence rather than an implication — collapses the analytic statement to the nonnegativity of a single integer polynomial,

$$D_k(u) = (7 + 4uT_k(u))^2 - 8(2u + 2T_k(u))^2, \quad \deg D_k = 2k + 2,$$

evaluated on the grid $u = \cos(2\pi j/n)$. Since $D_k(1) = -7$, the grid must clear a band at $u = 1$, and a fine grid cannot. Monotonicity, AM–GM and $1 - \cos \theta \leq \theta^2/2$ turn this into a bound holding for *every* k ,

$$n \leq \pi(2 + \sqrt{2})\sqrt{k^2 + 1},$$

with the sharper $n \leq 2\pi/\arccos u_k$ available for each individual k from the largest root u_k of D_k below 1. Since $D_k(-1) = -7$ for odd k and $+9$ for even k , there is a second band at $u = -1$ exactly when k is odd — but the grid meets $u = -1$ only when n is even, and there the sample is the trivial eigenvalue -3 of a bipartite graph, which the Ramanujan condition exempts. Hence a *parity law*: for odd k , odd n is cut off at *exactly half* the same constant, and bipartiteness

helps. The per- k forms are sharp in cases: the first is attained at $k = 2, 3, 4$ and the second at $k = 1, 5, 7, 9$.

Neither bound is uniform in k , so neither closes the question. Two further steps do. First, D_k is a difference of squares, so it factors over $\mathbb{Q}(\sqrt{2})$ into the two eigenvalue conditions $f_{\pm} = 7 + P \pm 2\sqrt{2}A$; these cannot both be negative, since $f_- + f_+ = 2(7 + P) \geq 6$, and under $x = \cos \frac{\pi j(k+1)}{n}$, $y = \cos \frac{\pi j(k-1)}{n}$ both collapse to the same quadratic form

$$Q(x, y) = 4x^2 + 4y^2 - 8\sqrt{2}|xy| + 3 \geq 0,$$

which *does not involve* k . The forbidden set is one fixed region — four congruent pieces at the corners of the square, each of area $\frac{3}{2} - \sqrt{2} + \frac{3}{8} \log \frac{1+\sqrt{2}}{3}$, together 0.4322% of it — and k survives only in the map into the square. Second, failure is forced whenever $\|j/n\|^2 + \|jk/n\|^2 < \tau/2\pi^2$, a *simultaneous Diophantine* condition, and Dirichlet's approximation theorem supplies such a j unconditionally: with $J = \lfloor \sqrt{n} \rfloor$ some $j \leq J$ has $\|jk/n\| \leq 1/(J+1)$, making the left side less than $2/n$. Hence $n \geq 231$ fails for *every* k , and since $1 \leq k < n/2$ the family is finite in both variables.

What remains is finite and is certified in exact arithmetic, never floating point: on the grid $A = c_1 + c_k$ and $P = c_{k+1} + c_{k-1}$ are values of integer Laurent polynomials at roots of unity, so reduction modulo Φ_m decides band-edge vanishing by integer arithmetic, and a separation bound for nonzero algebraic integers guards every sign. Exhausting the region gives the classification outright: exactly 460 pairs (n, k) satisfy the Riemann Hypothesis, 324 graphs up to isomorphism, from 13110 cases each decided exactly; the largest is $n = 112$, and no $P(n, k)$ with $k > 45$ qualifies at all. The classification is not monotone in k — $k = 9$ reaches $n = 90$ while $k = 8$ and $k = 10$ stop at 49 and 52 — because for even k an interior band binds first; the same asymmetry closes the list, since past $k = 25$ no even k survives at all.

None of this used $P(n, k)$ beyond “two-vertex base, cubic cover”. There are exactly two such bases, and the criterion carries to both: the two-loop base $B(a, b, c)$ by the *same* Q with $(a + b, a - b)$ in place of $(k + 1, k - 1)$, the edge voltage c playing no role at all, and the theta base by $\sum_{l < m} \cos \frac{2\pi j(c_l - c_m)}{n} \leq \frac{5}{2}$. For a base of any size, $M_1 \rightarrow M_0$ as $n \rightarrow \infty$ and M_0 has top eigenvalue 3, so some nontrivial eigenvalue exceeds $2\sqrt{2}$: *no infinite family of Ramanujan graphs is a cyclic cover of a fixed base*. That is the cyclic instance of the known principle that abelian covers do not expand, made explicit with a constant. Finally the *deficit* $\delta = \lambda_2 - 2\sqrt{2}$ measures what the no-go costs, and is bounded by $3 - 2\sqrt{2}$ throughout.

Part II: how degenerate the zero eigenvalue can be, over every matrix carrying the pattern. Over all symmetric matrices whose off-diagonal support is exactly $E(G)$ — Hamiltonians on that network — how degenerate can the zero eigenvalue be? That maximum, the maximum nullity $M(G)$, is the central quantity of the inverse eigenvalue problem of a graph, and determining it exactly is a statement about *every* matrix carrying the pattern. It is bounded by the zero forcing number, $M(G) \leq Z(G)$, so a matrix of large nullity and a small forcing set together pin down both.

We study them for *cyclic covers*: given a base graph B with edge voltages in $\mathbb{Z}_n$, the derived graph B^n on $V(B) \times \mathbb{Z}_n$. This is the setting of discrete Floquet theory, and the difficulty is that an answer must hold uniformly in the period n . Generalized Petersen graphs are the case of a two-vertex base with voltages 1, k and 0.

Our main structural result is that for such a cover the nullity is governed by a single scalar object determined by the voltages. For $P(n, k)$ we prove it for *every* matrix carrying the pattern, symmetric or not: eliminating the inner coordinates leaves one nine-term linear recurrence of order $2k + 2$, so

$$\text{null } A = \dim \ker(T - I) \leq 2k + 2,$$

T the monodromy, with equality iff $T = I$. The order $2k + 2$ is exactly the size of the forcing set produced by a rotation-bootstrap argument that gives $Z(P(n, k)) \leq 2k + 2$, so *the ceiling of the matrix-certificate method coincides with the upper bound it is trying to match*: either the method settles Z completely or nothing does. For a general cover the same reduction holds for

equivariant matrices, with the ceiling given by the degree span of $\det M(\zeta)$; and whenever the top coefficient of $\det M(\zeta)$ is a monomial in the edge weights, the ceiling holds for *every* matrix carrying the pattern of the cover, the kernel of the lifted operator on the infinite strip having dimension exactly the span (Theorem 91).

We then reach the ceiling. Under the rotation the singularity condition is a monic degree- $(k+1)$ polynomial in $s = 2\cos(2\pi m/n)$ carrying exactly three free parameters for every k ; three roots may be prescribed and no more, but a symbol with rational coefficients drags its whole Galois orbit onto the grid $\{2\cos(2\pi m/n)\}$. Choosing the symbol to be a product of minimal polynomials of $2\cos(2\pi/d)$ yields explicit *integer* certificates and hence

$$Z(P(n, 4)) = M(P(n, 4)) = 10, \quad Z(P(n, 5)) = M(P(n, 5)) = 12,$$

for every n divisible by 60, 70 or 90, respectively by 24. For $k \geq 4$ the only exact values previously known are $Z(P(2k+1, k)) = 6$ [6, Thm. 2.6], a single n for each k at which the bound $2k+2$ is far from attained; these are the first at which it is *attained*, and the first holding for infinitely many n . In the second case the certificate is the adjacency matrix itself, and we classify its nullity completely: with $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1$, $\text{null}(\text{Adj } P(n, k)) = \sum \varphi(d)$ over those d with $\Phi_d \mid P_k$ and $d \mid n$, a Conway–Jones phenomenon; in particular the adjacency matrix is never singular for $k = 3, 6, 9$.

Against this we prove a matching *obstruction*. Realisability of a certificate is the positivity (or nonvanishing) of $c^2 = a\alpha - \beta$, and because $L_k(-s) = (-1)^k L_k(s)$, for *every even* k a symbol with two roots $y, -y$ has $c^2 = 0$ identically, while for odd k no such obstruction exists. For $k = 2$, where the symbol has three roots and no residual conditions, this is the whole story in closed form:

$$c^2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3),$$

so the construction fails precisely when two prescribed roots are antipodal. Counting antipodal classes on the grid then settles $k = 2$ for all n : $Z(P(n, 2)) = M(P(n, 2)) = 6$ for every $n \geq 9$ with $n \neq 10$, by an explicit certificate at the three most negative grid values. The rational symbol Ψ_7 has $c^2 = -1$, so *that symbol* admits no symmetric certificate, while an integer non-symmetric matrix of nullity 6 exists for every n divisible by 7; since the main branch already covers $n \geq 9$, $n \neq 10$, the one value where the larger matrix class is needed rather than merely available is $n = 7$. The two exceptions are of opposite kinds — $Z(P(8, 2)) = 5$, so there the method is right to fail, and $Z(P(10, 2)) = 6$, so there it is the method that falls short.

The same analysis is what settles the remaining cases, because only $c^2 = 0$ is fatal: a symbol with $c^2 < 0$ admits no symmetric certificate but is realisable once the two spoke weights are allowed to differ. Redoing our exhaustive enumeration of rational symbols under the weaker condition turns two cases we had recorded as impossible into exact values. Both come from the *signed* adjacency matrix, the adjacency matrix of $P(n, k)$ with one set of spokes negated, whose nullity we classify by the polynomial $Q_k(z) = z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1$ — the unsigned P_k with the middle sign reversed. Q_k splits into cyclotomics only for $k = 2, 3, 7$ in $2 \leq k \leq 60$, and at $k = 3, 7$ it gives the ceiling:

$$Z(P(n, 3)) = 8 \quad (10 \mid n), \quad Z(P(n, 7)) = 16 \quad (120 \mid n),$$

by matrices of zeros and ± 1 , the second replacing a floating-point claim by an exact one. Conversely $k = 6, 8, 9, 10$ are *impossible* over $\mathbb{Q}$ in both classes, and now for an identified reason: every candidate clearing the family condition is one of at most 25 degenerate symbols, the bound coming from Niven’s theorem.

Breaking the symmetry to period d , the map from weights to the coefficients of the symbol has rank $\min(2d+1, k+1)$ in every case computed, so every symbol becomes reachable exactly when $d \geq \lceil k/2 \rceil$. This yields $Z(P(n, 6)) = 14$ at $n = 60, 72, 84, 90, 96, 108, 120$, $Z(P(96, 7)) = 16$ and $Z(P(96, 8)) = 18$, though these certificates are verified numerically rather than in exact arithmetic. Every exact value above holds for n in a divisibility class, and we prove that this is forced: a period-one symbol is a fixed polynomial, its roots are fixed algebraic numbers, and a fixed algebraic number lies on Γ_n only for n in one divisibility class — which contains at most

one prime. For $k \geq 3$ an n -dependent choice of roots is constrained by the identity $e_2 = -k$, a Conway–Jones relation on the grid; for $k = 2$ that condition is vacuous, which is exactly why $k = 2$ is general in n and no other k is.

Escaping this needs the non-invariant matrices, and the ceiling corollary supplies the criterion: $\text{null } A = 2k + 2$ if and only if the monodromy is the identity, a condition with no arithmetic in it. We certify that condition rigorously, not numerically, by recasting it as the bilinear system $A(w)K(X) = 0$ with K in graph form and applying a Krawczyk contraction test; the square subsystem must be chosen structurally, since $K^\top AK$ is symmetric and its antisymmetric part accounts for exactly $\binom{2k+2}{2}$ dependencies. This gives $Z(P(n, 3)) = M(P(n, 3)) = 8$ at $n = 17, 19, 21, 23, 25, 27$, the primes 17, 19, 23 among them — values that no period-one certificate can reach, and the first in this paper not lying on a congruence class. Finally we obtain a statement for *all* large n . Tiles carrying a fixed docking pattern at both ends have transfer products depending only on their interiors, so identity tiles of every length in $[L, 2L]$ give $Z(P(n, k)) = 2k + 2$ for every $n \geq L$ — a finite list of finite problems. The existence of those tiles is the one computer-assisted step in this paper, and it is exact: every bound in the contraction test is a rational number and the final inequality an integer comparison, so the step is a proof in the sense of the four-colour and Kepler proofs, not a numerical observation. Separately, and without finding any tile, an elliptic choice of dock and a single full-rank evaluation of the tile differential imply that identity tiles of *every* sufficiently large length exist (Theorem 117, hypotheses certified exactly for $k \leq 9$), and the rank condition is proved for every k by a second-order argument in $\mathfrak{sp}(2k + 2)$ (§20); together these give $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all sufficiently large n , for every $k \geq 2$, with an ineffective threshold. Carrying the explicit construction out and verifying all 77 tiles involved,

$$Z(P(n, 3)) = M(P(n, 3)) = 8 \quad (n \geq 17), \quad Z(P(n, 4)) = M(P(n, 4)) = 10 \quad (n \geq 29),$$

$$Z(P(n, 5)) = M(P(n, 5)) = 12 \quad (n \geq 162),$$

with no congruence condition. Together with exhaustive search on the finitely many smaller n , this determines $Z(P(n, k))$ for *every* n when $k = 2$, $k = 3$ and $k = 4$ — the last two for the first time, since the previously published value for $k = 3$ is false and no value at all was known for $k = 4$. In particular $Z(P(n, 3)) = 8$ for every $n \geq 13$, and 13 is optimal since $Z(P(11, 3)) = Z(P(12, 3)) = 7$; four exhaustive searches at $n = 13, \dots, 16$ close the gap below the tiling range. This proves *Conjecture 5* of [7], whose stated missing ingredient — “a lower-bound proof valid for all large n ” — is precisely what the tiling construction supplies, and fixes at 13 the stabilization threshold that note leaves open. The tile condition is a rational equation, but certifying it needs no interval arithmetic on rational maps: by the ceiling corollary the monodromy of the one-tile cyclic matrix *is* the tile product, so the same bilinear system certifies it.

We state throughout which claims are proved, which rest on exact certificates, which on interval certification, which are numerical, and which earlier claims of ours are corrected or withdrawn.

Main results

Part I — which $P(n, k)$ satisfy the Riemann Hypothesis (for the Ihara zeta function; equivalently, which are Ramanujan).

(I.1) If $P(n, k)$ satisfies the Riemann Hypothesis then $n \leq \pi(2 + \sqrt{2})\sqrt{k^2 + 1}$, and if k and n are both odd, $n \leq \frac{1}{2}\pi(2 + \sqrt{2})\sqrt{k^2 + 1}$. (Theorem 10)

(I.2) *Parity law.* For odd k and odd n , $n \leq \pi / \arccos(-v_k)$; no such restriction for even n or even k . (Theorem 15)

- (I.3) Exactly 460 pairs (n, k) qualify, in 324 isomorphism classes; no $k > 45$ occurs. All 13110 cases decided in exact arithmetic. (Theorem 18)

Part II — maximum nullity and zero forcing.

- (II.1) *Ceiling.* For every matrix carrying the $P(n, k)$ pattern, symmetric or not, $\text{null } A = \dim \ker(T - I) \leq 2k + 2$, with equality iff $T = I$. (Theorem 44)
- (II.2) $Z(P(n, k))$ is determined for *every* admissible n when $k = 2$, $k = 3$ and $k = 4$. (Theorem 110)
- (II.3) $Z(P(n, 3)) = 8$ for every $n \geq 13$, and 13 is optimal — this is Conjecture 5 of [7]. (Theorem 111)
- (II.4) $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all sufficiently large n , for every $2 \leq k \leq 9$; the threshold is ineffective. (Proposition 119)

Section 23 states what these do and do not contribute, and the Status of the claims section labels every claim by the standard it meets.

Part I

The Riemann Hypothesis for Generalized Petersen Graphs

1 The graph Riemann Hypothesis

Attached to a finite connected graph X is its *Ihara zeta function*,

$$\zeta_X(u) = \prod_{[C]} (1 - u^{\ell(C)})^{-1},$$

the product running over equivalence classes of primitive backtrackless tailless closed paths, $\ell(C)$ the length. It is a graph-theoretic analogue of the Riemann zeta function: it has an Euler product, a meromorphic continuation, a functional equation, and a prime number theorem counting closed geodesics. For a $(q + 1)$ -regular graph the substitution $u = q^{-s}$ makes the analogy exact, and one may ask the corresponding question.

Definition 1 (Riemann Hypothesis for a graph). A $(q + 1)$ -regular graph X satisfies the Riemann Hypothesis if $\zeta_X(q^{-s})$ has no poles with $0 < \text{Re } s < 1$ other than on the line $\text{Re } s = \frac{1}{2}$.

The point of the definition is the following equivalence, which converts an analytic statement about a zeta function into a finite spectral one. It is classical, following from the Ihara determinant formula; we take the statement from Terras [8, Exercise 9], where it appears in exactly this form.

Theorem 2 (Ihara determinant formula; see [8]). *A connected $(q + 1)$ -regular graph X satisfies the Riemann Hypothesis if and only if X is Ramanujan: every eigenvalue λ of the adjacency matrix with $|\lambda| \neq q + 1$ satisfies $|\lambda| \leq 2\sqrt{q}$.*

The term is due to Lubotzky, Phillips and Sarnak [10], and the bound is the best possible: $2\sqrt{q}$ is the spectral radius of the adjacency operator on the universal covering tree, so a Ramanujan graph is one whose nontrivial spectrum is as small as an infinite family can have it. Ramanujan graphs are optimal expanders, which is why they matter outside spectral graph theory – for communication networks, error-correcting codes, and derandomization.

Our graphs are cubic, $q + 1 = 3$, so the threshold is

$$2\sqrt{2} = 2.8284271247\dots,$$

and the trivial eigenvalues are 3, always, and -3 , exactly when the graph is bipartite. We ask which generalized Petersen graphs are on the right side of it.

Question 3. For which n and k does $P(n, k)$ satisfy the Riemann Hypothesis?

This is a natural question of a recognised kind. The analogue has been settled for other families: Droll classified the Ramanujan unitary Cayley graphs, and Le and Sander [12] classified the Ramanujan integral circulant graphs of prime power order, in both cases finding that only finitely many members qualify. For generalized Petersen graphs the spectrum has been completely known since Gera and Stănică [11], but we have not found Question 3 raised anywhere: [11] itself contains no occurrence of the words *Ramanujan*, *expander* or *spectral gap*.

We answer it completely, for every n and every k , in exact arithmetic. The answer has four parts: a criterion reducing the question to the sign of one integer polynomial; a change of variable after which the criterion is one fixed region not depending on k ; a Diophantine argument bounding n uniformly in k , so that the family is finite in both variables; and an exhaustive certification of the finitely many survivors.

2 The spectrum, and a criterion with no radicals

$P(n, k)$ is the cyclic cover of the two-vertex base of §18 – the same object this paper studies throughout – and the block decomposition used there is, on the zeta side, precisely the factorization of $\zeta_{P(n, k)}$ into Artin–Ihara L -functions over the characters of $\mathbb{Z}_n$. The spectral form we need is the following, which is Theorem 2.4 of [11].

Lemma 4. *The adjacency matrix of $P(n, k)$ is unitarily equivalent to $\bigoplus_{j=0}^{n-1} M_j$, where*

$$M_j = \begin{pmatrix} \alpha_j & 1 \\ 1 & \beta_j \end{pmatrix}, \quad \alpha_j = 2 \cos \frac{2\pi j}{n}, \quad \beta_j = 2 \cos \frac{2\pi jk}{n} = 2T_k(\alpha_j/2),$$

with T_k the Chebyshev polynomial of the first kind.

Proof. The rotation $\rho : i \mapsto i + 1$ acts on $P(n, k)$ with the adjacency matrix in its commutant, so the matrix decomposes over the characters $\chi_j(a) = \zeta_n^{ja}$ of $\mathbb{Z}_n$. On the χ_j isotypic component the outer cycle contributes $\zeta_n^j + \zeta_n^{-j} = \alpha_j$, the inner k -step cycle contributes $\zeta_n^{jk} + \zeta_n^{-jk} = \beta_j$, and the spoke contributes 1 off the diagonal. The identity $2 \cos k\theta = 2T_k(\cos \theta)$ gives the stated form of β_j . $\square$

Deciding Question 3 directly from Lemma 4 means comparing $(\alpha + \beta \pm \sqrt{(\alpha - \beta)^2 + 4})/2$ against $2\sqrt{2}$ – two nested irrationalities, and the comparison is decided at a band edge where floating point is worth nothing. Both radicals can be cleared at once, and what remains is a single polynomial with integer coefficients.

Theorem 5 (Criterion). *For $k \geq 1$ define*

$$D_k(u) = (7 + 4uT_k(u))^2 - 8(2u + 2T_k(u))^2 \in \mathbb{Z}[u], \quad \deg D_k = 2k + 2.$$

Then both eigenvalues of M_j lie in $[-2\sqrt{2}, 2\sqrt{2}]$ if and only if $D_k(u_j) \geq 0$, where $u_j = \cos \frac{2\pi j}{n}$. The polynomial has degree $2k + 2$ and leading coefficient 4^{k+1} .

Proof. Write $\alpha = \alpha_j$, $\beta = \beta_j$, $A = \alpha + \beta$, $P = \alpha\beta$. The characteristic polynomial of M_j is $\lambda^2 - A\lambda + (P - 1)$, with roots $\lambda_{\pm} = (A \pm \sqrt{(\alpha - \beta)^2 + 4})/2$.

For the upper constraint, $\lambda_+ \leq 2\sqrt{2}$ is $\sqrt{(\alpha - \beta)^2 + 4} \leq 4\sqrt{2} - A$. Since $|\alpha|, |\beta| \leq 2$ we have $|A| \leq 4 < 4\sqrt{2}$, so the right-hand side is positive and squaring is an equivalence, not merely an implication:

$$(\alpha - \beta)^2 + 4 \leq 32 - 8\sqrt{2}A + A^2.$$

Because $A^2 - (\alpha - \beta)^2 = 4P$, this simplifies to $4 \leq 32 - 8\sqrt{2}A + 4P$, that is $2\sqrt{2}A - P \leq 7$. Symmetrically $\lambda_- \geq -2\sqrt{2}$ is equivalent to $-2\sqrt{2}A - P \leq 7$. The two together say

$$2\sqrt{2}|A| \leq 7 + P.$$

Now $|P| = |\alpha\beta| \leq 4$, so $7 + P \geq 3 > 0$: the right-hand side is again positive, so squaring is again an equivalence, giving $8A^2 \leq (7 + P)^2$. Finally $\alpha = 2u$ and $\beta = 2T_k(u)$, so $A = 2u + 2T_k(u)$ and $P = 4uT_k(u)$, and $(7 + P)^2 - 8A^2 = D_k(u)$.

For the degree: T_k has degree k and leading coefficient 2^{k-1} , so $(7 + 4uT_k)^2$ has degree $2k + 2$ with leading coefficient $(4 \cdot 2^{k-1})^2 = 4^{k+1}$, while $8(2u + 2T_k)^2$ has degree only $2k$. Nothing cancels, so $\deg D_k = 2k + 2$ with leading coefficient 4^{k+1} . $\square$

Corollary 6 (Ramanujan, polynomially). *$P(n, k)$ satisfies the Riemann Hypothesis if and only if*

$$D_k\left(\cos \frac{2\pi j}{n}\right) \geq 0 \quad \text{for every } j \in \{1, \dots, n-1\} \text{ with } 2j \neq n \text{ or } k \text{ even.}$$

Proof. $P(n, k)$ is connected – the outer cycle is connected and every inner vertex carries a spoke – and cubic, so by Theorem 2 the question is whether every eigenvalue of modulus $\neq 3$ has modulus $\leq 2\sqrt{2}$. By Lemma 4 the spectrum is $\bigcup_j \text{spec}(M_j)$.

The block M_0 has $\alpha = \beta = 2$, hence eigenvalues 3 and 1. If n is even and k odd then $P(n, k)$ is bipartite and $M_{n/2}$ has $\alpha = \beta = -2$, hence eigenvalues -1 and -3 . No other block has ± 3 as an eigenvalue, since 3 and -3 have multiplicity one in a connected 3-regular graph (and -3 occurs only in the bipartite case). So the two excluded indices contribute the trivial eigenvalues together with 1 and -1 respectively, both of modulus $1 \leq 2\sqrt{2}$; every other index contributes only nontrivial eigenvalues. Apply Theorem 5 at each of those. $\square$

The reduction is worth pausing on. The side condition $7 + P \geq 0$ that licensed the second squaring is *never* binding, since $|4uT_k(u)| \leq 4$ on $[-1, 1]$ forces $7 + 4uT_k(u) \geq 3$. So the entire Riemann Hypothesis for $P(n, k)$ – an analytic statement about the poles of a zeta function – has become the assertion that one integer polynomial is nonnegative at $n - 1$ cyclotomic points. That is a statement our exact machinery can certify outright, and §5 does.

3 The criterion without k

Theorem 5 gives one polynomial per k , of degree growing with k . There is a change of variable after which k leaves the geometry entirely, and what is left is a single fixed region that does not depend on k at all.

Theorem 7 (Corner criterion). *Put*

$$x = \cos \frac{\pi j(k+1)}{n}, \quad y = \cos \frac{\pi j(k-1)}{n}, \quad Q(x, y) = 4x^2 + 4y^2 - 8\sqrt{2}|xy| + 3.$$

Both eigenvalues of M_j lie in $[-2\sqrt{2}, 2\sqrt{2}]$ if and only if $Q(x, y) \geq 0$. Hence $P(n, k)$ satisfies the Riemann Hypothesis if and only if $Q(x, y) \geq 0$ at every nontrivial index. The form Q does not involve k .

Proof. $D_k = (7 + P)^2 - 8A^2$ is a difference of squares, so over $\mathbb{Q}(\sqrt{2})$ it factors as $D_k = f_- f_+$ with

$$f_{\pm} = 7 + P \pm 2\sqrt{2}A,$$

each of degree $k+1$ in u . These factors are not new objects: the proof of Theorem 5 produced exactly them, $f_- \geq 0$ being $\lambda_+ \leq 2\sqrt{2}$ and $f_+ \geq 0$ being $\lambda_- \geq -2\sqrt{2}$. What is required is that *both* be nonnegative. They cannot both be negative, since

$$f_- + f_+ = 2(7 + P) \geq 6 > 0$$

by $|P| \leq 4$; so the product is nonnegative exactly when both factors are, which is why the single inequality $D_k \geq 0$ was equivalent to the pair.

Now set $t = 2\pi j/n$, so $u = \cos t$ and $x = \cos \frac{(k+1)t}{2}$, $y = \cos \frac{(k-1)t}{2}$. From $2 \cos t \cos kt = \cos(k+1)t + \cos(k-1)t$ and $\cos(k \pm 1)t = 2 \cos^2 \frac{(k \pm 1)t}{2} - 1$,

$$P = 4 \cos t \cos kt = 2((2x^2 - 1) + (2y^2 - 1)) = 4x^2 + 4y^2 - 4,$$

and from $\cos t + \cos kt = 2 \cos \frac{(k+1)t}{2} \cos \frac{(k-1)t}{2}$,

$$A = 2(\cos t + \cos kt) = 4xy.$$

Hence $7 + P = 3 + 4x^2 + 4y^2$ and

$$f_{\mp} = 3 + 4x^2 + 4y^2 \mp 8\sqrt{2}xy,$$

whose minimum over the two signs is $4x^2 + 4y^2 - 8\sqrt{2}|xy| + 3 = Q(x, y)$. Both are nonnegative exactly when $Q \geq 0$. $\square$

The content of the theorem is that k has moved out of the region and into the map. For every k the question is asked about the same subset of the square.

Corollary 8 (The forbidden set).

$$Q = 4(|x| - (\sqrt{2} + 1)|y|)(|x| - (\sqrt{2} - 1)|y|) + 3,$$

so $\{Q < 0\}$ is bounded by a hyperbola with asymptotes $|x| = (\sqrt{2} \pm 1)|y|$. Its intersection with the square $[-1, 1]^2$ is four congruent regions, one at each corner $(\pm 1, \pm 1)$, each meeting the two sides through its corner in segments of length exactly $\frac{3}{2} - \sqrt{2}$ and having area

$$\frac{3}{2} - \sqrt{2} + \frac{3}{8} \log \frac{1 + \sqrt{2}}{3} = 0.004321924508 \dots,$$

so the four together occupy 0.4322% of the square.

Proof. The factorisation is an expansion. The quadratic form $4x^2 + 4y^2 - 8\sqrt{2}xy$ has matrix of determinant $16 - 32 < 0$, so it is indefinite and the conic $Q = 0$ is a hyperbola; $\{Q < 0\}$ is therefore *unbounded* in the plane, running out along the diagonals, and it is the square that cuts it into four pieces. In the first quadrant $Q < 0$ reads $4y^2 - 8\sqrt{2}xy + (4x^2 + 3) < 0$, i.e. y strictly between $y_{\pm}(x) = \sqrt{2}x \pm \frac{1}{2}\sqrt{4x^2 - 3}$. Now $y_+(x) > 1$ throughout $[\frac{\sqrt{3}}{2}, 1]$, since $y_+(x) = 1$ would force $\sqrt{2}x \leq 1$; and $y_-(x) \leq 1$ exactly when $x^2 - 2\sqrt{2}x + \frac{7}{4} \leq 0$, i.e. $x \geq \sqrt{2} - \frac{1}{2}$. So inside the square the region is $\{\sqrt{2} - \frac{1}{2} \leq x \leq 1, y_-(x) \leq y \leq 1\}$, which meets the side $y = 1$ in a segment of length $1 - (\sqrt{2} - \frac{1}{2}) = \frac{3}{2} - \sqrt{2}$, and by the symmetry of Q likewise on $x = 1$. Its area is $\int_{\sqrt{2}-1/2}^1 (1 - y_-(x)) dx$, which evaluates to the stated closed form. $\square$

Figure 1 is the criterion: the square, the four regions, and the $n - 1$ image points. It also makes the parity law of Theorem 15 visible. When n is even and k odd the excluded index $j = n/2$ lands on the corner $(-1, +1)$ — inside a forbidden region — and is forgiven, because the eigenvalue it carries is the trivial -3 of a bipartite graph. The exemption of one point is the whole of the parity law.

4 Only finitely many, and a parity law

Both of the following come from the same observation: D_k is negative at the two points $u = \pm 1$ where the grid $\{\cos(2\pi j/n)\}$ accumulates, so the grid must *clear* a band, and a fine grid cannot.

Lemma 9. *For every $k \geq 1$,*

$$D_k(1) = -7, \quad D_k(0) \geq 17, \quad D_k(-1) = \begin{cases} -7, & k \text{ odd}, \\ 9, & k \text{ even}. \end{cases}$$

In particular D_k has a root in $(0, 1)$, and for odd k also one in $(-1, 0)$.

Proof. $T_k(1) = 1$ gives $D_k(1) = 11^2 - 8 \cdot 4^2 = 121 - 128 = -7$. $T_k(-1) = (-1)^k$ gives $D_k(-1) = (7 - 4(-1)^k)^2 - 8(2(-1)^k - 2)^2$, which is (-7) for k odd and $3^2 - 0 = 9$ for k even. Finally $T_k(0) \in \{0, \pm 1\}$, so $D_k(0) = 49 - 32T_k(0)^2 \in \{49, 17\}$. The sign changes give the roots. $\square$

Theorem 10 (Uniform finiteness, for every k). *Let $\tau = 3 - 2\sqrt{2} = (\sqrt{2} - 1)^2$. If $P(n, k)$ satisfies the Riemann Hypothesis then*

$$n \leq \pi(2 + \sqrt{2})\sqrt{k^2 + 1},$$

and if in addition k and n are both odd, $n \leq \frac{1}{2}\pi(2 + \sqrt{2})\sqrt{k^2 + 1}$.

Proof. Write $x = \cos t$, $y = \cos kt$, and $a = 1 - x$, $b = 1 - y$, $s = a + b$. Suppose $s < \tau$; then $a, b < \tau < 1$, so $x, y > 0$ and in particular $x + y > 0$. Hence $|A| = 2(x + y)$ and $P = 4xy$, and

$$G_k(t) = 4\sqrt{2}(x + y) - 4xy = (8\sqrt{2} - 4) - (4\sqrt{2} - 4)s - 4ab.$$

By AM–GM, $4ab \leq s^2$, so $G_k(t) > 7$ is implied by

$$s^2 + (4\sqrt{2} - 4)s < 8\sqrt{2} - 11.$$

The left-hand side is increasing for $s \geq 0$, and at $s = \tau$ it equals

$$(17 - 12\sqrt{2}) + (20\sqrt{2} - 28) = 8\sqrt{2} - 11$$

exactly. So $s < \tau$ suffices.

Now $1 - \cos \theta \leq \theta^2/2$ for every real θ , whence $s \leq (1 + k^2)t^2/2$, and therefore

$$0 < t < \frac{\sqrt{2}\tau}{\sqrt{k^2+1}} = \frac{2-\sqrt{2}}{\sqrt{k^2+1}} \implies G_k(t) > 7,$$

using $\sqrt{2}\tau = \sqrt{2}(\sqrt{2}-1) = 2-\sqrt{2}$. The grid's smallest nonzero angle is $2\pi/n$, so a Ramanujan n needs $2\pi/n \geq (2-\sqrt{2})/\sqrt{k^2+1}$, i.e. $n \leq 2\pi\sqrt{k^2+1}/(2-\sqrt{2}) = \pi(2+\sqrt{2})\sqrt{k^2+1}$.

For the second bound take k odd and put $t = \pi - \varphi$. Then $x = -\cos \varphi$ and, since k is odd, $y = \cos(k\pi - k\varphi) = -\cos k\varphi$. With $a' = 1 + x = 1 - \cos \varphi$ and $b' = 1 + y = 1 - \cos k\varphi$ and $s' = a' + b' < \tau$ we get $x, y < 0$, so $|A| = 2(2 - s')$ and $P = 4(-1 + a')(-1 + b')$, and expanding gives the same expression $(8\sqrt{2} - 4) - (4\sqrt{2} - 4)s' - 4a'b'$. The rest of the argument is verbatim, and $s' \leq (1 + k^2)\varphi^2/2$ as before. For odd n the grid's nearest point to π is at $\varphi = \pi/n$, which gives the stated bound with π/n in place of $2\pi/n$ — exactly half. $\square$

Remark 11. An earlier draft of this work stated the same constant $\pi(2 + \sqrt{2})$, obtained by expanding λ_+ to second order in t at fixed k . That derivation is *invalid*: the critical t scales like $1/k$, so kt is of order one and the expansion was used outside its domain. The constant survived numerical testing only because the bound is loose. The proof above uses no expansion in kt at all — only monotonicity, AM–GM, and $1 - \cos \theta \leq \theta^2/2$ — and holds for every k . The constant was right; the argument was not.

Theorem 12 (Finiteness). *Let u_k be the largest root of D_k in $(-1, 1)$. If $P(n, k)$ satisfies the Riemann Hypothesis then*

$$n \leq \frac{2\pi}{\arccos u_k} =: B_k.$$

In particular, for each k only finitely many n qualify.

Proof. By Lemma 9, $D_k(1) = -7 < 0$, and u_k is the largest root below 1, so $D_k < 0$ throughout $(u_k, 1]$. Suppose $n > B_k$. Then $t = 2\pi/n < \arccos u_k$, so $\cos t > u_k$ and hence $D_k(\cos t) < 0$. The index $j = 1$ is never one of the two excluded indices of Corollary 6 (that would need $n = 2$), so Corollary 6 fails at $j = 1$. $\square$

Theorem 10 is uniform but not sharp; for a *given* k the exact root of D_k does better, and that is what the classification uses.

Theorem 10 bounds n for each k , but the bound grows with k , so it leaves open whether the family contains Ramanujan members of arbitrarily large k . It does not, and the reason is that the failure condition behind Theorem 10 is a *simultaneous Diophantine* one, which Dirichlet's theorem satisfies unconditionally.

Theorem 13 (Absolute finiteness). *If $n \geq 231$ then $P(n, k)$ fails the Riemann Hypothesis for every k . Hence only finitely many generalized Petersen graphs satisfy it, and all of them have $n \leq 230$.*

Proof. The proof of Theorem 10 shows that if

$$s = \left(1 - \cos \frac{2\pi j}{n}\right) + \left(1 - \cos \frac{2\pi jk}{n}\right) < \tau = 3 - 2\sqrt{2}$$

for even one nontrivial index j , then $P(n, k)$ is not Ramanujan. Writing $\|\theta\|$ for the distance from θ to the nearest integer, $1 - \cos 2\pi\theta = 2\sin^2 \pi\theta \leq 2\pi^2\|\theta\|^2$, so it is enough that

$$\left\|\frac{j}{n}\right\|^2 + \left\|\frac{jk}{n}\right\|^2 < \frac{\tau}{2\pi^2} = 0.0086919\dots$$

Let $J = \lfloor \sqrt{n} \rfloor$. Dirichlet's approximation theorem, applied to the real number k/n , provides an integer j with $1 \leq j \leq J$ and $\|jk/n\| \leq 1/(J+1)$. For that j we have $J \leq \sqrt{n} < J+1$, hence

$$\left\| \frac{j}{n} \right\|^2 \leq \left(\frac{J}{n} \right)^2 \leq \frac{1}{n}, \quad \left\| \frac{jk}{n} \right\|^2 \leq \frac{1}{(J+1)^2} < \frac{1}{n},$$

so the left-hand side is less than $2/n$. Therefore $2/n < \tau/(2\pi^2)$ suffices, that is

$$n > \frac{4\pi^2}{\tau} = 230.097 \dots$$

The index is legitimate: $1 \leq j \leq \sqrt{n} < n/2$ once $n \geq 5$, so j is neither 0 nor the index $n/2$ that Corollary 6 excludes. $\square$

Remark 14. The threshold 231 is not sharp — no Ramanujan $P(n, k)$ has n above 112 — and it does not need to be. Its only job is to be uniform in k , which is exactly what the per- k bound of Theorem 12 is not. With it the classification becomes a finite problem in *both* variables: since $1 \leq k < n/2$, everything lives in $3 \leq n \leq 230$, a region that can simply be exhausted.

Theorem 15 (Parity law). *Let k be odd and let v_k be the smallest root of D_k in $(-1, 1)$. If n is odd and $P(n, k)$ satisfies the Riemann Hypothesis then*

$$n \leq \frac{\pi}{\arccos(-v_k)}.$$

No such restriction holds for even n , nor for even k .

Proof. For k odd, $D_k(-1) = -7 < 0$ by Lemma 9, so $D_k < 0$ on $[-1, v_k)$. The grid points nearest $t = \pi$ are $t = \pi(n \pm 1)/n$ when n is odd, at distance π/n , with $\cos t = -\cos(\pi/n)$; and neither index is excluded, since the excluded index $2j = n$ needs n even. So the criterion demands $-\cos(\pi/n) \geq v_k$, i.e. $\pi/n \geq \arccos(-v_k)$.

When n is even the grid contains $t = \pi$ exactly, at $j = n/2$ — and that is precisely the index Corollary 6 excludes, because the sample there is the trivial eigenvalue -3 of the bipartite graph. The band at $u = -1$ is therefore invisible to even n . For even k , $D_k(-1) = 9 > 0$ and there is no band at $u = -1$ at all. $\square$

The parity law is the reason the classification looks the way it does: for odd k the odd values of n stop at roughly half the bound, and the tail of the classification is even. It is also a small structural surprise — *being bipartite helps*. Bipartiteness exempts the eigenvalue -3 from the Ramanujan condition, and for odd k that exemption is exactly what lets n run twice as far.

5 The classification, certified exactly

Theorems 12 and 15 leave a finite computation, but a finite computation is not yet a proof: the decision at a band edge is a comparison of algebraic numbers, and that is where floating point fails silently. We certify every case in exact arithmetic.

The mechanism is the one Corollary 6 makes available. Writing $c_m = 2 \cos(2\pi jm/n)$ we have, by $4 \cos a \cos b = 2 \cos(a+b) + 2 \cos(a-b)$,

$$A = c_1 + c_k, \quad P = c_{k+1} + c_{k-1},$$

so A , P and hence D are values at $z = \zeta_n^j$ of Laurent polynomials with integer coefficients. Since ζ_n^j is a primitive m -th root of unity for $m = n/\gcd(n, j)$, each value is an algebraic integer of $\mathbb{Z}[\zeta_m]$, and reducing modulo the cyclotomic polynomial Φ_m puts it in an exact integer coordinate vector. That vector vanishes if and only if the value is exactly zero – so the band-edge cases are decided by integer arithmetic, with no estimate anywhere.

For the sign of a nonzero value we use the separation bound available to any nonzero algebraic integer: the product of its conjugates is a nonzero rational integer, so $|D| \geq 1/\prod_{\sigma \neq 1} |\sigma(D)| \geq 249^{-(\deg - 1)}$, using $|D| \leq (7 + 4)^2 + 8 \cdot 4^2 = 249$ on the whole conjugate set. We evaluate at a precision set from that bound and *refuse* to report a sign unless the computed value clears the guard, so a wrong sign is excluded rather than improbable.

Theorem 16 (Classification, per k). *For every k and every n , whether $P(n, k)$ satisfies the Riemann Hypothesis is decided by Table 1: for each k listed there, the n shown are exactly those that qualify, and no $n > B_k$ qualifies, by Theorem 12. In particular there are exactly 247 such graphs with $k \leq 10$, from 458 cases below the bound each decided in exact arithmetic. That no k beyond the table admits any n at all is Theorem 13, and the total is Theorem 18.*

The bound is not merely finite but frequently sharp: at $k = 2, 3, 4$ the largest Ramanujan n equals $\lfloor B_k \rfloor$ exactly, so the band at $u = 1$ is the only obstruction there, and the parity bound of Theorem 15 is attained at $k = 1, 5, 7, 9$.

A floating-point classifier was run alongside the exact one throughout, as a control with no authority: a disagreement is recorded as a failure of the run rather than resolved in favour of either. It earned its place. At $k = 1$ the exponents of the Laurent polynomials collide ($k = 1$ and $k - 1 = -(k - 1) = 0$), and an implementation that built them as a dictionary literal silently discarded the duplicated terms; the exact classifier then declared every $P(n, 1)$ Ramanujan, including $P(9, 1)$, whose spectrum contains $2 \cos(8\pi/9) - 1 = -2.879$. The control caught all four affected values.

Remark 17. The classification is genuinely irregular in k . The largest Ramanujan n is not monotone — $k = 9$ reaches $n = 90$ while $k = 8$ and $k = 10$ stop at 49 and 52 — because for even k a third band, interior to $(0, \pi)$ rather than at an endpoint, binds well before the band at $u = 1$. Locating those interior bands for general k is exactly the question of which cyclotomic reals avoid a semialgebraic set, and is where the Galois-orbit machinery of Part II applies.

6 The classification for every k

Theorem 13 confines the whole family to $3 \leq n \leq 230$, and with $1 \leq k < n/2$ that region is finite. Exhausting it settles the question completely.

Theorem 18 (Complete classification). *There are exactly 460 pairs (n, k) with $n \geq 3$ and $1 \leq k < n/2$ for which $P(n, k)$ satisfies the Riemann Hypothesis. They fall into 324 isomorphism classes. The largest is $n = 112$, attained at $k = 41$; the largest k that occurs at all is 45; and the values of k that occur are $1, \dots, 25$ together with the odd numbers from 27 to 45. Every one of the 13110 cases in the region was decided in exact arithmetic.*

Two features of the answer are worth isolating. The first is that the list is finite in *both* variables, which Theorem 10 alone does not give: for each k it allows n up to $\pi(2 + \sqrt{2})\sqrt{k^2 + 1}$, a bound that grows without limit, and only the Diophantine argument rules out large k . The second is the parity asymmetry at the top of the range. Above $k = 25$ no even k occurs at all, while odd k

persists as far as 45; this is the parity law of Theorem 15 still operating at the end of the list, where the exemption of the trivial -3 is the only thing keeping a member alive.

Remark 19 (Counting graphs rather than parameters). 460 counts parameter pairs. $P(n, k) \cong P(n, \ell)$ exactly when $\ell \equiv \pm k^{\pm 1} \pmod{n}$, and quotienting by that relation leaves 324 graphs up to isomorphism. We report both because the first is what the criterion decides and the second is what the classification is *about*; conflating them would overstate the count by a third.

7 Towards a closed form

Theorem 18 is a certified list, not a formula, and it is worth being precise about what a formula would have to be. This section gives the structure the list has; it stops short of a closed form, and §7 ends by saying exactly where.

The first observation is that the criterion is *local at the divisors of n* . Call m the *level* of the index j if $m = n / \gcd(n, j)$; then $\cos \frac{2\pi j}{n}$ is a primitive m -th value, and as j ranges over all indices of level m the point ranges over the full conjugate set $\{\cos \frac{2\pi i}{m} : \gcd(i, m) = 1\}$. Say m is *bad for k* if some primitive level- m point violates $Q \geq 0$.

Theorem 20 (Locality). *$P(n, k)$ satisfies the Riemann Hypothesis if and only if no divisor $m \geq 3$ of n is bad for k . Moreover badness at level m depends on k only through $k \bmod m$.*

Proof. Every index $j \in \{1, \dots, n-1\}$ has a level $m \mid n$, and $m \geq 3$ unless $j = 0$ or $2j = n$ — the two indices Corollary 6 excludes. Level $m = 2$ never obstructs in any case: there $u = -1$, and $D_k(-1) = 9 > 0$ for even k by Lemma 9, while for odd k the cover is bipartite and the point carries the exempt trivial -3 . So the criterion holds at every index exactly when it holds at every primitive point of every level $m \geq 3$ dividing n . The last claim holds because the point $(\cos \frac{\pi j(k+1)}{m}, \cos \frac{\pi j(k-1)}{m})$ is unchanged when k is replaced by $k + m$. $\square$

Corollary 21 (Divisor closure). *If $P(n, k)$ satisfies the Riemann Hypothesis and $d \mid n$ with $d > 2k$, then $P(d, k)$ does too. For each k the set of Ramanujan n is closed downwards under division, so it is determined by its minimal bad divisors — and by Theorem 12 only those below B_k can matter.*

Proof. The divisors of d are divisors of n , so if none of the latter is bad for k then none of the former is; apply Theorem 20 twice. $\square$

That reduces the classification to knowing which $(m, k \bmod m)$ are bad. The next lemma is what converts that into arithmetic, and it is where the gcd enters.

Lemma 22 (The minimum is a gcd). *Let $\|\cdot\|$ denote distance to the nearest integer. If $m \nmid c$ then*

$$\min \left\{ \left\| \frac{jc}{m} \right\| : \gcd(j, m) = 1 \right\} = \frac{\gcd(c, m)}{m}.$$

If $m \mid c$ the minimum is 0.

Proof. Write $g = \gcd(c, m)$, $c = ga$, $m = gb$ with $\gcd(a, b) = 1$, and $b > 1$ since $m \nmid c$. Then $jc \equiv g(ja) \pmod{m}$, so $\|jc/m\| = \|ja/b\|$ scaled: the attainable residues are gt with $t \equiv ja \pmod{b}$. As j runs over the units mod m , $j \bmod b$ runs over the units mod b , and a is a unit there, so $ja \bmod b$ runs over every unit of b — in particular over 1. The smallest nonzero residue attainable is therefore g , giving g/m , and 0 is not attainable because $b \nmid ja$ for j a unit. $\square$

Now the geometry pays. Corollary 8 showed the forbidden region meets the sides of the square at distance $\frac{3}{2} - \sqrt{2}$ from each corner, so $Q < 0$ forces *both* $|x|$ and $|y|$ to be at least $\sqrt{2} - \frac{1}{2}$. Since $|\cos \pi t| \geq \sqrt{2} - \frac{1}{2}$ exactly when $\|t\| \leq \gamma$ for

$$\gamma = \frac{\arccos(\sqrt{2} - \frac{1}{2})}{\pi} = 0.1328095\dots, \quad \frac{1}{\gamma} = 7.5295\dots,$$

a level is safe as soon as one coordinate cannot come close enough — and by Lemma 22 that is a divisibility condition. Because $m/\gcd(k \pm 1, m)$ is an integer and $1/\gamma$ is just above 7, it is a condition on small integers.

Theorem 23 (Gcd safety). *If $m/\gcd(k+1, m) \in \{2, \dots, 7\}$ or $m/\gcd(k-1, m) \in \{2, \dots, 7\}$, then m is not bad for k .*

Proof. Suppose $m/\gcd(k+1, m) = b$ with $2 \leq b \leq 7$. Then $m \nmid k+1$, so Lemma 22 gives $\|j(k+1)/m\| \geq \gcd(k+1, m)/m = 1/b \geq 1/7 > \gamma$ for every j coprime to m . Hence $|x| = |\cos \frac{\pi j(k+1)}{m}| < \sqrt{2} - \frac{1}{2}$ at every primitive level- m point, and $Q \geq 0$ there by Corollary 8. The other case is identical. $\square$

The value $b = 1$ is excluded for a reason: $m \mid k+1$ pins $|x| = 1$, which is the worst case rather than the best, and Lemma 22 gives minimum 0.

Theorem 20 already gives the classification a pure form for *every* k : $P(n, k)$ is Ramanujan iff no divisor $m \geq 3$ of n is bad for $k \bmod m$, and badness of a level is witnessed by a single index j with $Q < 0$, an exact one-line check. Since $m \leq 231$ by Theorem 13, the whole classification is one theorem plus a finite table of minimal bad levels (a few thousand pairs $(m, k \bmod m)$, generated and stored with the verification record); the list of 460 pairs is a consequence of that table, not the primary object. For small k the two bounds already in hand turn out to be the whole answer, and this can be *proved*, not merely checked: everything depends on where D_k is negative, and that is a question about the real roots of one integer polynomial, which Sturm's theorem decides in integer arithmetic.

Lemma 24 (Band structure). *Let u_k be the largest root of D_k in $(-1, 1)$. For $k \in \{1, 2, 3, 4, 5, 7, 9\}$ the set where $D_k < 0$ on $[-1, 1]$ is exactly*

$$(u_k, 1] \quad (k \text{ even}), \quad [-1, -u_k) \cup (u_k, 1] \quad (k \text{ odd}).$$

For $k = 6, 8$ and for every k with $10 \leq k \leq 45$, D_k has a further root in $(-1, 1)$ and is negative on an interval in the interior.

Proof. For odd k the Chebyshev polynomial T_k is odd, so $uT_k(u)$ is even and $(2u + 2T_k(u))^2$ is even; hence $D_k(-u) = D_k(u)$ and the roots come in pairs $\pm u$, with $v_k = -u_k$ in the notation of Theorem 15. For even k , $D_k(-1) = 9 > 0$ by Lemma 9, so there is no band at -1 . The number of real roots of D_k in $[-1, 1]$, and the sign of D_k between consecutive roots, are decided by Sturm's theorem, a finite computation in integer arithmetic on the coefficients of $D_k \in \mathbb{Z}[u]$; carried out exactly for $k \leq 45$ (`verification/interior_bands.py`, which isolates the roots over $\mathbb{Q}$ and evaluates D_k at rational points between them), it finds one root in $(-1, 1)$ for $k = 2, 4$, two for $k = 1, 3, 5, 7, 9$, and at least three for every other $k \leq 45$, with the signs stated. For $k = 2$ the whole computation fits on a page: $D_2(u) = 64u^6 - 192u^4 - 16u^3 + 112u^2 + 8u + 17$ has $D_2(-1) = 9$, $D_2(0.96) = 0.765\dots > 0$ and $D_2(1) = -7$; Sturm's theorem gives D_2 no root in $[-1, 0.96]$, and D'_2 no root in $(0.96, 1)$, so D_2 is positive on $[-1, 0.96]$ and crosses zero exactly once in $(0.96, 1)$, at $u_2 = 0.9641\dots$ $\square$

Theorem 25 (Closed form for $k \leq 9$). *Let $B_k = 2\pi / \arccos u_k$ be the bound of Theorem 12 and $P_k = \pi / \arccos u_k = B_k/2$ that of Theorem 15. Then*

$$\begin{aligned} k = 2, 4 : \quad & P(n, k) \text{ is Ramanujan} \iff 2k < n \leq B_k; \\ k = 1, 3, 5, 7, 9 : \quad & P(n, k) \text{ is Ramanujan} \iff 2k < n \leq B_k \text{ and } (n \text{ even or } n \leq P_k). \end{aligned}$$

Equivalently, for odd k : $2k < n \leq B_k$ and every odd divisor of n is at most P_k .

Proof. By Theorem 5, $P(n, k)$ is Ramanujan iff $D_k(\cos \frac{2\pi j}{n}) \geq 0$ for every j with $1 \leq j \leq n-1$ and $j \neq n/2$. By Lemma 24, $D_k(\cos t) < 0$ exactly when $\cos t > u_k$, or k is odd and $\cos t < -u_k$.

The band at 1. Among $1 \leq j \leq n-1$ the largest value of $\cos \frac{2\pi j}{n}$ is $\cos \frac{2\pi}{n}$, at $j = 1$ and $j = n-1$. So some grid point lies in $(u_k, 1]$ iff $\cos \frac{2\pi}{n} > u_k$ iff $n > B_k$.

The band at -1 (odd k). Write $d = n-2j$, so that $\cos \frac{2\pi j}{n} = -\cos \frac{\pi d}{n}$, and $d \neq 0$ because $j = n/2$ is the excluded trivial index. A grid point lies in $[-1, -u_k)$ iff $\cos \frac{\pi d}{n} > u_k$ for some admissible d , and the admissible $|d|$ are the positive integers of the same parity as n . The smallest is 2 when n is even, giving the condition $\cos \frac{2\pi}{n} > u_k$ again, i.e. $n > B_k$; and 1 when n is odd, giving $\cos \frac{\pi}{n} > u_k$, i.e. $n > P_k$.

The lower bound $2k < n$ is the standing hypothesis $k < n/2$. For the equivalent form: an even $n \leq B_k$ has every odd divisor at most $n/2 \leq P_k$, and an odd n is its own odd divisor. $\square$

For even k there is no band at $u = -1$ at all, which is why $k = 2$ and $k = 4$ are unbroken runs; for odd k the second band contributes exactly the constraint that n carry no large odd divisor, which by Corollary 21 is the natural form for a divisor-closed set to take. The thresholds themselves — $\lfloor B_k \rfloor$ and $\lfloor P_k \rfloor$ — are $\lfloor B_k \rfloor = 23, 32, 42, 50, 70, 90$ at $k = 2, 3, 4, 5, 7, 9$ (Table 1); deciding whether $\cos \frac{2\pi}{n} \leq u_k$ at the boundary n is one exact comparison in $\mathbb{Z}[\zeta_n]$, as in §5.

Remark 26 (What a closed form still needs, and why we do not have one). Three things are missing. Theorem 23 is *sufficient and one-sided*: it certifies safety and says nothing about badness, and a closed form needs both directions. Theorem 25 stops at $k = 6, 8$ and at every $k \geq 10$ — precisely where Lemma 24 finds an interior band, neither at $u = 1$ nor at $u = -1$, and locating which grid points fall into those bands is the question of which cyclotomic reals avoid a semialgebraic set that §5 already flagged. And nothing here explains the shape of the answer at the top of the range: we can prove no $P(n, k)$ with $k > 45$ is Ramanujan, but only by exhausting the finitely many cases Theorem 13 leaves, and we cannot say why the cut falls at 45 rather than near it. The list is complete and certified; it is not yet explained.

8 Every cubic cyclic cover

Nothing in §3 used $P(n, k)$ beyond two facts: the base has two vertices, and the cover is cubic. The criterion therefore belongs to the setting of §18 rather than to this family, and saying so turns a classification of one family into a statement about a class of covers.

A cubic base on two vertices is one of exactly two things. A loop absorbs two of the three half-edges at its vertex, so a vertex carries at most one loop, and a vertex with a loop has exactly one non-loop edge. Either both vertices carry loops — the *two-loop base* $B(a, b, c)$, with a loop of voltage a at u , a loop of voltage b at v , and an edge $u-v$ of voltage c — or neither does, leaving three parallel edges, the *theta base* $T(c_1, c_2, c_3)$. We have $P(n, k) = B(1, k, 0)^n$, and the covers of a theta base are the cubic cyclic Haar graphs, which are bipartite.

Theorem 27 (Corner criterion on a two-loop base). *Let $X = B(a, b, c)^n$ be simple and connected. Then X satisfies the Riemann Hypothesis if and only if*

$$Q\left(\cos \frac{\pi j(a+b)}{n}, \cos \frac{\pi j(a-b)}{n}\right) \geq 0$$

at every nontrivial index, with Q the form of Theorem 7. In particular the spectrum, and hence the answer, does not depend on the edge voltage c at all.

Proof. By Theorem 65 the character block is $M_j = \begin{pmatrix} \alpha_j & \omega^{jc} \\ \omega^{-jc} & \beta_j \end{pmatrix}$ with $\alpha_j = 2 \cos \frac{2\pi ja}{n}$ and $\beta_j = 2 \cos \frac{2\pi jb}{n}$. Its characteristic polynomial is $\lambda^2 - (\alpha + \beta)\lambda + (\alpha\beta - |\omega^{jc}|^2)$, and $|\omega^{jc}| = 1$, so it is $\lambda^2 - A\lambda + (P - 1)$ exactly as in Lemma 4 — with no c in it. The derivation of Theorem 5 used only $|\alpha|, |\beta| \leq 2$, so it applies verbatim, and the substitution of Theorem 7 applies with $a + b$ and $a - b$ in place of $k + 1$ and $k - 1$, because $\alpha + \beta = 4 \cos \frac{\pi j(a+b)}{n} \cos \frac{\pi j(a-b)}{n}$ and $\alpha\beta = 2\left(\cos \frac{2\pi j(a+b)}{n} + \cos \frac{2\pi j(a-b)}{n}\right)$. $\square$

Theorem 28 (Theta base). *$T(c_1, c_2, c_3)^n$ satisfies the Riemann Hypothesis if and only if*

$$\sum_{l < m} \cos \frac{2\pi j(c_l - c_m)}{n} \leq \frac{5}{2} \quad \text{for every } j \neq 0.$$

Proof. The block is $\begin{pmatrix} 0 & S_j \\ \overline{S_j} & 0 \end{pmatrix}$ with $S_j = \sum_l \omega^{jc_l}$, so its eigenvalues are $\pm |S_j|$ and the cover is bipartite. Now $|S_j|^2 = 3 + 2 \sum_{l < m} \cos \frac{2\pi j(c_l - c_m)}{n}$, and the Ramanujan condition on a nontrivial index is $|S_j|^2 \leq 8$. $\square$

Theorem 29 (Uniform finiteness on both bases). *Writing voltages by their representatives of least absolute value, and d_1, d_2, d_3 for the pairwise differences $c_l - c_m$,*

$$\begin{aligned} B(a, b, c)^n \text{ Ramanujan} &\implies n \leq \pi(2 + \sqrt{2})\sqrt{a^2 + b^2}, \\ T(c_1, c_2, c_3)^n \text{ Ramanujan} &\implies n \leq 2\pi\sqrt{d_1^2 + d_2^2 + d_3^2}. \end{aligned}$$

Proof. The first is the proof of Theorem 10 with $\cos \frac{2\pi ja}{n}$ and $\cos \frac{2\pi jb}{n}$ in place of $\cos t$ and $\cos kt$; the estimate $1 - \cos \theta \leq \theta^2/2$ now gives $s \leq (a^2 + b^2)\theta^2/2$, and setting $\theta = 2\pi/n$ gives the bound. Specialising to $(a, b) = (1, k)$ returns Theorem 10 exactly. For the second, the condition of Theorem 28 fails as soon as $\sum_l (1 - \cos \frac{2\pi jd_l}{n}) < \frac{1}{2}$, and that sum is at most $\theta^2 \sum_l d_l^2/2$; at $j = 1$ this forces failure once $2\pi/n < 1/\sqrt{\sum d_l^2}$. $\square$

The second bound is sometimes attained exactly: for $T(0, 1, 2)$ it reads $n \leq 2\pi\sqrt{6} = 15.39\dots$, and $n = 15$ is Ramanujan.

Both bases are therefore finite, and the reason is not special to two-vertex bases.

Theorem 30 (No infinite family is a cyclic cover). *Let B be a fixed finite connected cubic base with integer voltages. Then B^n is Ramanujan for only finitely many n . Explicitly, writing L for $\sqrt{2} \sum |v_e|$ over the non-loop edges and R for $\sum v_e^2$ over the loops, B^n is not Ramanujan as soon as*

$$\frac{2\pi}{n}L + \left(\frac{2\pi}{n}\right)^2 R < \tau = 3 - 2\sqrt{2}.$$

Proof. M_0 is the adjacency matrix of the base, which is 3-regular, so its largest eigenvalue is 3. A non-loop edge of voltage v moves two entries of M_1 by $|\omega^v - 1| \leq 2\pi|v|/n$ each, and a loop of voltage v moves one entry by $|2\cos \frac{2\pi v}{n} - 2| \leq (2\pi v/n)^2$; summing gives $\|M_1 - M_0\|_2 \leq \|M_1 - M_0\|_F \leq \frac{2\pi}{n}L + (\frac{2\pi}{n})^2R$. By Weyl's inequality $\lambda_{\max}(M_1) \geq 3 - \|M_1 - M_0\|_2$, and $\lambda_{\max}(M_1)$ is a nontrivial eigenvalue, since in a connected cover the eigenvalue 3 has multiplicity one and occurs at $j = 0$. So once the displayed quantity is below $3 - 2\sqrt{2}$ the cover has a nontrivial eigenvalue exceeding $2\sqrt{2}$. $\square$

Remark 31 (What this is, and is not). Theorem 30 says that no construction of the kind this paper studies can produce an infinite family of Ramanujan graphs — the no-go is a property of the class, not a limitation of our search. We claim no novelty for the principle: it is the cyclic instance of the familiar fact that abelian covers do not expand, which is precisely why the Lubotzky–Phillips–Sarnak construction [10] is built on PGL_2 and not on a cyclic group. What is new here is the explicit constant, and the two exact criteria of Theorems 27 and 28 for the two-vertex bases, which decide the finitely many surviving cases rather than merely bounding them.

9 The deficit: what the no-go costs

A classification that ends in a finite list tells a designer that beyond a certain size there is nothing optimal to build. That is not by itself useful advice. The useful quantity is how much is lost, and the criterion supplies it in closed form.

Definition 32. For a cover with nontrivial spectral radius λ_2 , the *deficit* is $\delta = \lambda_2 - 2\sqrt{2}$, and for a given size, $\delta^*(n) = \min_{1 \leq k < n/2} \delta(n, k)$.

$\delta^*(n) \leq 0$ exactly when some $P(n, k)$ is Ramanujan, which by Theorem 18 happens for exactly 80 values of n , the largest being 112.

Proposition 33. $\delta(n, k) < 3 - 2\sqrt{2}$ for every n and k , and $\sup_{n,k} \delta(n, k) = 3 - 2\sqrt{2}$, approached as $n \rightarrow \infty$.

Proof. $\lambda_2 < 3$ because 3 is simple in a connected cubic graph, which gives the bound. For the supremum, at $j = 1$ the block M_1 tends to M_0 as $n \rightarrow \infty$, so its top eigenvalue tends to 3, and it is nontrivial. $\square$

So the loss is bounded, and the bound is exactly the gap τ that the whole argument has been about: an engineer who builds at a size where no optimal member exists gives up at most $3 - 2\sqrt{2} = 0.1716\dots$ in nontrivial spectral radius, and that is the worst case rather than a typical one. Figure 2 plots δ^* . Its two branches are the parity law once more: for one parity of n the exemption of the trivial -3 is available and for the other it is not.

Part II

Maximum Nullity and Zero Forcing on Cyclic Covers

10 Background

Given a set S of *filled* vertices of a graph G , the colour change rule says: if a filled vertex has exactly one unfilled neighbour, that neighbour becomes filled. Write $\text{cl}(S)$ for the closure of S under repeated application. S is a *zero forcing set* if $\text{cl}(S) = V$, and $Z(G)$ is the least size of one. $Z(G)$ bounds the maximum nullity over the symmetric matrices with G 's off-diagonal pattern, and arose independently in the control of quantum systems and in sensor placement for power networks.

The families we treat all have $2n$ or $4n$ vertices, are cubic, and carry the rotation $\rho : i \mapsto i + 1$ acting on index subscripts.

- $P(n, k)$: outer $u_i u_{i+1}$, inner $v_i v_{i+k}$, spokes $u_i v_i$.
- $I(n, j, k)$: outer $u_i u_{i+j}$, inner $v_i v_{i+k}$, spokes $u_i v_i$; thus $P(n, k) = I(n, 1, k)$.
- $DP(n, k)$: two outer cycles $u_i u_{i+1}$ and $x_i x_{i+1}$, inner edges $w_i y_{i+k}$ and $y_i w_{i+k}$, spokes $u_i w_i$ and $x_i y_i$.

Rashidi, Shajareh Poursalavati and Tavakkoli [6] proved $Z(P(n, k)) \leq 2k+2$ and $Z(P(n, 2)) = 6$ for $n \geq 10$. Their Theorem 3.6, claiming $Z(P(n, 3)) = 8$ for $n \geq 12$, is false: Krishnan [7] exhibits $Z(P(12, 3)) = 7$, the flaw being a case analysis that never excludes 7-vertex sets. That note states as its Conjecture 5 that $Z(P(n, 3)) = 8$ for every $n \geq 13$, identifying the missing ingredient as “a lower-bound proof valid for all large n ”, and closes by recording that

“Determining $Z(P(n, k))$ for general k — the stabilization threshold and a rigorous lower bound for $k \geq 4$ — remains open.”

We settle all three. Theorem 111 proves Conjecture 5 and so fixes the threshold at 13; Theorems 69 and 109 give rigorous lower bounds, indeed exact values, for $k = 4$, and determine the whole $k = 4$ sequence.

11 The rotation bootstrap

The following two lemmas are elementary; the point is that together they reduce an “all n ” statement to one finite check.

Lemma 34 (Bootstrap). *Let ρ be an automorphism of a finite graph G and let $S \subseteq V(G)$ satisfy $\rho(S) \subseteq \text{cl}(S)$. Then $\rho(\text{cl}(S)) = \text{cl}(S)$.*

Proof. Closure commutes with automorphisms, and is monotone and idempotent, so $\rho(\text{cl}(S)) = \text{cl}(\rho(S)) \subseteq \text{cl}(\text{cl}(S)) = \text{cl}(S)$. As ρ permutes the finite set $V(G)$, the two sets have equal cardinality, hence are equal. $\square$

Lemma 35 (Closing). *Let G be cubic with vertex set $O \sqcup I$ such that every vertex of I has exactly one neighbour in O , every vertex of O has exactly one neighbour in I , and the remaining neighbours of each vertex of O lie in O . If $O \subseteq \text{cl}(S)$ then $\text{cl}(S) = V(G)$.*

Proof. Suppose some $w \in I$ is unfilled and let $o \in O$ be its unique neighbour in O . The other two neighbours of o lie in O , hence are filled, so w is the unique unfilled neighbour of the filled vertex o and is forced. $\square$

To apply these to a family it suffices to exhibit one set S , described uniformly in n , and verify $\rho(S) \subseteq \text{cl}(S)$ by an explicit finite forcing chain. Since ρ 's orbit of any outer vertex is all of O , Lemmas 34 and 35 then give $\text{cl}(S) = V$ and so $Z(G) \leq |S|$, for every n in range.

12 I-graphs

Theorem 36. *For all $j, k \geq 1$ and $n \geq 2(j + k) + 1$, $Z(I(n, j, k)) \leq 2(j + k)$.*

Proof. Put $m = 2(j + k)$ and $S = \{u_0, \dots, u_{m-1}\}$, so $|S| = m$. Since $n \geq m + 1$ the indices $0, \dots, m$ are pairwise distinct mod n .

Step 1. The outer neighbours of u_i are $u_{i \pm j}$, both in S precisely when $j \leq i \leq m - 1 - j$. For such i the spoke neighbour v_i is therefore the unique unfilled neighbour of u_i and is forced. This fills $v_j, \dots, v_{m-1-j}$, that is $v_j, \dots, v_{j+2k-1}$.

Step 2. Consider v_{j+k} . Its neighbours are $u_{j+k} \in S$ (as $j + k \leq m - 1$), v_j , filled in Step 1, and v_{j+2k} , which is not: Step 1 filled indices at most $j + 2k - 1$. So v_{j+k} forces v_{j+2k} .

Step 3. Consider $u_{j+2k} = u_{m-j}$. Its neighbours are $u_{2k} \in S$ (as $2k \leq m - 1$, using $j \geq 1$), the vertex v_{j+2k} filled in Step 2, and u_m , which is unfilled. So u_{j+2k} forces u_m .

Hence $\rho(S) = \{u_1, \dots, u_m\} \subseteq \text{cl}(S)$. By Lemma 34, $\text{cl}(S)$ is ρ -invariant; it contains u_0 , whose ρ -orbit is all of $O = \{u_0, \dots, u_{n-1}\}$, so $O \subseteq \text{cl}(S)$, and Lemma 35 gives $\text{cl}(S) = V$. $\square$

Corollary 37. *For $k \geq 1$ and $n \geq 2k + 3$, $Z(P(n, k)) \leq 2k + 2$.*

Proof. $P(n, k) = I(n, 1, k)$; take $j = 1$ in Theorem 36. $\square$

Remark 38. The corollary is [6, Thm. 2.2]. We note it because that paper is known to contain a false theorem, and because the corrected argument given in [7] is written only for $k = 3$; the proof above is uniform in j and k . Every step of the chain has been verified by computer for $1 \leq j \leq 4$, $1 \leq k \leq 5$ and six values of n per pair (106 cases).

Remark 39 (Scope). Theorem 36 should not be read as $2(j + k)$ new bounds. If $\gcd(n, j) = 1$ then multiplying indices by $j^{-1} \bmod n$ is an isomorphism $I(n, j, k) \cong I(n, 1, j^{-1}k) = P(n, j^{-1}k)$, so those graphs are generalized Petersen graphs and the corollary already applies. We verified this in five cases, e.g. $I(15, 2, 4) \cong P(15, 2)$ with $Z = 6$ on both sides and $I(13, 2, 5) \cong P(13, 4)$ with $Z = 8$. The genuinely new content of Theorem 36 is therefore confined to $\gcd(n, j) > 1$ and $\gcd(n, k) > 1$.

In that regime the bound is usually far from sharp: among the cases we computed it is attained only at $I(18, 2, 3)$ ($Z = 10 = 2(2 + 3)$), while e.g. $I(20, 5, 8)$ has $Z = 10$ against a bound of 26. The reason is structural. When $\gcd(n, j) = d > 1$ the outer vertices form d cycles of length n/d rather than one cycle, and the set $\{u_0, \dots, u_{m-1}\}$ used in the proof spreads across all of them; a smaller, better-adapted set suffices. We record the bound as proved and its sharpness in this regime as open.

13 Double generalized Petersen graphs

Theorem 40. *For all $k \geq 1$ and $n \geq 2k + 3$, $Z(DP(n, k)) \leq 4k + 4$.*

Proof. Put $S = \{u_0, \dots, u_{2k+1}\} \cup \{x_0, \dots, x_{2k+1}\}$, so $|S| = 4k + 4$.

Step 1. For $1 \leq i \leq 2k$ the outer neighbours $u_{i\pm 1}$ of u_i lie in S , so u_i forces w_i ; symmetrically x_i forces y_i . This fills $w_1, \dots, w_{2k}$ and $y_1, \dots, y_{2k}$.

Step 2. The neighbours of w_{k+1} are $u_{k+1} \in S$, y_1 (filled) and y_{2k+1} (not), so w_{k+1} forces y_{2k+1} . Symmetrically y_{k+1} forces w_{2k+1} .

Step 3. The neighbours of u_{2k+1} are $u_{2k} \in S$, w_{2k+1} (filled in Step 2) and u_{2k+2} , so u_{2k+1} forces u_{2k+2} ; symmetrically x_{2k+1} forces x_{2k+2} .

Hence $\rho(S) \subseteq \text{cl}(S)$. The outer set here is $O = \{u_i\} \cup \{x_i\}$, on which ρ acts with two orbits, both met by $\text{cl}(S)$; Lemmas 34 and 35 finish as before. $\square$

No zero forcing results for $DP(n, k)$ were found in the literature; the family is studied for Hamiltonicity, super-connectivity and its automorphism group. Every step above was verified by computer for $1 \leq k \leq 5$ and eight values of n per k .

14 Computed values

All values are exact, by exhaustive search over vertex subsets with a 128-bit mask closure, reduced by ρ : a minimum forcing set may be assumed to contain a fixed orbit representative. The implementation reproduced every published value before use — Krishnan's table of $Z(P(n, 3))$ for $7 \leq n \leq 20$ (including the corrected $Z(P(12, 3)) = 7$ and the non-monotone values at $n = 10, 11$), $Z(P(n, 2)) = 6$ for $10 \leq n \leq 16$, and $Z(P(2k + 1, k)) = 6$ for $k = 5, 6, 7$ — and was independently cross-checked against a brute-force implementation and two exact solver formulations.

Conjecture 41. *For each fixed k and each $d \mid k$, the values $Z(P(n, k))$ over n with $\gcd(n, k) = d$ are eventually constant and equal to $2k + 2$.*

An earlier form of this conjecture, asserting a single threshold $n_0(k) = 5k - 2$ beyond which $Z(P(n, k)) = 2k + 2$ for all n , is false. It fits $k = 3, 4, 5$, and predicted that $n = 28$ would be the first n with $Z(P(n, 6)) = 14$; that prediction, recorded before the computation finished, was borne out. But $Z(P(29, 6)) = 12$, and $29 > 28$, so 28 is not a stabilization point and the single-threshold form fails at $k = 6$.

The explanation is $\gcd(n, k)$. Sorting the $k = 6$ data by $d = \gcd(n, 6)$:

d	n	$Z(P(n, 6))$
1	13, 17, 19, 23, 25, 29	6, 8, 8, 10, 10, 12
2	14, 16, 20, 22, 26, 28	8, 9, 10, 12, 12, 14
3	15, 21, 27	9, 12, 13
6	18, 24	9, 12

Each class climbs on its own schedule: the class $d = 2$ attains 14 at $n = 28$ while the coprime class has only reached 12 by $n = 29$. At $k = 4$ and $k = 5$ all classes had converged within the computed range, which is why a single threshold appeared to fit; at $k = 6$ they separate. The thresholds 10, 13, 18, 23 for $k = 2, 3, 4, 5$ should therefore be read as the point at which the slowest class happened to arrive, not as values of a formula.

For DP , our computed values give $Z(DP(n, 1)) = 8 = 4 \cdot 1 + 4$ for $6 \leq n \leq 12$ and $Z(DP(n, 2)) = 12 = 4 \cdot 2 + 4$ for $n = 10, 11, 14, 15$, so Theorem 40 is attained at both $k = 1$ and $k = 2$: the bound is sharp there, not merely an upper estimate.

15 Towards a sharper bound for I-graphs

Theorem 36 is stated in terms of j and k , but the outer and inner edge sets of $I(n, j, k)$ decompose into $d_1 = \gcd(n, j)$ and $d_2 = \gcd(n, k)$ cycles respectively, and it is these that appear to govern Z . In the regime where Theorem 36 carries new content (Remark 39), it is markedly non-optimal: $I(24, 9, 10)$ has $Z = 10$ against $2(j + k) = 38$. The proof uses $2(j + k)$ *consecutive* outer vertices, which is wasteful exactly when j is large but $\gcd(n, j)$ is small, since consecutive indices stride across all d_1 outer cycles.

Observation 42. Among the 20 graphs we computed with $d_1, d_2 > 1$, every one satisfies $Z(I(n, j, k)) \leq 2(d_1 + d_2)$, with equality at $I(18, 2, 3)$, $I(18, 3, 8)$, $I(24, 3, 10)$ and $I(24, 9, 10)$.

This must be stated as an observation confined to that regime, and *not* as a bound in general. Taking $j = 1$ gives $d_1 = 1$, and if additionally $\gcd(n, k) = 1$ then $2(d_1 + d_2) = 4$, whereas $Z(P(n, k)) = 2k + 2$ grows without bound: $Z(P(11, 3)) = 7$ and $Z(P(17, 4)) = 9$ both exceed 4. So any bound in terms of d_1, d_2 alone is false, and the true quantity must retain a dependence on j and k . We flag this because the restricted observation is nonetheless suggestive: the four equality cases all have $2(j + k)$ slack by between 12 and 28, so Theorem 36 is certainly not optimal in the regime where it carries new content.

A supporting structural fact, found by computing minimum *bootstrapping* sets (those S with $\rho(S) \subseteq \text{cl}(S)$, which is what a uniform-in- n proof requires): for $I(18, 3, 8)$ the minimum such set has size $10 = 2(d_1 + d_2)$ against $2(j + k) = 22$, and decomposes as three consecutive vertices within each of the $d_1 = 3$ outer cycles together with one inner vertex. Three consecutive vertices in an outer cycle is exactly the configuration that lets the middle one force its spoke, so a proof adapted to the cycle decomposition rather than to consecutive indices should beat Theorem 36. We have not carried this out.

Remark 43 (Limits of verification). Sharpness of Theorem 36 can be tested only for very small (j, k) . The hypothesis $n \geq 2(j + k) + 1$ forces n upward while exhaustive search costs $\binom{2n-1}{2(j+k)-1}$, so the two constraints compound. On our hardware $(j, k) = (2, 3)$ takes minutes, $(3, 4)$ about ninety, $(2, 5)$ roughly a day, and $(3, 5)$ some 10^4 years. We therefore report sharpness at $(2, 3)$ — where $Z(I(n, 2, 3)) = 10 = 2(j + k)$ for $n = 18, 24$ — and note that the question is computationally out of reach beyond small parameters, rather than merely unfinished.

16 The exact ceiling of the certificate method

Every technique in Section 22 is combinatorial and fails. A matrix-theoretic route succeeds, and produces certificates. Before constructing any, we determine exactly how far the method can possibly go.

Let A be a real matrix indexed by $V(P(n, k))$ with $A_{xy} \neq 0$ iff $xy \in E$ for $x \neq y$, the diagonal unrestricted. We do *not* assume A symmetric: the bound $\text{null } A \leq Z(G)$ needs only that every edge carries a nonzero entry in each direction. Indeed if $Ax = 0$ and x vanishes on a set B , and $u \in B$ has exactly one neighbour $w \notin B$, then row u of $Ax = 0$ reads $A_{uu}x_u + \sum_{v \sim u} A_{uv}x_v = 0$, every term vanishes except $A_{uw}x_w$, and $A_{uw} \neq 0$; so $x_w = 0$, and induction along a forcing process gives $x = 0$ on $\text{cl}(B)$. Hence $x \mapsto x|_S$ is injective on $\ker A$ for every zero forcing set S , so

$$\text{null } A \leq Z(G). \tag{1}$$

Write $M(G)$ for the maximum nullity over symmetric such A and $M_{\text{cs}}(G)$ for the maximum over all of them, so $M \leq M_{\text{cs}} \leq Z$. Set

$$b_i = A_{u_i u_{i+1}}, \quad b'_i = A_{u_{i+1} u_i}, \quad c_i = A_{u_i v_i}, \quad c'_i = A_{v_i u_i}, \quad e_i = A_{v_i v_{i+k}}, \quad e'_i = A_{v_{i+k} v_i},$$

with diagonals a_i at u_i and d_i at v_i ; all of b, b', c, c', e, e' are nonzero.

Theorem 44 (Reduction). *Let $n \geq 2k+3$. For $x \in \mathbb{R}^{\mathbb{Z}_n}$ put $(Lx)_i = -(b'_{i-1}x_{i-1} + a_i x_i + b_i x_{i+1})/c_i$. Then $x \mapsto (x, Lx)$ is a linear bijection from*

$$\left\{ x : \sum_{p=-k-1}^{k+1} \gamma_{i,p} x_{i+p} = 0 \text{ for all } i \in \mathbb{Z}_n \right\}$$

onto $\ker A$, where $\gamma_{i,p} \neq 0$ only for $p \in \{-k-1, -k, -k+1\} \cup \{-1, 0, 1\} \cup \{k-1, k, k+1\}$ — nine positions — and

$$\gamma_{i,k+1} = -\frac{e_i b_{i+k}}{c_{i+k}} \neq 0, \quad \gamma_{i,-k-1} = -\frac{e'_{i-k} b'_{i-k-1}}{c_{i-k}} \neq 0.$$

Consequently, with $T = T_{n-1} \cdots T_0 \in \text{GL}_{2k+2}(\mathbb{R})$ the monodromy of this recurrence on the state $(x_{i-k-1}, \dots, x_{i+k})$,

$$\text{null } A = \dim \ker(T - I), \quad \det T = \frac{(\prod_i b'_i)(\prod_i e'_i)}{(\prod_i b_i)(\prod_i e_i)},$$

so $\det T = 1$ whenever A is symmetric. In particular $\text{null } A \leq 2k+2$, with equality if and only if $T = I$.

Proof. Each spoke weight c_i is nonzero, so the equation of $\ker A$ at u_i solves uniquely for $y_i = (Lx)_i$, a 3-term expression in x . Substituting into the equation at v_i , namely $e'_{i-k} y_{i-k} + d_i y_i + e_i y_{i+k} + c'_i x_i = 0$, gives the displayed relation: the three terms contribute the index triples $\{i-k-1, i-k, i-k+1\}$, $\{i-1, i, i+1\}$ and $\{i+k-1, i+k, i+k+1\}$, and the two extreme coefficients are as stated, both nonzero. Conversely any x satisfying all the relations yields the kernel vector (x, Lx) . Since both end coefficients are nonzero the relation is a recurrence of order exactly $2k+2$, solvable forward and backward, so its solutions on $\mathbb{Z}$ form a $(2k+2)$ -dimensional space on which T acts, and the n -periodic ones are exactly the fixed vectors. Each step matrix T_i is the companion matrix of the relation, with determinant $\gamma_{i,-k-1}/\gamma_{i,k+1}$; taking the product over $i \in \mathbb{Z}_n$, every index set runs over a full residue system and the stated formula follows. $\square$

Corollary 45. $M(P(n, k)) \leq M_{\text{cs}}(P(n, k)) \leq 2k+2$ for $n \geq 2k+3$.

Corollary 45 is not a new bound — it follows from (1) and Theorem 36 — but it locates the method precisely, and the identification is exact rather than approximate. The recurrence has order $2k+2$ because a nonzero kernel vector cannot vanish on $2k+2$ consecutive outer indices; and $\{u_0, \dots, u_{2k+1}\}$ is exactly the zero forcing set that the rotation bootstrap of Theorem 36 produces. So *the order of the recurrence equals the size of that forcing set*, and the ceiling of the certificate method coincides with the upper bound the method is trying to match, with no slack on either side. A certificate can prove $Z(P(n, k)) \geq 2k+2$ and, by Theorem 44, it does so precisely when the monodromy is the identity. Our earlier description of the method as “capped near 6–8” confused a property of the constructions we had tried with a property of the method.

Proposition 46 (Product form). *For symmetric A with $C = \text{diag}(c_i)$, put $U = A[O, O]$ and $S = C^{-1}A[I, I]C^{-1}$. Then $\text{null } A = \dim \ker(SU - I)$. Here U and S are symmetric with the outer-cycle and inner-cycle patterns respectively, free diagonals and nonzero prescribed off-diagonal entries, and every such pair arises. Hence $M(P(n, k))$ is the largest possible geometric multiplicity of the eigenvalue 1 in a product of two such matrices.*

Proof. $\ker A$ is cut out by $Ux + Cy = 0$ and $Cx + A[I, I]y = 0$; eliminating y gives $(C - A[I, I]C^{-1}U)x = 0$, and factoring out the invertible C turns this into $(I - SU)x = 0$. $\square$

Theorem 44 and Proposition 46 were verified computationally before use: over 63 pairs (n, k) with $2 \leq k \leq 7$ and three random matrices each (189 matrices), the nine-position sparsity, both end-coefficient formulas and the identity $\text{null } A = \dim \ker(T - I) \leq 2k + 2$ hold in every case; over 30 further pairs and both matrix classes, so do the determinant formula, the product form, and the statement that no kernel vector vanishes on $u_0, \dots, u_{2k+1}$. See `verification/recurrence_order.py`.

17 Period-one certificates and cyclotomy

Take $A \in S(P(n, k))$ invariant under ρ , normalised so that the outer weight is 1:

$$U = aI + (P + P^{-1}), \quad V = dI + e(P^k + P^{-k}), \quad C = cI,$$

with $c, e \neq 0$. Diagonalising P splits A into the n blocks $M_m = \begin{pmatrix} a+s_m & c \\ c & d+et_m \end{pmatrix}$ with $s_m = 2\cos(2\pi m/n)$ and $t_m = 2\cos(2\pi km/n)$. Each singular block has nullity exactly 1, since $c \neq 0$, so $\text{null } A = \#\{m : \det M_m = 0\}$.

Proposition 47 (The symbol). *Let $L_k \in \mathbb{Z}[s]$ be the monic polynomial of degree k with $L_k(z + z^{-1}) = z^k + z^{-k}$ (so $L_1 = s$, $L_2 = s^2 - 2$, $L_k = sL_{k-1} - L_{k-2}$). Then $t_m = L_k(s_m)$ exactly, and with $\alpha = d/e$, $\beta = (ad - c^2)/e$,*

$$\det M_m = 0 \iff F(s_m) = 0, \quad F(s) = (s + a)L_k(s) + \alpha s + \beta.$$

F is monic of degree $k + 1$ and carries exactly three free parameters (a, α, β) , for every k . Writing $\Gamma_n = \{2\cos(2\pi m/n)\}$, each root of F in $\Gamma_n \cap (-2, 2)$ contributes 2 to $\text{null } A$ and each root at ± 2 contributes 1, because $m \mapsto n - m$ fixes s_m and is free elsewhere. Finally $c^2 = a\alpha - \beta$ after scaling $e = 1$, so the certificate is admissible precisely when $a\alpha - \beta > 0$.

Remark 48 (A claim of ours, corrected). We previously asserted that this period-one construction “never exceeds 6” for every $k \geq 2$, reasoning that only three roots can be prescribed. The premise is right and the conclusion is wrong. At a root r the condition is *linear*,

$$aL_k(r) + \alpha r + \beta = -rL_k(r), \tag{2}$$

so three distinct roots determine (a, α, β) and no fourth can be imposed. But prescribing three roots also *determines the other $k - 2$* , and those may lie in Γ_n of their own accord. They do exactly when arithmetic puts them there: if F has rational coefficients its roots form Galois orbits, and the Galois orbit of $2\cos(2\pi m/n)$ consists again of elements of Γ_n . A single rational F can therefore place all $k + 1$ roots on the grid at once. This raises the period-one ceiling from 6 to $2k + 2$, and makes the more elaborate period-two construction we reported unnecessary.

When realisability fails, exactly

The parameter count says what can be prescribed; realisability says what survives. The condition $c^2 = a\alpha - \beta \neq 0$ turns out to be a *factorisation* condition on the symbol, and this settles the question completely.

Theorem 49 (Realisability). *For a period-one matrix with symbol F ,*

$$c^2 = 0 \iff F = (s + a)(L_k(s) + t) \text{ for some constant } t.$$

Consequently the matrix is admissible in the combinatorially symmetric class if and only if F does not factor in this way, and in the symmetric class if and only if in addition $c^2 > 0$.

Proof. $c^2 = a\alpha - \beta$, so $c^2 = 0$ iff $\beta = a\alpha$, and then

$$F = (s + a)L_k + \alpha s + a\alpha = (s + a)L_k + \alpha(s + a) = (s + a)(L_k + \alpha).$$

Conversely if $F = (s + a')(L_k + t)$ then comparing the coefficient of s^k gives $a' = a$, and expanding gives $\alpha = t$, $\beta = at$, so $c^2 = at - at = 0$. The two admissibility statements are Proposition 47: the product of the two spoke weights is c^2 , which must be nonzero for the pattern to be that of $P(n, k)$, and must be a square for the matrix to be symmetric. $\square$

Over the rationals the degenerate symbols form a finite, explicit list, and the reason is Niven's theorem [14]: the only rational values of $2\cos(\pi q)$ with $q \in \mathbb{Q}$ are $0, \pm 1, \pm 2$.

Corollary 50 (The degenerate rational symbols). *Let $F \in \mathbb{Q}[s]$ be monic of degree $k + 1$ with every root in Γ_n , and suppose $c^2 = 0$. Then $F = (s + a)(L_k + t)$ with*

$$a, t \in \{0, \pm 1, \pm 2\}.$$

There are therefore at most 25 degenerate rational symbols for each k .

Proof. By Theorem 49, $F = (s + a)(L_k + t)$, and a is the coefficient of s^k in F , hence rational; t is then rational too. Now $-a$ is a root of F , so $-a = 2\cos\theta$ for some rational multiple θ of 2π , and a rational forces $a \in \{0, \pm 1, \pm 2\}$ by Niven's theorem. Likewise any root r of $L_k + t$ lies in Γ_n , so $r = 2\cos\varphi$ with φ a rational multiple of 2π and $t = -L_k(r) = -2\cos(k\varphi)$, again rational, so $t \in \{0, \pm 1, \pm 2\}$. $\square$

Enumerating the 25 pairs and keeping those for which $(s + a)(L_k + t)$ splits into distinct Ψ_d with every root interior gives 7, 8, 7, 6, 9 degenerate symbols for $k = 2, \dots, 6$, agreeing exactly with the count of $c^2 = 0$ cases found by the independent exhaustive enumeration in each case (`verification/cs_exhaustive.py`).

One special case of Theorem 49 is worth isolating, because it is visible on the root set alone and depends only on the parity of k .

Corollary 51 (Antipodal obstruction). *Let k be even and suppose the symbol F has two roots y and $-y$ with $y \neq 0$. Then $c^2 = a\alpha - \beta = 0$, so the matrix is not admissible in the symmetric class and not admissible in the combinatorially symmetric class either. Hence for even k a period-one certificate requires a symbol whose root set contains no antipodal pair. For odd k no such obstruction exists: there*

$$c^2 = \frac{L_k(y)(y^2 - a^2)}{y},$$

which is not identically zero.

Proof. From $L_k(z + z^{-1}) = z^k + z^{-k}$ we get $L_k(-s) = (-1)^k L_k(s)$, since $s \mapsto -s$ corresponds to $z \mapsto -z$. Let k be even, so $L_k(-y) = L_k(y)$. Writing out $F(y) = 0$ and $F(-y) = 0$ using $F(s) = (s + a)L_k(s) + \alpha s + \beta$ gives

$$aL_k(y) + \alpha y + \beta = -yL_k(y), \quad aL_k(y) - \alpha y + \beta = yL_k(y).$$

Adding gives $\beta = -aL_k(y)$; subtracting and dividing by $2y \neq 0$ gives $\alpha = -L_k(y)$. Therefore $c^2 = a\alpha - \beta = -aL_k(y) + aL_k(y) = 0$. Admissibility requires the spoke weights to be nonzero, and their product is c^2 by Proposition 47, which rules out both classes. For odd k , $L_k(-y) = -L_k(y)$, the same two equations become $aL_k(y) + \alpha y + \beta = -yL_k(y)$ and $-aL_k(y) - \alpha y + \beta = -yL_k(y)$, whence $\beta = -yL_k(y)$ and $\alpha = -aL_k(y)/y$, and $c^2 = a\alpha - \beta = L_k(y)(y^2 - a^2)/y$. $\square$

For $k = 2$ the obstruction is the *only* one, because the symbol then has exactly three roots and the $k - 2$ residual conditions of Theorem 63 are vacuous.

Proposition 52 ($k = 2$ in closed form). *For $k = 2$, prescribing three distinct roots s_1, s_2, s_3 determines $a = -e_1$, $\alpha = e_2 + 2$, $\beta = -e_3 - 2e_1$ with e_i the elementary symmetric functions, and*

$$c^2 = e_3 - e_1e_2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3).$$

So $c^2 = 0$ if and only if two of the roots are antipodal, and $c^2 > 0$ if and only if an odd number of the three pairwise sums is negative.

Proof. Expanding $F = (s + a)(s^2 - 2) + \alpha s + \beta = s^3 + as^2 + (\alpha - 2)s + (\beta - 2a)$ and comparing with $\prod_i (s - s_i) = s^3 - e_1s^2 + e_2s - e_3$ gives the three formulas, and then $a\alpha - \beta = -e_1(e_2 + 2) + e_3 + 2e_1 = e_3 - e_1e_2$. The identity $(s_1 + s_2)(s_1 + s_3)(s_2 + s_3) = e_1e_2 - e_3$ is elementary. The two consequences follow because the three factors are the pairwise sums. $\square$

Write $\Gamma_n^\circ = \Gamma_n \cap (-2, 2)$, with $M = \lfloor (n - 1)/2 \rfloor$ elements $s_1 > \dots > s_M$. For odd n , Γ_n° contains no antipodal pair, since $s_i = -s_j$ forces $i + j = n/2$. For even n the antipodal classes number $n/4$ when $4 \mid n$ and $(n - 2)/4$ otherwise, so three pairwise non-antipodal values exist exactly when n is odd with $n \geq 7$ or n is even with $n \geq 12$. This settles $k = 2$ completely.

Theorem 53 ($k = 2$, all n). *Take the three most negative elements of Γ_n° , that is s_M, s_{M-1}, s_{M-2} . Then*

$$s_{M-1} + s_{M-2} = \begin{cases} -4 \cos(4\pi/n) \cos(\pi/n), & n \text{ odd}, \\ -4 \cos(5\pi/n) \cos(\pi/n), & n \text{ even}, \end{cases}$$

which is negative exactly for odd $n \geq 9$ and for even $n \geq 12$. In those cases all three pairwise sums are negative, so $c^2 > 0$ by Proposition 52 and

$$Z(P(n, 2)) = M(P(n, 2)) = 6 \quad (n \geq 9, n \neq 10).$$

For n divisible by 7 the symbol $\Psi_7 = s^3 + s^2 - 2s - 1$ gives $c^2 = -1$: no symmetric certificate, but the integral combinatorially symmetric matrix

$$A = \begin{pmatrix} I + P + P^{-1} & I \\ -I & P^2 + P^{-2} \end{pmatrix}$$

has nullity exactly 6, so $Z(P(n, 2)) = M_{\text{cs}}(P(n, 2)) = 6$ whenever $7 \mid n$. The remaining n are finite in number and exhaustive computation gives $Z(P(5, 2)) = 5$, $Z(P(6, 2)) = 4$, $Z(P(8, 2)) = 5$ and $Z(P(10, 2)) = 6$.

Proof. For odd n , $M = (n - 1)/2$, so $2\pi(M - 1)/n = \pi - 3\pi/n$ and $2\pi(M - 2)/n = \pi - 5\pi/n$, giving $s_{M-1} + s_{M-2} = -2(\cos(3\pi/n) + \cos(5\pi/n)) = -4 \cos(4\pi/n) \cos(\pi/n)$, negative iff $\cos(4\pi/n) > 0$ iff $n > 8$. For even n , $M = n/2 - 1$ and the same computation with $4\pi/n, 6\pi/n$ gives $-4 \cos(5\pi/n) \cos(\pi/n)$, negative iff $n > 10$. Since s is decreasing, $s_M \leq s_{M-2}$, so $s_M + s_{M-1} \leq s_{M-1} + s_{M-2} < 0$ and $s_M + s_{M-2} \leq s_{M-1} + s_{M-2} < 0$: all three pairwise sums are negative, their

product is negative, and $c^2 > 0$. Proposition 47 then gives $\text{null } A = 6$, and $6 \leq M(P(n, 2)) \leq Z(P(n, 2)) \leq 6$ by the rotation bootstrap. The Ψ_7 claim is Theorem 63 with c^2 negative, read in the combinatorially symmetric class where the two spoke weights need only have the prescribed product. $\square$

Remark 54 (Exact but stiff). The certificate of Theorem 53 is exact for every admissible n , but it is numerically delicate for large n , and it is worth saying why so that a floating-point check is not misread. The three prescribed roots are $-2\cos(\pi/n)$, $-2\cos(3\pi/n)$, $-2\cos(5\pi/n)$ for odd n , all within $O(1/n^2)$ of -2 : at $n = 1001$ they span 2.4×10^{-4} . So F is close to having a triple root, the matrix is ill-conditioned, and grid values merely *near* the roots produce small singular values — at $n = 1001$ the true roots give 9×10^{-16} and the nearest non-roots 4×10^{-11} , so a fixed relative threshold over-counts the nullity. None of this touches the proof, which is symbolic: $c^2 > 0$ follows from the product-to-sum identity and the nullity is exactly 6 because F has exactly three distinct interior roots on Γ_n , each met by two Fourier indices.

Remark 55 (What the two exceptions mean). At $n = 8$ and $n = 10$ the grid is too small to avoid antipodal pairs — $\Gamma_8^\circ = \{\sqrt{2}, 0, -\sqrt{2}\}$ and Γ_{10}° has four elements in two antipodal classes, so by pigeonhole every triple contains such a pair. The two cases then behave differently: $Z(P(8, 2)) = 5$, so there the method is right to fail, while $Z(P(10, 2)) = 6$, so there it is the method that falls short and not the bound. This is the sharpest available statement of the limitation, and it is a statement about Γ_n alone.

That reading is now confirmed rather than inferred. Maximising the nullity directly over all symmetric matrices carrying the $P(10, 2)$ pattern — 30 edge weights and 20 diagonal entries, by gradient descent on the sum of squares of the six eigenvalues nearest zero — produces a matrix of nullity exactly 6, with the next eigenvalue at 0.40 of the matrix scale and every edge weight at least 0.30. So

$$M(P(10, 2)) = Z(P(10, 2)) = 6 :$$

there is no strict gap at $n = 10$, and the period-one construction fails there for the reason the remark gives — the grid, not the bound.

This is now *certified*, not numerical. The optimiser’s witness has been replaced by an exact one. A period-one matrix cannot attain nullity 6 here at all: its 2×2 Fourier blocks pair under conjugation, the inner cosine $2\cos(4\pi j/10)$ takes each of its values twice on $0 \leq j \leq 5$, so any three blocks forced to be singular contain two sharing that value, and subtracting their determinant conditions forces either an edge weight or the spoke to vanish — which is why the search was hard, and why it stalls at 5. A period-two matrix can, and only as $6 = 2 + 1 + 1$ across the blocks over the fifth roots of unity, since a block at $j = 1, 2$ has nullity at most 1 and the block at $j = 0$ at most 2. Imposing that, and requiring the entries to be rational so that the $j = 1$ and $j = 2$ conditions are Galois conjugate over $\mathbb{Q}(\sqrt{5})$, leaves $st + 2sf + 2et - ef = 0$ with $a^2 = -c^2 d^2 ef / ((s + 2e)(t + 2f)(sf + et - ef))$. Solving it and clearing denominators gives an *integer* matrix on the $P(10, 2)$ pattern with entries in $\{-375, -125, -25, -5, -1, 5, 25\}$ whose rank over $\mathbb{Z}$ is 14, hence nullity exactly 6 (`verification/p10_2_exact.py`).

Remark 56 (A search failure is not an impossibility, again). The first run of that search reached nullity 5 easily and stalled at 6×10^{-6} for nullity 6, across sixty restarts — which looks exactly like a strict gap, and on a solver that found nullity 6 instantly at $P(9, 2)$, $P(11, 2)$ and $P(12, 2)$. It was not one: two further random seeds out of six found the nullity-6 basin at 10^{-18} . The $n = 10$ optimum is simply harder to locate than its neighbours’, which is the same fact about Γ_{10} that makes the symbol construction fail there. We flag it because it is precisely the failure mode this

paper warns about elsewhere — a failed search measures effort spent, not impossibility — and we walked into it on our own graph.

Corollary 57 (Excluded cyclotomic symbols, even k). *Let k be even. The roots of Ψ_{2d} are the negatives of the roots of Ψ_d for odd d , and the root set of Ψ_d is closed under negation whenever $4 \mid d$ and $\varphi(d) \geq 4$. Hence in Theorem 63 the factor set $\mathcal{D}$ may contain neither both d and $2d$ for any odd d , nor any d divisible by 4 with $\varphi(d) \geq 4$.*

Corollary 57 in fact accounts for *all* of the rejections in the exhaustive enumeration of Proposition 71. At $k = 2$ that enumeration found 17 factor sets in the three-parameter family, of which 5 had $c^2 > 0$ and 12 were rejected. Those 12 split exactly:

$$\underbrace{(3, 8), (3, 12), (4, 8), (4, 12), (6, 8), (6, 12), (3, 4, 6)}_{c^2=0, \text{ every one antipodal}} \quad \underbrace{(7), (9), (4, 5), (5, 6), (6, 10)}_{c^2=-1, \text{ none antipodal}}.$$

The seven on the left are dead in both matrix classes, and Theorem 51 identifies them without computing c^2 at all. The five on the right carry $c^2 = -1$, so each of them *is* realisable in the combinatorially symmetric class. The admissible count at $k = 2$ is therefore 5 symmetrically and 10 in the larger class. The extra symbols cover every n divisible by 7 or by 9; of these only $n = 7$ is out of reach of Theorem 53, and it is precisely the case the larger class was needed for. The antipodal obstruction is the only thing standing in the way of any rational period-one symbol for $k = 2$. Every prediction of Theorem 51 and Corollary 57 was checked against 50 candidate factor sets at $k = 2, 4$ with no violation, and Proposition 52 was verified by exact arithmetic in $\mathbb{Z}[x]/\Phi_n(x)$ over all $\binom{M}{3}$ triples for every $7 \leq n \leq 60$ — 11,000 triples, with $c^2 = 0$ and “contains an antipodal pair” agreeing in every one. See `verification/antipodal_obstruction.py`.

The larger matrix class, and $k = 3$

Theorem 49 separates $c^2 = 0$ from $c^2 < 0$, and that separation matters: only the first is fatal. A symbol with $c^2 < 0$ admits no symmetric certificate, but in the combinatorially symmetric class the two spoke weights are independent and need only have product c^2 , so it is admissible there. Redoing the exhaustive enumeration of Proposition 71 with the weaker condition $c^2 \neq 0$ therefore changes the answer. The case $k = 3$ changes the most.

Theorem 58 ($k = 3$). *Let $F = \Psi_5\Psi_{10} = (s^2+s-1)(s^2-s-1) = s^4-3s^2+1$. Then $F = sL_3(s)+1$, so $a = \alpha = 0$, $\beta = 1$ and $c^2 = -1$. Hence the integer matrix*

$$A = \begin{pmatrix} P + P^{-1} & I \\ -I & P^3 + P^{-3} \end{pmatrix}$$

has zero diagonal, carries exactly the pattern of $P(n, 3)$, and has nullity exactly $8 = 2k + 2$ for every n divisible by 10. Therefore

$$Z(P(n, 3)) = M_{\text{cs}}(P(n, 3)) = 8 \quad (10 \mid n).$$

Proof. $\Psi_5 = s^2+s-1$ and $\Psi_{10} = s^2-s-1$, whose product is $s^4-3s^2+1 = s(s^3-3s)+1 = sL_3(s)+1$. Matching against $(s+a)L_3+\alpha s+\beta$ gives $a = 0$, $\alpha = 0$, $\beta = 1$, so $c^2 = a\alpha - \beta = -1$. By Theorem 49 the symbol is non-degenerate, and taking the two spoke weights to be 1 and -1 realises $c^2 = -1$ with both nonzero, so the zero–nonzero pattern is exactly that of $P(n, 3)$. Every root of F is a root of Ψ_5 or Ψ_{10} , hence lies in Γ_n whenever $10 \mid n$, and all four roots are interior, so each contributes 2: null $A = 8$. The upper bound $Z(P(n, 3)) \leq 8$ is the rotation bootstrap, and $M_{\text{cs}} \leq Z$ holds because the forcing argument uses only the zero–nonzero pattern. $\square$

Remark 59 (Two things this certificate illustrates). First, A is the adjacency matrix of $P(n, 3)$ with one set of spokes negated, and the sign matters entirely: the unsigned adjacency matrix has symbol $sL_3 - 1$ and nullity 0 for every n divisible by 10, consistent with Theorem 64. Second, the four roots of F are $\pm\varphi$ and $\pm\varphi^{-1}$ — *two antipodal pairs*. By Corollary 51 such a symbol would be degenerate for every even k ; because 3 is odd it is not. So the parity corollary does not only rule things out, it says where to look.

The same enumeration in the larger class yields five further symbols at $k = 2$ (covering n divisible by 7, 9, 20 or 30), five more at $k = 3$ beyond Theorem 58 (n divisible by 24, 40 or 42), three at $k = 4$ (60, 70, 90) and two at $k = 5$ (126); all have $c^2 \in \{-1, -2\}$. At $k = 6$ it yields nothing, and now for a reason.

The signed adjacency matrix, and $k = 7$

Both the $k = 3$ certificate and one of the $k = 7$ symbols found by this enumeration have $a = \alpha = 0$ and $\beta = 1$, that is, symbol $F = sL_k(s) + 1$. That is not a coincidence, and identifying what it means gives an exact value at $k = 7$.

With $a = \alpha = 0$ both diagonals vanish, so A is the adjacency matrix of $P(n, k)$ with one set of spokes negated. Substituting $s = z + z^{-1}$, $L_k = z^k + z^{-k}$ and clearing denominators turns $F(s) = 0$ into

$$Q_k(z) := z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1 = 0,$$

which is exactly the adjacency polynomial $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1$ of Theorem 64 with the sign of the middle term reversed — the spoke sign flip, visible in the symbol.

Theorem 60 (Signed adjacency classification). *Let A^- be the adjacency matrix of $P(n, k)$ with one set of spokes negated. Then*

$$\text{null } A^- = \sum \varphi(d) \quad \text{over those } d \text{ with } \Phi_d \mid Q_k \text{ and } d \mid n,$$

by the argument of Theorem 64 applied to Q_k in place of P_k . In particular $\text{null } A^- = 2k + 2$ for suitable n exactly when Q_k is a product of cyclotomic polynomials.

So the question of which k admit an *integer* certificate of this shape becomes: for which k does Q_k split into cyclotomics? This is decidable for each k , and the answer is rigid. For $2 \leq k \leq 60$ it happens only for

$$k = 2 : Q_2 = \Phi_5\Phi_6, \quad k = 3 : Q_3 = \Phi_5\Phi_{10}, \quad k = 7 : Q_7 = \Phi_5\Phi_{10}\Phi_{24},$$

with $\sum \varphi(d)$ equal to 6, 8, 16 — the ceiling $2k + 2$ in every case — and lcm equal to 30, 10, 120. The unsigned P_k splits only for $k = 2$ and $k = 5$ (Theorem 64), so between them the two sign choices give integer adjacency-type certificates exactly at $k \in \{2, 3, 5, 7\}$.

Theorem 61 ($k = 7$). $Q_7 = \Phi_5\Phi_{10}\Phi_{24}$, so for every n divisible by 120 the integer matrix

$$A = \begin{pmatrix} P + P^{-1} & I \\ -I & P^7 + P^{-7} \end{pmatrix}$$

carries the pattern of $P(n, 7)$ and has nullity exactly $16 = 2k + 2$. Hence

$$Z(P(n, 7)) = M_{\text{cs}}(P(n, 7)) = 16 \quad (120 \mid n).$$

This replaces a numerical claim by an exact one. Section 19 reports $Z(P(96, 7)) = 16$ via a period-4 certificate verified in floating point; Theorem 61 gives the same value for infinitely many n from a matrix of zeros and ± 1 , with the nullity computed by a cyclotomic factorisation rather than by an eigenvalue. Confirmed numerically: the matrix has nullity 16 at $n = 120$, and 6, 8 at the corresponding n for $k = 2, 3$ (`verification/signed_adjacency.py`).

Theorem 62 ($k = 6$ is impossible over $\mathbb{Q}$). *Of the 225 products $\prod_{d \in \mathcal{D}} \Psi_d$ of total degree 7 with every Ψ_d of interior roots, 216 do not lie in the three-parameter family, and all 9 that do are degenerate in the sense of Theorem 49: each is $(s + a)(L_6 + t)$ with $a \in \{0, \pm 1\}$ and $t \in \{0, \pm 1\}$, namely Ψ_d -products over $\mathcal{D} = (3, 8, 24), (4, 8, 24), (6, 8, 24), (3, 9, 18), (4, 9, 18), (6, 9, 18), (3, 36), (4, 36), (6, 36)$. Hence no rational period-one symbol attains 14 for $k = 6$, in either matrix class.*

This is stronger than the statement it replaces. Where we previously reported that a search found nothing at $k = 6$, the obstruction now identifies every candidate that survives the family condition and shows each one is degenerate, and Corollary 50 bounds the degenerate list *a priori* at 25 polynomials. The $k \geq 6$ certificates of Section 19 therefore have to break the symmetry, and they do.

For $d \geq 3$ let $\Psi_d \in \mathbb{Z}[s]$ be the minimal polynomial of $2 \cos(2\pi/d)$, of degree $\varphi(d)/2$, whose roots are the primitive values $2 \cos(2\pi j/d)$ with $\gcd(j, d) = 1$; put $\Psi_1 = s - 2$, $\Psi_2 = s + 2$. Every root of Ψ_d lies in Γ_n as soon as $d \mid n$.

Theorem 63 (Cyclotomic certificates). *Let $\mathcal{D}$ be a set of distinct positive integers with $\sum_{d \in \mathcal{D}} \deg \Psi_d = k + 1$ and put $F = \prod_{d \in \mathcal{D}} \Psi_d$. Write $G = F - sL_k$, let a be the coefficient of s^k in G , and suppose $G - aL_k$ has degree at most 1, say $G - aL_k = \alpha s + \beta$; that is $k - 2$ linear conditions with integer coefficients. Put $e = 1$, $d = \alpha$, $c^2 = a\alpha - \beta$, and suppose $c^2 > 0$. Then for every n divisible by $\text{lcm}(\mathcal{D})$ with $n \geq 2k + 3$, the matrix*

$$A = \begin{pmatrix} aI + P + P^{-1} & cI \\ cI & \alpha I + P^k + P^{-k} \end{pmatrix}$$

lies in $S(P(n, k))$ and has nullity $2 \sum_{d \in \mathcal{D}, d \geq 3} \deg \Psi_d + \#\{d \in \mathcal{D} : d \leq 2\}$. If every $d \in \mathcal{D}$ is at least 3 this equals $2k + 2$, and then

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2.$$

Proof. By construction $F = (s + a)L_k + \alpha s + \beta$ as an identity in $\mathbb{Z}[s]$, so F is the symbol of A by Proposition 47, and $A \in S(P(n, k))$ because $c^2 > 0$ makes c real and nonzero while the edge weights 1 are nonzero. Each $d \in \mathcal{D}$ divides n , so all $k + 1$ roots of F lie in Γ_n ; they are distinct because the Ψ_d are distinct irreducible polynomials. Counting multiplicities as in Proposition 47 gives the stated nullity, and the last line combines it with (1) and Theorem 36. $\square$

A search over $k \leq 12$ and $d \leq 210$ (`verification/cyclotomic_certificates.py`) gives the following. Every certificate listed has $a, \alpha, c \in \mathbb{Z}$, hence is an *integer* matrix, and its nullity was confirmed by exact rational Gaussian elimination.

Two of these deserve to be written out. For $k = 2$ and $k = 5$ the parameters are $a = d = 0$, $c = 1$: the certificate is the *adjacency matrix* of $P(n, k)$, and the symbol is $sL_k(s) - 1$, with

$$sL_2 - 1 = (s + 1)(s^2 - s - 1) = \Psi_3 \Psi_{10}, \quad sL_5 - 1 = (s - 1)(s + 1)(s^4 - 4s^2 + 1) = \Psi_6 \Psi_3 \Psi_{24}.$$

Equivalently, since $\det M_m = s_m t_m - 1$,

$$\text{null}(\text{Adj } P(n, k)) = \#\left\{m \in \mathbb{Z}_n : 2 \cos \frac{2\pi(k+1)m}{n} + 2 \cos \frac{2\pi(k-1)m}{n} = 1\right\},$$

a rational linear relation among cosines of rational angles — the setting of the Conway–Jones classification. That question has a complete answer, given in Theorem 64 below. (An earlier version of this paper asserted that “in the computed range $k \leq 10$ only $k = 2, 5, 8$ ” admit solutions. That was drawn from a scan filtered at nullity ≥ 6 and is wrong: $k = 4, 7, 10, 11, 12$ also admit solutions, with smaller nullity.) For $k = 5$ the count is $12 = 2k + 2$ whenever $24 \mid n$: the plain adjacency matrix of $P(n, 5)$ has nullity 12, which settles Z for that family. For $k = 4$ the symbol is $(s + 1)L_4(s) - s - 2 = s(s^4 + s^3 - 4s^2 - 4s + 1) = \Psi_4\Psi_{30}$, giving the integer certificate with $a = 1$, $d = -1$, $c = 1$.

The adjacency nullity, classified completely

Since $\det M_m = s_m t_m - 1$, the adjacency matrix is singular at m exactly when $z = \omega^m$ satisfies $z^{k+1} + z^{-(k+1)} + z^{k-1} + z^{-(k-1)} = 1$. Clearing denominators gives an integer polynomial, and the question becomes which of its roots are roots of unity.

Theorem 64. *Put $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1 \in \mathbb{Z}[z]$. Then*

$$\text{null}(\text{Adj } P(n, k)) = \sum_{\substack{d: \Phi_d \mid P_k \\ d \mid n}} \varphi(d),$$

the sum over those d whose cyclotomic polynomial divides P_k . The list of such d is finite and computable once per k .

Proof. m contributes iff ω^m is a common root of P_k and $z^n - 1$, i.e. a root of $\gcd(P_k, z^n - 1)$. The roots of unity among the roots of P_k are exactly the roots of its cyclotomic factors Φ_d , and Φ_d ’s roots are n -th roots of unity precisely when $d \mid n$; each such Φ_d contributes its $\varphi(d)$ roots. Finiteness is immediate from $\deg P_k = 2k + 2$, and is the elementary shadow of the Conway–Jones theorem on rational linear relations among cosines of rational angles. $\square$

k	$2k + 2$	cyclotomic factors of P_k	lcm	nullity when $\text{lcm} \mid n$
2	6	Φ_3, Φ_{10}	30	$6 = 2k + 2$
3	8	none	—	0 for every n
4	10	Φ_3	3	2
5	12	$\Phi_3, \Phi_6, \Phi_{24}$	24	$12 = 2k + 2$
6	14	none	—	0 for every n
7	16	Φ_3, Φ_6	6	4
8	18	Φ_3, Φ_{10}	30	6

So the adjacency matrix of $P(n, k)$ is *never* singular for $k = 3, 6, 9$, whatever n ; and it attains the full ceiling $2k + 2$ only for $k = 2$ and $k = 5$. The formula was checked against a direct count for $k = 2, 5, 8$ at every $13 \leq n < 200$, with no discrepancy (`verification/adjacency_classification.py`).

18 The general setting: cyclic covers

Nothing in Section 17 used the particular shape of $P(n, k)$ beyond the presence of the rotation. It is worth saying what the natural setting is.

Let B be a finite graph whose edges carry *voltages* in $\mathbb{Z}_n$ (no edge of voltage 0 at a loop, since that would lift to a self-loop). The *derived graph* B^n has vertex set $V(B) \times \mathbb{Z}_n$, with (x, i) joined to $(y, i + v)$ for each edge $x - y$ of voltage v . This is the standard cyclic cover of topological graph

theory. Taking B to be two vertices u, v with a loop of voltage 1 at u , a loop of voltage k at v , and an edge $u-v$ of voltage 0 recovers $B^n = P(n, k)$ exactly.

Theorem 65 (Cover reduction). *Let A carry the pattern of B^n and commute with the $\mathbb{Z}_n$ action. Then A block-diagonalises into n blocks of size $|V(B)|$,*

$$M_m[x, y] = \sum_{\text{edges } x \rightarrow y \text{ of voltage } v} w_{xy} \omega^{mv} + \delta_{xy} a_x,$$

and $\text{null } A = \sum_m \text{null } M_m$. Suppose $\det M(\zeta)$, a Laurent polynomial in ζ , is not identically zero. Then $\text{null } A$ is at most the number of roots of $\det M(\zeta)$ among the n th roots of unity counted with multiplicity, hence at most the degree span of $\det M(\zeta)$, which is at most the generic span D determined by the voltages of B alone.

The hypothesis is not decorative. The nullity of M_m is at most the multiplicity of ω^m as a root of $\det M(\zeta)$, which is what the count uses; but $\det M(\zeta)$ can vanish identically for a legal choice of weights, and then every block is singular and $\text{null } A \geq n$. Section 21 exhibits exactly this on a cubic cover of K_4 . An earlier version of this theorem omitted the hypothesis, and the omission propagated into a conjecture we have since refuted (Theorem 129).

For $P(n, k)$ the degree span is $2k+2$, recovering Corollary 45 in the equivariant case. The reduction and the ceiling were verified against independently constructed full matrices, with weights tuned so that the nullity is genuinely nonzero, on seven base graphs on two to five vertices with assorted voltages (`verification/cyclic_covers.py`).

Two caveats. Theorem 65 is stated for *equivariant* A ; the stronger statement of Theorem 44, which bounds the nullity of *every* matrix carrying the pattern, we have proved only for $P(n, k)$, where the nonzero spoke weights permit the elimination. Earlier versions of this paper left open whether the same holds over an arbitrary base graph — whether the ceiling is a property of the cover rather than of equivariant matrices on it. It is not, and Theorem 85 settles it in the negative. Second, we have tested the non-equivariant bound on general covers by *maximising* the nullity rather than sampling — random matrices are generically nonsingular, so sampling proves nothing. Optimising over fully non-equivariant weights — one free weight per lift of each base edge and a free diagonal per lift — on ten connected covers, the largest attainable nullity never exceeded the degree span. The bases include two-loop, theta and four-edge theta bases, a triangle, a C_4 , K_4 , a path base with loops, and a star base with a loop, so the test spans sizes two to four, Hamiltonian and non-Hamiltonian, with and without parallel edges and loops. Tree bases are excluded rather than passed: a tree has trivial cycle space, so its cover is n disjoint copies and the question does not apply.

That search is reported here because of how it failed, not because of what it found. It is an optimisation, so it establishes only that a search did not exceed the span; and on the star base it did not exceed the span while a matrix exceeding it exists — by an unbounded margin, as Theorem 85 shows. Three separate weaknesses produced that false negative. The optimiser was Nelder–Mead, which is hopeless at this dimension; edge weights were held above a floor by a penalty, which excludes part of the admissible set rather than merely discouraging it; and the statistic was the *sum* of the d smallest squared singular values, which reports success when one is at 10^{-10} and the rest at 10^{-6} . Replacing the statistic by σ_d itself and the optimiser by a gradient method flagged the star base immediately; the counterexample was then written down by hand and verified over $\mathbb{Z}$.

We leave the record of the failed search in place deliberately. A search that does not find something is evidence about the search. The K_4 base was in the same list, and the same search

missed the matrices of Theorem 129 as well: they lie on the locus where three of the four fibre blocks are singular, which a local optimiser started from generic weights does not reach.

The construction on a general cover

Section 17 prescribes a symbol for $P(n, k)$ and solves for the weights. Nothing in that used the base graph either. For any cover, writing $\det M(\zeta)$ in $\sigma = \zeta + \zeta^{-1}$ gives a polynomial of degree $D/2$ whose roots in Γ_n are the singular blocks, so the same move applies: prescribe $D/2$ interior grid values, solve for the weights, and the nullity is D .

Theorem 66 (Attainment on a cover). *Let B^n be a cyclic cover and suppose the prescribed-symbol construction yields A carrying the pattern of B^n with null $A = \nu$. Then*

$$Z(B^n) \geq \nu,$$

by $M \leq Z$. In particular $Z(B^n) \geq D$ whenever the construction attains the ceiling.

This is a statement about all cyclic covers, with $P(n, k)$ — base graph two loops of voltages 1 and k and one edge of voltage 0, $D = 2k + 2$ — as the special case. We have verified attainment of the ceiling on four different base graphs: the $P(n, 2)$ base; the theta graph with voltages $(0, 1, 3)$, a different cubic family; a three-vertex path base with $D = 12$, whose cover has $3n$ vertices; and a four-vertex star base with $D = 12$. In each the constructed matrix has nullity exactly D with every required entry a substantial fraction of the matrix scale.

We also tested the ceiling sharply, which the earlier check did not. Asking whether $\ker A$ injects into a window of D consecutive fibre coordinates is vacuous when the nullity is 2 and D is 12; it bites only at the ceiling. Repeating it on certificates of nullity exactly D , injectivity holds in every case — including the star base, which has *no* Hamiltonian path, so the ordering hypothesis under which the $P(n, k)$ elimination generalises is sufficient but not necessary.

Remark 67 (The hypothesis is needed). Theorem 66 is conditional on the construction succeeding, and that condition is not vacuous. For the $P(n, 2)$ base, $D = 6$, the construction attains 6 at $n = 7, 9, 11, 12$ but fails at $n = 8$ and $n = 10$; and $Z(P(8, 2)) = 5 < 6$ by exhaustive search, so the unconditional statement $Z(B^n) \geq D$ is *false*. What does hold unconditionally is $Z(B^n) \geq \nu$ for the attained ν , which we checked against exhaustively computed zero forcing numbers on covers with 14 to 20 vertices, with no violation. Whether the construction succeeds is not a dimensional question: $n = 7$ attains the ceiling with no slack in the choice of roots while $n = 10$ fails with slack to spare. For the $P(n, 2)$ base the criterion is now known exactly — it is Proposition 52, and $n = 8, 10$ fail because every triple on those grids contains an antipodal pair. For a general base we do not have the analogue.

Remark 68 (Where this sits). The reduction is a discrete Floquet theory, and we claim no novelty for that layer. Decomposing an operator on a cyclic cover into a family indexed by the characters of the deck group is standard in spectral graph theory and in the analysis of periodic and magnetic Laplacians on covering graphs; see [5] and its references. The monodromy of Theorem 44 plays the role of the Hill discriminant for a periodic Jacobi operator, and the condition $T = I$ — all Floquet multipliers roots of unity — is the statement that the operator has purely periodic spectrum at the relevant energy. The parameter $M(G)$ itself belongs to the minimum rank programme for graphs [1, 2], and $Z(G)$ arose independently in the controllability of quantum spin networks and in sensor placement for power grids.

The nearest prior work we are aware of on a family of this shape is [3], which studies Z and M for circulant graphs. We flag it as the place a reader should look first for overlap: that is the

same pair of parameters on a cyclically symmetric family. Two caveats, and we state them because the alternative is to overclaim in either direction. First, $P(n, k)$ is a 2×2 block circulant rather than a circulant, so results about circulants do not apply to it directly. Second, we have not been able to obtain the full text of [3] — the publisher’s site refused automated access — so we do not characterise its methods here and do not assert either that it does or does not anticipate any part of our symbol layer. That is a gap in our literature review rather than a claim, and it should be closed by reading the paper. What we have not found anywhere is the use of a transfer matrix or monodromy to force nullity on $P(n, k)$, nor a certified-tiling argument of the kind in Theorem 107; those, in combination, are what we take to be new here.

Certificates beyond the rational ones

Theorem 63 is sufficient but not necessary: F need not have rational coefficients, only real ones, and its roots need only lie in Γ_n , not form full Galois orbits. By (2) every period-one symbol with three or more grid roots arises from some triple of them, so running over all triples of Γ_n finds the true period-one optimum for each (n, k) . Candidates so found are re-checked in exact arithmetic in $\mathbb{Q}(\zeta_n)$, representing $2 \cos(2\pi m/n)$ as $x^m + x^{n-m}$ modulo $\Phi_n(x)$: the membership identity, the reality of a, α, β, c^2 (they are fixed by $x \mapsto x^{n-1}$) and $c^2 \neq 0$ are all decided exactly, and the sign of the nonzero algebraic number c^2 is then read off at high precision. This is not pedantry — see the remark on numerical traps below.

Theorem 69. *Each of the following holds, with an explicit period-one certificate verified exactly in $\mathbb{Q}(\zeta_n)$:*

$$\begin{aligned} Z(P(n, 3)) &= M(P(n, 3)) = 8 && \text{for } n = 40, 60, 80, 120, \\ Z(P(n, 4)) &= M(P(n, 4)) = 10 && \text{for } n = 60, 70, 90, 120, \\ Z(P(n, 5)) &= M(P(n, 5)) = 12 && \text{for } n = 24, 48, 72, 96, 120, \\ Z(P(n, 2)) &= M(P(n, 2)) = 6 && \text{for } 9 \leq n \leq 130, n \neq 10. \end{aligned}$$

The $k = 4$ and $k = 5$ lines are the first exact values of $Z(P(n, k))$ for $k \geq 4$ at which the bound $2k + 2$ is attained. The one previously known family for $k \geq 5$ is $Z(P(2k + 1, k)) = 6$ [6, Thm. 2.6] — one n per k , with $6 \ll 2k + 2$ — and before this we had only $8 \leq Z(P(n, 4)) \leq 10$. The $k = 3$ line is a matrix proof of values [7] obtains combinatorially. Some certificates are irrational: at $(n, k) = (40, 3)$ one has $a = -\sqrt{2}$, $d = 0$, $c^2 = 1$, and at $(60, 3)$, $c^2 = 1/\varphi$ with φ the golden ratio. Full output is in `results/zero_forcing/certified_lower_bounds.txt`.

Remark 70 (Why exact arithmetic, twice over). Counting grid roots by $|F(r)| < \varepsilon$ is not safe. With $\varepsilon = 10^{-8}$ the scan reported nullity 7 for $k = 2$ at $n = 126, 128, 130$ — impossible, since $2k + 2 = 6$ — because Γ_n becomes dense and near-roots pass the test. Separately, a certificate at $(n, k) = (48, 7)$ with $|a|$ enormous against the edge weights made a threshold relative to $\max |A|$ report nullity 48. Both were caught by Corollary 45, which bounds the nullity by $2k + 2$ and so serves as an automatic correctness check on any claimed certificate, exactly as the inequality $Z \geq M$ did for the earlier period-two search. The fixes are to test each Fourier block against its own scale, and to certify in $\mathbb{Q}(\zeta_n)$ rather than in floating point.

Where the method stops

Running the exhaustive triple scan for $2 \leq k \leq 8$ and $n \leq 130$ gives the period-one optimum exactly (`verification/period1_optimum.py`):

Proposition 71 (The rational search is complete). *Fix k . A period-one symbol attaining $2k + 2$ must be squarefree with all $k + 1$ roots interior and on Γ_n ; if in addition it has rational coefficients it is a product of distinct Ψ_d with $\sum \deg \Psi_d = k + 1$. Every factor then satisfies $\deg \Psi_d \leq k + 1$, i.e. $\varphi(d) \leq 2k + 2$, which admits only finitely many d , all listable. Hence for each k the search over rational symbols is a finite complete enumeration. Carrying it out (*verification/cyclotomic_exhaustive.py*):*

k	$2k + 2$	candidates	rejected		admissible	
			outside family	$c^2 = 0$	sym	cs
2	6	17	0	7	5	5 more
3	8	35	21	8	0	6
4	10	67	54	7	3	3 more
5	12	127	118	6	1	2 more
6	14	225	216	9	0	0
7	16	385	378	6	0	1
8	18	651	644	7	0	0
9	20	1065	1057	8	0	0
10	22	1705	1698	7	0	0

The last two columns are the point. The original enumeration reported only the “sym” column and recorded $k = 3$ and $k = 7$ as impossible for rational symbols; they are not. By Theorem 49 only the $c^2 = 0$ column is fatal, and every remaining rejection has $c^2 \in \{-1, -2\}$, hence is realisable in the combinatorially symmetric class. This is what yields Theorem 58 at $k = 3$ and Theorem 61 at $k = 7$. Note also that the $c^2 = 0$ counts are 7, 8, 7, 6, 9, 6, 7, 8, 7, never more than nine, exactly as Corollary 50 predicts from its bound of 25.

Two further things follow. First, $k = 6, 8, 9$ and 10 really are impossible over $\mathbb{Q}$ in both classes, and now for an identified reason rather than a failed search: every candidate that clears the family condition is one of the finitely many degenerate symbols. Second, the rejections are overwhelmingly of one kind — at $k = 7$, 378 of the 385 candidates fail because F is *not in the three-parameter family*. The binding constraint is therefore the $k - 2$ linear membership conditions imposed on a 3-dimensional family, and it tightens as k grows. That identifies the fix: enlarge the family.

How far must one enlarge it? A period- d matrix has $5d$ weights, but most of that freedom is gauge: conjugating by a positive diagonal matrix preserves both the pattern and the nullity. What matters is the dimension of the set of symbols actually reachable. Writing G for the degree- $(k + 1)$ polynomial in $\sigma = \zeta + \zeta^{-1}$ obtained from $\det M_\ell$, we computed the rank of the Jacobian of the map (weights) $\mapsto$ (monic coefficients of G) at random points, for $3 \leq k \leq 13$ and $1 \leq d \leq 7$ (*verification/period_d_rank.py*). In every case

$$\text{rank} = \min(2d + 1, k + 1).$$

So the reachable symbols form a $(2d + 1)$ -dimensional family — not the $5d - 1$ -dimensional one a naive parameter count suggests — and the map is a submersion, so that every G is locally reachable, precisely when

$$2d + 1 \geq k + 1, \quad \text{that is} \quad d \geq \lceil k/2 \rceil.$$

At $d = 1$ this returns the 3 of Proposition 47, which is a check on the computation. We record the law as an observation supported by computation, not as a theorem: we have not proved the rank formula, only verified it over that range. Its consequence, if it holds in general, is that the period- d construction should reach $2k + 2$ for every k once $d \geq \lceil k/2 \rceil$ and n/d is large enough to

host $k + 1$ grid roots — a construction whose modulus is d rather than an lcm of cyclotomic orders, and therefore one that would cover a fixed congruence class of n for every k .

So the construction settles $k \leq 5$ on those residue classes and says nothing useful for $k \geq 6$; whether that is a limitation of period one or of $n \leq 130$ we do not know. For n outside the classes above, no certificate of nullity $2k + 2$ was found at all. This is not for want of parameters. Dropping equivariance

entirely leaves $5n$ free weights in the symmetric class and $8n$ in the combinatorially symmetric one, whereas a nullity of $r = 2k + 2$ costs $r(r + 1)/2 = (k + 1)(2k + 3)$ conditions in the first case and r^2 in the second — budgets that are met comfortably at every graph we tried. A two-stage search (many random starts on $\sum_{j \leq r} \lambda_j^2$ with analytic gradients, then a Gauss–Newton polish of the bilinear system $A(w)X = 0$, $X^\top X = I$; `verification/fast_nullity.py`) attains nullity 6 at machine precision on $P(12, 2)$ and $P(14, 2)$ but plateaus near 10^{-4} for $r = 7, 8$ on $P(12, 3)$ over hundreds of independent starts, in both matrix classes. We therefore record, as an observation rather than a theorem, that the obstruction to $2k + 2$ at general n appears to be arithmetic — whether Γ_n contains a compatible configuration of $k + 1$ points — and not a matter of dimension. Two guardrails are worth stating, since we tripped on both: the combinatorially symmetric class *contains* the symmetric one, so any search reporting a smaller answer for it is faulty (ours did, on $P(12, 2)$, until the larger search was seeded with the symmetric certificate); and any claimed nullity above $2k + 2$ is a bug by Corollary 45.

Why every result so far lives on a congruence class

Each exact value in this section holds for n in a divisibility class, not for all large n . That is not an artefact of our search. It is forced, and the following says exactly how.

Theorem 72 (Period-one certificates are bound to congruence classes). *Fix $k \geq 1$ and a period-one matrix A with parameters (a, α, β) and symbol F .*

- (i) *Suppose $\text{null } A = 2k + 2$ for some $n > 2k$. Then every root of F is $2\cos(2\pi j/d)$ with $\gcd(j, d) = 1$ and $d \geq 3$, and such a root lies in Γ_m if and only if $d \mid m$. Hence the set of n at which this same A attains the ceiling is exactly $\{n : \text{lcm}(\mathcal{D}) \mid n\}$, where $\mathcal{D}$ is the multiset of those d . It is therefore a single divisibility class of density $1/\text{lcm}(\mathcal{D}) \leq 1/3$, and it contains at most one prime.*
- (ii) *Consequently, to attain the ceiling for infinitely many n outside one divisibility class — in particular for infinitely many primes, or for all large n — the weights must depend on n .*
- (iii) *For $k \geq 3$ that dependence is constrained by one universal identity: the coefficient of s^{k-1} in F equals $-k$ for every (a, α, β) , so the roots always satisfy*

$$e_2(r_1, \dots, r_{k+1}) = -k, \quad \text{equivalently} \quad \sum_i r_i^2 = e_1^2 + 2k.$$

For $k = 2$ the coefficient of $s^{k-1} = s$ is $\alpha - 2$, which is free, so no such condition exists.

Proof. (i) Attaining $2k + 2$ needs all $k + 1$ roots interior and in Γ_n (Proposition 47), and an element of $\Gamma_n \cap (-2, 2)$ is $2\cos(2\pi m/n)$, which written in lowest terms is $2\cos(2\pi j/d)$ with $d \mid n$, $d \geq 3$. Conversely $2\cos(2\pi j/d) \in \Gamma_m$ iff $d \mid m$. The roots of F do not move with n , since F depends only on (a, α, β) ; so all of them lie on Γ_n exactly when every d divides n . A set $\{n : L \mid n\}$ with $L \geq 3$ contains a prime only if L is itself prime, and then only L .

(ii) Immediate from (i): a fixed A covers one class, and a union of finitely many classes still omits all but finitely many primes.

(iii) Write $(s + a)L_k = sL_k + aL_k$. Since L_k has nonzero coefficients only in degrees congruent to k modulo 2, aL_k has no s^{k-1} term, and the s^{k-1} coefficient of sL_k is the s^{k-2} coefficient of L_k , namely $-k$. The terms αs and β reach degree $k - 1$ only when $k \leq 2$. For a monic polynomial of degree $k + 1$ the coefficient of s^{k-1} is e_2 , and $\sum r_i^2 = e_1^2 - 2e_2$. $\square$

Part (iii) is what separates $k = 2$ from everything else. With no residual condition, Theorem 53 is free to choose the roots *as functions of n* — the three most negative elements of Γ_n° — and that is precisely why $k = 2$ is general in n while no other k is. For $k \geq 3$ an n -dependent choice must satisfy $e_2 = -k$ identically along the family. Substituting $r_i = 2\cos(2\pi m_i/n)$ and expanding e_1^2 by the product-to-sum formula turns this into a rational linear relation among cosines of rational angles set equal to a rational number. Relations of that shape are the subject of Conway–Jones [13], and they are arithmetically rigid: they hold not on open families but on sparse, classifiable configurations. That rigidity is the mechanism behind every divisibility condition in this paper, and it is the same mechanism as in Theorem 64.

We verified the identity of (iii) against all 26 certificates found in this paper: $e_2 = -k$ in every case with $k \geq 3$, and e_2 takes six different values among the $k = 2$ certificates, as (iii) predicts (`verification/general_n.py`).

Remark 73 (What this licenses, and what it forecloses). Theorem 72 should be read as a demarcation rather than a limitation. It says the entire cyclotomic layer of this paper — every result in Sections 17 and 19 — is *provably* incapable of producing a statement valid for all large n , so the congruence conditions are a boundary of the method and not a shortfall in its execution. It also says where to go instead: Corollary 45 characterises the ceiling for *every* matrix on the pattern, not only the invariant ones, by the condition $T = I$ on the monodromy, and that condition carries no arithmetic in it.

Exactly which n a rational symbol can reach

Theorem 63 builds certificates from products of the Ψ_d and Theorem 72 says every period-one certificate lives on one divisibility class. Together they make the rational case *classifiable*: there are finitely many candidate root sets, each is admissible or not by a finite check, and the union of the admissible classes is exactly the reach of the method. We carry this out.

Theorem 74 (Classification of rational-symbol certificates). *Fix $k \geq 2$. A symmetric rotation-invariant matrix in $S(P(n, k))$ whose symbol has rational coefficients and nullity $2k + 2$ exists if and only if $n > 2k$ and $L \mid n$ for some L in the finite list $\mathcal{L}_k$ below:*

k	admissible root sets $\mathcal{D}$ (orbits O_d)	$\mathcal{L}_k$
2	$\{7\}, \{9\}, \{14\}, \{18\}, \{3, 5\}, \{3, 10\}, \{4, 5\}, \{4, 10\}, \{5, 6\}, \{6, 10\}$	7, 9, 15, 20
3	$\{5, 10\}, \{5, 8\}, \{8, 10\}, \{3, 14\}, \{6, 7\}, \{3, 6, 8\}$	10, 24, 42
4	$\{4, 15\}, \{4, 30\}, \{5, 14\}, \{7, 10\}, \{5, 18\}, \{9, 10\}$	60, 70, 90
5	$\{3, 6, 24\}, \{7, 18\}, \{9, 14\}$	24, 126
6	<i>none</i>	<i>none</i>
7	$\{5, 10, 24\}$	120

In particular the classes $24 \mid n$ and $42 \mid n$ at $k = 3$ and $126 \mid n$ at $k = 5$ are new, the lists 60, 70, 90 at $k = 4$ and 24 at $k = 5$ were already complete, and $k = 6$ admits no rational-symbol certificate at all, which recovers Theorem 62.

Proof. Necessity. Let A be rotation-invariant with symbol $F \in \mathbb{Q}[s]$ and nullity $2k + 2$. By Theorem 72(i) every root of F is $2\cos(2\pi j/d)$ with $\gcd(j, d) = 1$ and $d \geq 3$, and F has $k + 1$ distinct such roots, each counted twice on Γ_n . Since F has rational coefficients its root set is stable under $\text{Gal}(\mathbb{Q}(\zeta_d)/\mathbb{Q})$, hence a union of full orbits $O_d = \{2\cos(2\pi j/d) : \gcd(j, d) = 1\}$ of sizes $\varphi(d)/2$ summing to $k + 1$; so $F/\text{lc}(F) = \prod_{d \in \mathcal{D}} \Psi_d$ for a set $\mathcal{D}$ of integers $d \geq 3$, and $d \mid n$ for each. Writing the symbol as $F = e[(s + a)(L_k(s) + \alpha) - c_0]$ with L_k monic of degree k (the block determinant with inner weight e), $\prod \Psi_d - (s + a)L_k$ must have degree at most 1, where a is the coefficient of s^k in $\prod \Psi_d$: these are the $k - 2$ linear conditions of Theorem 63, and they are now *necessary*. Finally $c^2 = e c_0$ must be nonzero, so $c_0 \neq 0$.

Sufficiency. Conversely, given $\mathcal{D}$ with $\sum \varphi(d)/2 = k + 1$ satisfying the degree condition and $c_0 \neq 0$, take $e = \text{sign}(c_0)$, $c = \sqrt{|c_0|}$, $d = ea$; then A is symmetric, carries the pattern, and its symbol vanishes exactly at the $k + 1$ roots, each hit twice whenever $\text{lcm}(\mathcal{D}) \mid n$.

Finiteness and the table. $\varphi(d)/2 \leq k + 1$ bounds d , so there are finitely many $\mathcal{D}$; the degree condition and $c_0 \neq 0$ are exact computations in $\mathbb{Z}[s]$. The enumeration (`verification/period_one_classification.py`) produces the table; the suite rebuilds the lists on every run and checks the nullity of each listed certificate on the first two members of its class and its absence just off the class. The minimal elements of the lcm's under divisibility are $\mathcal{L}_k$. $\square$

Remark 75. The theorem is sharp about the method, not about Z : $Z(P(n, 3)) = 8$ holds for every $n \geq 13$ (Theorem 111), but a *rotation-invariant, rational-symbol* proof of it exists only when 10, 24 or 42 divides n . Irrational real symbols reach more — at $k = 2$ they reach every $n \geq 9$ except 10 (Theorem 53) while the rational ones reach only the classes of 7, 9, 15, 20 — and the tiles of §20 reach everything. The table is the exact boundary of what cyclotomy alone can do.

19 Period- d certificates, and the first values for $k \geq 6$

Proposition 71 leaves $k = 6$ and $k \geq 8$ untouched: no rational period-one symbol attains $2k + 2$ there in either matrix class, and the exhaustive triple scan of Section 17 reaches only 8, 10, 8 for $k = 6, 7, 8$ over $n \leq 130$. ($k = 7$ is settled exactly by Theorem 61; what follows gives it, and $k = 6$ and $k = 8$, for further n .) The rank law says why, and what to do about it.

Prescribing the symbol

For $d \geq \lceil k/2 \rceil$ the map (weights) $\mapsto$ (monic coefficients of G) is a submersion, so instead of *searching* for a certificate one can *prescribe* one: choose $k + 1$ interior values of Γ_m , $m = n/d$, form the monic G^* with exactly those roots, and solve $G(w) = G^*$ for the weights. This is $k + 1$ equations in $5d$ unknowns with a full-rank Jacobian, so the solution set is a manifold of dimension $5d - (k + 1)$ — eight-dimensional for $k = 6$, $d = 3$. Each root is interior, hence met by two Fourier indices ℓ and $m - \ell$, so all $2k + 2$ of the corresponding blocks are singular and the nullity is $2k + 2$ by Corollary 45.

Two conditions must both hold, and they are independent: $m \geq 2k + 4$, so that Γ_m contains $k + 1$ distinct *interior* values at all; and $d \geq \lceil k/2 \rceil$, so that the symbol is reachable. For $k = 6$ this forces $d \geq 3$ and $n \geq 48$.

A degeneracy, and how it is escaped

Solving $G(w) = G^*$ without further constraint reliably returns a point at which one required-nonnzero weight — in every case a spoke — has collapsed to zero. Such a point is a genuine solution

of $G(w) = G^*$, because with a spoke weight zero the matrix block-diagonalises and G factors as $\det U_\ell \cdot \det V_\ell$; but the matrix then carries the pattern of a *different* graph and certifies nothing. At these points the Jacobian rank drops from 7 to 6: they are singular points of the map, which is why repeated random starts return to them.

Neither a realisability barrier nor continuation escapes them. Driving the vanishing spoke away from zero while re-solving loses the symbol at the first step ($\tau = 0.02$, error 9×10^{-5}). What does escape is *slack in the choice of roots*. At $n = 48$, Γ_{16} has exactly 7 interior values against the 7 required, so the root set is unique and there is nothing to vary; at $n = 54$ there are 8 interior values, giving 8 root sets, and all eight degenerate. At $n = 60$ and $n = 72$ there are 9 and 11 interior values, hence 36 and 330 root sets, and realisable points exist.

Theorem 76.

$$\begin{aligned} Z(P(60, 6)) &= M(P(60, 6)) = 14, & Z(P(72, 6)) &= M(P(72, 6)) = 14, \\ Z(P(96, 7)) &= M(P(96, 7)) = 16, & Z(P(96, 8)) &= M(P(96, 8)) = 18. \end{aligned}$$

In each case the period is the one the rank law prescribes: $d = 3$ for $k = 6$ and $d = 4$ for $k = 7, 8$, i.e. $d = \lceil k/2 \rceil$.

These are the first exact values of $Z(P(n, k))$ for $k \geq 6$ at which the bound $2k + 2$ is attained; $Z(P(2k + 1, k)) = 6$ [6, Thm. 2.6] is exact but far below it. That the required period was predicted in advance, and that $d = 3$ then failed for $k = 7, 8$ while $d = 4$ succeeded, is the strongest evidence we have that the rank law is describing something real rather than fitting the cases it was measured on. The certificates are period-three matrices found by the construction above and verified on the full $2n \times 2n$ matrix, independently of the Fourier blocks used to find them. For $n = 60$ with roots at $\ell \in \{2, 3, 4, 5, 6, 8, 9\}$ the 120×120 matrix has its 14 smallest eigenvalues below 4×10^{-15} and its fifteenth at 5.1×10^{-2} — a separation of thirteen orders of magnitude — every vertex carries exactly three off-diagonal nonzeros, and the smallest required entry is 0.188 of the matrix scale. For $n = 72$ with $\ell \in \{1, 4, 6, 7, 9, 10, 11\}$ the corresponding figures are nullity 14, first nonzero eigenvalue 1.2×10^{-3} of scale, and smallest required entry 0.162. For $k = 7$ at $n = 96$, $d = 4$, $\ell \in \{3, 5, 6, 7, 8, 9, 10, 11\}$: nullity 16, sixteen eigenvalues below 1.4×10^{-14} , smallest required entry 0.107. For $k = 8$ at $n = 96$, $d = 4$, $\ell \in \{2, 3, 4, 5, 6, 7, 8, 9, 11\}$: nullity 18, first nonzero eigenvalue 1.25×10^{-2} of scale, smallest required entry 0.260. Several further root sets at each n give independent certificates — three at $k = 8$ alone, the best with smallest required entry 0.362 — so these are not isolated points but regions.

Why an exact certificate is hard here

The $k \geq 6$ certificates are numerical because their weights are algebraic numbers of no recognised form. The obvious remedy — find *rational* weights, so that the nullity can be confirmed by exact rational elimination as for $k = 4, 5$ — runs into a structural obstruction.

Proposition 77. *Let A be a period- d matrix on $P(n, k)$ with rational weights, and $m = n/d$. The Fourier blocks have entries in $\mathbb{Q}(\zeta_m)$, and the Galois automorphism $\sigma_j : \zeta_m \mapsto \zeta_m^j$ carries M_ℓ to $M_{j\ell}$. Since σ_j fixes the weights, $\sigma_j(\det M_\ell) = \det M_{j\ell}$, so*

$$\det M_\ell = 0 \implies \det M_{j\ell} = 0 \quad \text{for every } j \text{ coprime to } m.$$

*Hence the set of singular blocks is a union of full Galois orbits of Γ_m , and a rational certificate can only realise a **Galois-closed** root set.*

This is checked numerically in the form it can be: for rational weights the product of $\det M_\ell$ over a Galois orbit — the norm — comes out exactly rational, to machine zero, at $(k, d, n) = (6, 3, 60)$, $(6, 3, 72)$, $(6, 4, 64)$, $(5, 1, 24)$ and $(7, 4, 96)$.

The obstruction is now visible: exactness requires a Galois-closed root set, and Galois-closed root sets are precisely the ones that degenerate. All eight at $(k, d, n) = (6, 3, 72)$ hit the symbol to 10^{-15} with nullity 14 and all eight collapse a weight; both at $(6, 3, 60)$ likewise. So for these parameters *we found no rational certificate*. We state it that way deliberately: the degenerate points are singular points of the map and therefore attractors, which is why repeated random starts return to them, so a search collapsing every time is exactly the kind of negative evidence we caution against reading as a non-existence proof elsewhere in this section.

Relaxing to a subfield does not obviously help, and it is worth being precise about why. If the weights lie in $K = \mathbb{Q}(\zeta_m)^H$ then the singular set need only be H -stable, and at $(6, 3, 72)$ all three index-two subgroups admit realisable root sets. But that implication runs one way: an H -stable root set being realisable shows only that the necessary condition is not violated, and gives no reason to expect a certificate for it to have weights in K . Indeed integer-relation detection finds no such representation for the certificates we have, a search parametrised inside $\mathbb{Q}(\sqrt{2})$ from the start returns nothing, and at $(6, 3, 60)$ every index-two set degenerates outright. We therefore record exactness for $k \geq 6$ as **open**, with the evidence leaning negative, rather than as a route in progress. The remaining candidate is a computer-assisted existence proof (interval Newton) for an irrational certificate, which the ceiling makes tractable in principle, since certifying the equation $G(w) = G^*$ exactly delivers the nullity as a discrete consequence.

How many n does this reach?

The construction needs a divisor d of n with

$$\lceil k/2 \rceil \leq d \leq \frac{n}{2k+4},$$

the left inequality from the rank law and the right from requiring $\Gamma_{n/d}$ to contain $k+1$ distinct interior values. This is a real restriction and it cuts both ways.

Proposition 78 (Reachability). *The set of n admitting such a divisor has density 1, but its complement contains every prime, and also every $n = 2p$ once $k \geq 6$ (since then $d \geq 3$ while the only small divisor is 2). Convergence is slow: at $k = 6$ the reachable proportion of n is 54% up to 240, 71% up to 1000 and 79% up to 5000; at $k = 8$, 36%, 62% and 73%.*

In particular **prime n lies outside the period- d method entirely**, because d must divide n and $d = 1$ gives rank 3. That is not a gap in the search but a structural feature: for prime n the only available construction is the fully non-equivariant one, i.e. solving $T = I$ directly in the $5n$ weights, which is the regime where our searches plateau. Since $Z(P(23, 5)) = 12 = 2k + 2$ with 23 prime, such n certainly can attain the ceiling; we simply cannot certify them this way.

Larger d is not merely permitted but helpful. The rank law requires only $d \geq \lceil k/2 \rceil$, and taking d strictly larger enlarges the solution manifold from dimension $5d - (k+1)$, giving more room to avoid the degenerate locus. The effect is visible at the tightest configurations. At $k = 6$, $d = 3$, $n = 48$ the grid Γ_{16} offers exactly 7 interior values for the 7 required — a unique root set, no slack — and every solution collapses a spoke. At $k = 6$, $d = 4$, $n = 64$ the grid is again Γ_{16} with the same unique root set, but now the manifold has dimension $20 - 7 = 13$ rather than $15 - 7 = 8$, and a realisable certificate is found on the first attempt (smallest required entry 0.086, gap 6.8×10^{-2}). So the degeneracy of the previous subsection is a symptom of working at the minimum period, not an obstruction inherent to tight root sets. Certificates were likewise found at $k = 6$, $d = 6$, $n = 96$.

Within the reachable set, the construction appears to work essentially always. Sweeping n for $k = 6$ at $d = 3$ produced a realisable certificate at *every* n tested:

$$n = 60, 72, 84, 90, 96, 108, 120,$$

most of them on the first root set tried, with smallest required entries between 0.12 and 0.46. So

$$Z(P(n, 6)) = M(P(n, 6)) = 14 \quad \text{for each of these } n,$$

and the earlier appearance of a scattered list of congruence classes was an artefact of the cyclotomic route, not of the problem. We state this as a computation rather than a theorem: we have no proof that a realisable root set exists for every reachable n , only that one was found in every case examined.

Remark 79 (Status of Theorem 76). Unlike the cyclotomic certificates of Section 17, which are integer matrices whose nullity is confirmed by exact rational elimination, these are found by a Newton solve and their weights are algebraic numbers of no recognised form. We therefore record Theorem 76 as **numerically certified but not yet verified in exact arithmetic**. The evidence is strong — a thirteen-order spectral separation, the correct zero pattern with every required entry a fifth of the matrix scale, and the nullity never exceeding the ceiling $2k + 2$, which is an automatic check on the whole pipeline — but it is not of the same kind as the $k = 4$ and $k = 5$ results, and we do not claim that it is. The natural route to closing the gap is to choose the root set to be a union of full Galois orbits of Γ_m , which makes G^* a product of the Ψ_e and hence puts the system $G(w) = G^*$ over $\mathbb{Q}$; that is in progress.

20 Beyond congruence classes: a certified value at a prime

Theorem 72 says the cyclotomic layer cannot produce a value valid for all large n , and points at the replacement: Corollary 45 characterises the ceiling for *every* matrix on the pattern by the condition $T = I$, which carries no arithmetic. This section carries that out at a prime n , where by Theorem 72(i) no period-one certificate can reach.

A better system to certify

Optimising $T = I$ is convenient but bad to certify: the map $w \mapsto T$ has a 51-dimensional image inside the 64-dimensional space of 8×8 matrices at $k = 3$, so $\{T = I\}$ is cut out by 13 local equations we cannot identify and no square subsystem is available. We certify the nullity directly instead.

Now null $A \geq r$ if and only if some full-rank $K \in \mathbb{R}^{2n \times r}$ has $AK = 0$. Fixing a row permutation P and writing K in graph form $K = P \begin{pmatrix} I_r \\ X \end{pmatrix}$ makes rank $K = r$ automatic, and

$$f(w, X) := A(w) K(X) = 0$$

is *bilinear*: quadratic of total degree 2, with rational coefficients. Its Jacobian is affine, so the Lipschitz constant of J is an explicit integer, and a Krawczyk contraction bound needs only three norms rather than an interval matrix product.

The choice of which equations to solve is not free. The system $AK = 0$ has $2nr$ equations of rank $2nr - \binom{r}{2}$, and the deficiency is structural: $K^\top AK$ is symmetric because A is, and $\binom{r}{2}$ is exactly its antisymmetric part. Concretely, if the $(2n - r)r$ equations on the non-pivot rows $P[r:]$ vanish

then $K^\top AK$ equals the $r \times r$ block $(AK)[P[:r]]$, which must therefore be symmetric, so setting its upper triangle to zero forces the lower triangle as well. Hence

$$\underbrace{(AK)[P[r+m], t] = 0}_{(2n-r)r \text{ equations}} \quad \text{and} \quad \underbrace{(AK)[P[s], t] = 0, s \leq t}_{\binom{r+1}{2} \text{ equations}}$$

is *exactly equivalent* to $AK = 0$. Selecting rows by pivoting instead would leave $\binom{r}{2}$ equations unproved.

The reduction is not special to $P(n, k)$

Theorem 44 is stated for $P(n, k)$, but its proof uses only one structural fact: the row at an outer vertex meets *exactly one* inner variable, whose coefficient is a spoke weight and so is nonzero. Nothing in it requires the outer cycle to have voltage 1. Let $B_{p,q,w}$ denote the two-vertex base with a loop of voltage p at u , a loop of voltage q at v , and a single edge uv of voltage w ; its cover is cubic on $2n$ vertices, and $P(n, k) = B_{1,k,0}^n$.

Theorem 80 (Reduction on every two-vertex cover). *Let $n > 2(p+q)$ and let $A \in \mathcal{S}(B_{p,q,w}^n)$. Eliminating the inner coordinates leaves a single linear recurrence in the outer coordinates supported on the nine offsets*

$$\{-(p+q), -q, -(q-p), -p, 0, p, q-p, q, p+q\},$$

whose two extreme coefficients are nonzero. Consequently

$$\text{null } A = \dim \ker(T - I) \leq 2(p+q),$$

which is the degree span of $\det M(\zeta)$, with equality if and only if $T = I$. Theorem 44 is the case $p = 1, w = 0$, where $2(p+q) = 2k+2$.

Proof. The neighbours of u_i are $u_{i \pm p}$ and v_{i+w} , so the row at u_i is $b'_{i-p}x_{i-p} + a_i x_i + b_i x_{i+p} + c_i y_{i+w} = 0$ with $c_i \neq 0$. Writing $i = j - w$ solves y_j as a three-term expression in x at offsets $-w-p, -w, -w+p$ from j . Substituting into the row at v_j , namely $e'_{j-q}y_{j-q} + d_j y_j + e_j y_{j+q} + c'_{j-w}x_{j-w} = 0$, and reindexing by $m = j - w$ gives exactly the nine listed offsets. The coefficient at $+(p+q)$ is $-e_j b_{j+q-w}/c_{j+q-w}$, a product of an inner edge weight, an outer edge weight and a spoke weight, all nonzero; symmetrically at $-(p+q)$. So the relation is a recurrence of order exactly $2(p+q)$, solvable in both directions, its solutions on $\mathbb{Z}$ form a space of that dimension on which T acts, and the n -periodic ones are its fixed vectors. If $p = q$ the nine offsets collapse to five, but the extreme coefficients are unchanged and so is the order. $\square$

The elimination was verified symbolically for (p, q, w) with $p \leq 3, q \leq 5$ and $w \leq 2$ — the offsets and the extreme coefficients come out as stated in every case — and the resulting bijection $x \mapsto (x, Lx)$ onto $\ker A$ was confirmed numerically, with $\text{null } A = \text{null } R$ and a lift residual at machine precision. See `verification/general_reduction.py`.

The hypothesis that each outer row meets exactly one inner variable is not, however, the real requirement. It makes the elimination *single-step*; what the argument actually needs is that the system can be advanced one fibre at a time. On a base with several u - v edges the elimination takes two steps instead of one, and still closes.

Theorem 81 (Reduction on theta covers). *Let B have two vertices joined by edges of voltages $t_1 < \dots < t_m$ and no loops, put $L = t_m - t_1$, and let $n > 2L$. For every $A \in \mathcal{S}(B^n)$ the kernel equations form a first-order recurrence on the state*

$$s_i = (x_i, \dots, x_{i+L-1}, y_i, \dots, y_{i+L-1}) \in \mathbb{R}^{2L},$$

so that $\text{null } A = \dim \ker(T - I) \leq 2L$, which is again the degree span of $\det M(\zeta)$.

Proof. The row at u_i is $a_i x_i + \sum_r w_i^{(r)} y_{i+t_r} = 0$; its highest-index inner variable is y_{i+t_m} , whose coefficient $w_i^{(r=m)}$ is an edge weight and so nonzero, and solving for it expresses y_{i+t_m} through x_i and the earlier y 's. The row at v_j is $d_j y_j + \sum_r w_{j-t_r}^{(r)} x_{j-t_r} = 0$; its highest-index outer variable is x_{j-t_1} , again with a nonzero coefficient. Shifting t_1 to 0, the two together carry s_i to s_{i+1} , so T_i exists and is invertible, and the n -periodic solutions are the fixed vectors of $T = T_{n-1} \cdots T_0$. Finally, with $f(\zeta) = \sum_r w^{(r)} \zeta^{t_r}$,

$$\det M(\zeta) = ad - f(\zeta)f(\zeta^{-1}),$$

whose degree span is $2(t_m - t_1) = 2L$. □

Verified on eight bases, including four-edge ones, with $\text{null } A = \dim \ker(T - I)$ and the state dimension equal to the degree span in every case; see `verification/theta_reduction.py`. Together with Theorem 80 this settles the ceiling for two-vertex bases of both shapes — loops with a single connecting edge, and several connecting edges with no loops.

Both of those are two-vertex statements. The elimination in Theorem 80 is not, however, limited to two vertices: it chains, and chaining it settles the ceiling on bases of arbitrary size.

Theorem 82 (Reduction on a path-with-loops cover). *Let B have vertices $x_1, \dots, x_m$, a loop of voltage $p_t \geq 1$ at each x_t , and a single edge $x_t - x_{t+1}$ of voltage w_t for $1 \leq t < m$. Put $P = \sum_{t=1}^m p_t$, let $n > 2P$, and let $A \in \mathcal{S}(B^n)$ have all edge weights nonzero. Then eliminating the coordinates of the fibres over $x_2, \dots, x_m$ leaves a single linear recurrence in the fibre over x_1 whose extreme offsets are $\pm P$ with nonzero coefficients. Consequently*

$$\text{null } A = \dim \ker(T - I) \leq 2P,$$

which is the degree span of $\det M(\zeta)$, with equality if and only if $T = I$. Theorem 80 is the case $m = 2$.

Proof. Write X_t for the fibre over x_t and $S_t = 2(p_1 + \cdots + p_t)$, $S_0 = 0$. We show by induction that X_{t+1} is expressible in the coordinates of X_1 across a window of offsets of span S_t .

For $t = 1$: the neighbours of (x_1, i) are $(x_1, i \pm p_1)$ and $(x_2, i + w_1)$, so the row there reads $b'_{i-p_1} X_1(i - p_1) + a_i X_1(i) + b_i X_1(i + p_1) + c_i X_2(i + w_1) = 0$ with $c_i \neq 0$. Solving for $X_2(i + w_1)$ expresses X_2 in X_1 across offsets of span $2p_1 = S_1$.

For the step, let $1 < t < m$. The row at (x_t, i) is

$$b'_{i-p_t} X_t(i - p_t) + a_i X_t(i) + b_i X_t(i + p_t) + \gamma_i X_{t-1}(i - w_{t-1}) + c_i X_{t+1}(i + w_t) = 0,$$

again with $c_i \neq 0$, so it solves for $X_{t+1}(i + w_t)$. By induction X_t has span S_{t-1} in X_1 ; the two loop terms shift that window by $\pm p_t$ and so contribute span $S_{t-1} + 2p_t = S_t$, while X_{t-1} contributes only $S_{t-2} < S_t$. Hence X_{t+1} has span S_t , as claimed.

The row at (x_m, i) has no further vertex to solve for: it reads $b'_{i-p_m} X_m(i - p_m) + a_i X_m(i) + b_i X_m(i + p_m) + \gamma_i X_{m-1}(i - w_{m-1}) = 0$, and substituting gives a relation in X_1 alone of span $S_{m-1} + 2p_m = S_m = 2P$, the term in X_{m-1} again contributing less. The coefficient at the offset $+P$ is a product of the loop weights b at each x_t , the connecting weights, and the inverses of the connecting weights used to solve at each step; every factor is nonzero, so it is nonzero, and symmetrically at $-P$. The relation is therefore a recurrence of order exactly $2P$, solvable in both directions. Its solutions on $\mathbb{Z}$ form a space of dimension $2P$ on which the monodromy T of one

period acts, and the n -periodic solutions — which is to say $\ker A$ — are the fixed vectors of T . That $2P$ is the degree span of $\det M(\zeta)$ is immediate: $M(\zeta)$ is tridiagonal with diagonal entries $d_t + b_t(\zeta^{p_t} + \zeta^{-p_t})$ and off-diagonal pairs contributing $\zeta^{w_t}\zeta^{-w_t} = 1$, so the extreme exponents come from $\prod_t(\zeta^{p_t} + \zeta^{-p_t})$ alone. $\square$

Theorem 83 (Cycle bases). *Let B be the cycle $x_0-x_1-\dots-x_{m-1}-x_0$ with edge voltages $w_0, \dots, w_{m-1}$ and holonomy $W = \sum_t w_t$. If $W \neq 0$ then for every $A \in \mathcal{S}(B^n)$,*

$$\text{null } A \leq 2 \gcd(|W|, n) \leq 2|W|,$$

and $2|W|$ is the degree span of $\det M(\zeta)$.

Proof. Every vertex of B has degree 2, hence so does every vertex of B^n : the cover is a disjoint union of cycles. Lifting the base cycle starting at (x_0, i) returns to $(x_0, i + W)$, so the fibre over x_0 is partitioned into the orbits of $i \mapsto i + W$ on $\mathbb{Z}_n$, of which there are $\gcd(|W|, n)$; the cover is therefore $\gcd(|W|, n)$ cycles, each of length $mn / \gcd(|W|, n)$. A matrix supported on a disjoint union of g cycles is block diagonal with g blocks, each supported on a single cycle, and $M(C_N) = 2$ since $\text{mr}(C_N) = N - 2$; so $\text{null } A \leq 2g$. Finally $\gcd(|W|, n) \leq |W|$ because $W \neq 0$.

For the span: $M(\zeta)$ is tridiagonal with corners. In the Leibniz expansion every permutation other than the two m -cycles pairs each off-diagonal entry $c_t \zeta^{w_t}$ with its transpose $c_t \zeta^{-w_t}$ and so contributes no power of ζ ; the two m -cycles contribute $\mp \prod_t c_t \zeta^{\pm W}$. The extreme exponents are therefore $\pm W$, and the span is $2|W|$. $\square$

Remark 84. The hypothesis $W \neq 0$ is not a technicality. If $W = 0$ the lift of the base cycle closes after one circuit, the cover is n disjoint m -cycles — and in particular disconnected — while $\det M(\zeta)$ is constant, so the span is 0 and the bound would assert $\text{null } A = 0$, which is false. This is the same degeneracy that excludes tree bases: a cover whose voltages do not generate $\mathbb{Z}_n$ is disconnected, and the ceiling is a statement about connected covers.

Theorem 83 is worth contrasting with Theorem 82. It closes the cycle case, which the chained elimination cannot reach — but it closes it for an entirely different reason, and an easier one: on a cycle base the cover degenerates to a union of cycles, where the bound follows from $M(C_N) = 2$ rather than from any transfer-matrix argument. What that tells us is that the obstruction we named was an obstruction to the *method*, not evidence against the statement.

Theorem 85 (The cover ceiling fails on a general base). *There is a connected cyclic cover and a matrix carrying its pattern whose nullity exceeds the degree span of $\det M(\zeta)$ by an amount growing with n . Explicitly, let B have vertices $0, 1, 2, 3$, edges $0-1, 0-2, 0-3$ of voltages $0, 1, 2$ and a loop of voltage 1 at vertex 1, so the degrees are $(3, 3, 1, 1)$ and $D = 2$. Give every edge of B^n weight 1, the fibres over 0 and 1 diagonal entries 1 and 3, and the fibres over 2 and 3 diagonal entry 0. Then B^n is connected and*

$$\text{null } A = n \quad \text{for every } n, \quad \text{against } D = 2.$$

Proof. Vertices 2 and 3 have degree 1 in B , so their fibres are pendant vertices of B^n . The diagonal is unconstrained in $\mathcal{S}(G)$, so the assignment is legitimate. The row at a pendant $(2, j)$ reads $0 \cdot x_{(2,j)} + x_{(0,j-1)} = 0$, forcing $x_{(0,i)} = 0$ for every i , and likewise from the fibre over 3. With the fibre over 0 annihilated, the row at $(1, j)$ becomes $x_{(1,j-1)} + 3x_{(1,j)} + x_{(1,j+1)} = 0$, a three-term recurrence on a cycle of length n whose only solution is $x_{(1,\cdot)} = 0$ because $3 > 2$. What remains is the row at $(0, i)$, namely $x_{(1,i)} + x_{(2,i+1)} + x_{(3,i+2)} = 0$, which with $x_{(1,\cdot)} = 0$ is one linear relation between the two pendant values attached to $(0, i)$. That leaves $2n$ pendant coordinates subject to n independent relations, so $\text{null } A = n$. The ranks were also computed over $\mathbb{Z}$ for $n = 5, 6, 7$ and for several edge weights (`verification/cover_ceiling_counterexample.py`). $\square$

Proposition 86 (The independent-set obstruction). *Let G be a graph and $S \subseteq V(G)$ an independent set. Then*

$$M(G) \geq |S| - |N(S)|.$$

Proof. Choose $A \in \mathcal{S}(G)$ with $a_{vv} = 0$ for every $v \in S$ and with all off-diagonal entries nonzero; the diagonal is unconstrained in $\mathcal{S}(G)$, so this is legitimate. Consider the vectors x supported on S . For $v \in S$ the row at v reads $a_{vv}x_v + \sum_{w \sim v} a_{vw}x_w$, and every neighbour w of v lies outside S because S is independent, so $x_w = 0$ and the row reduces to $a_{vv}x_v = 0$, which holds identically. For $u \notin S$ the row reads $\sum_{w \sim u, w \in S} a_{uw}x_w = 0$, which is vacuous unless $u \in N(S)$. So the vectors supported on S that lie in $\ker A$ are cut out by at most $|N(S)|$ linear conditions on $|S|$ coordinates, and $\text{null } A \geq |S| - |N(S)|$. $\square$

The bound is elementary and we do not claim it is new; what matters here is that it is exactly the mechanism behind Theorem 85. On the star-with-a-loop cover take S to be the two pendant fibres: $|S| = 2n$, $N(S)$ is the fibre over the centre, $|N(S)| = n$, and the bound gives $\text{null } A \geq n$ — which is precisely the nullity exhibited there. On the $K_{2,3}$ cover, S the three degree-two fibres gives $|S| - |N(S)| = 3n - 2n = n$, against $D = 4$.

And it is also exactly what regularity forbids.

Corollary 87 (Why regularity is the right hypothesis). *If G is d -regular then $|N(S)| \geq |S|$ for every independent set S , so Proposition 86 is vacuous on a regular graph.*

Proof. Count the edges leaving S . Each $v \in S$ has d of them and, S being independent, every one lands in $N(S)$; each vertex of $N(S)$ receives at most d . Hence $d|S| \leq d|N(S)|$. $\square$

So the particular obstruction that kills the ceiling on the star and on $K_{2,3}$ is available exactly when some independent set fails to expand, and in a regular graph no independent set fails to expand. It rules out one mechanism, not all of them; and the mechanism it rules out is the degenerate case of a larger one that regularity does not block.

Proposition 88 (The low-rank-subgraph obstruction). *For every $S \subseteq V(G)$,*

$$M(G) \geq |V(G)| - \text{mr}(G[S]) - 2|V(G) \setminus S|,$$

where $\text{mr}(G[S])$ is the minimum rank of the induced subgraph. Proposition 86 is the case of S independent, where $\text{mr}(G[S]) = 0$.

Proof. Choose $A \in \mathcal{S}(G)$ whose principal submatrix on S attains $\text{mr}(G[S])$: the diagonal is free and the off-diagonal support on S is $E(G[S])$, so this is legitimate; give every other edge a nonzero weight. Write $S^c = V(G) \setminus S$. The rows of A indexed by S form the block $[A[S] \ A[S, S^c]]$, of rank at most $\text{rank } A[S] + \text{rank } A[S, S^c] \leq \text{mr}(G[S]) + |S^c|$, and the rows indexed by S^c number $|S^c|$. So $\text{rank } A \leq \text{mr}(G[S]) + 2|S^c|$. $\square$

Regularity bounds $|N(S)|$ from below for an independent S and says nothing about $\text{mr}(G[S])$ for a general S . A regular cover whose fibres contain triangles has induced subgraphs of very low minimum rank, and that is what happens on K_4 in Section 21.

Remark 89 (The obvious repair fails too). One might demand minimum degree 2 in the base, so that the cover has no pendant vertices. That is not enough. Take $B = K_{2,3}$ with parts $\{0, 1\}$ and $\{2, 3, 4\}$ and voltages 0, 1, 2 and 1, 0, 1; the degrees are $(3, 3, 2, 2, 2)$ and $D = 4$. Zeroing the diagonal on the three degree-2 fibres gives $\text{null } A = 6$ at $n = 5$ and 8 at $n = 7$, both connected covers, again computed exactly over $\mathbb{Z}$.

Regularity does not survive either. Every *regular* base we attacked — C_3 , the theta and four-edge theta bases, the two-loop base, K_4 , $K_{3,3}$ and the prism — held under both the diagonal-zeroing construction above and a gradient search over fully non-equivariant weights, and on that evidence we conjectured the ceiling for regular bases. The conjecture is false, and K_4 was in the list: Theorem 129 gives nullity n against $D = 6$ on a cubic cover of K_4 , by Proposition 88 rather than by either attack. What does survive is the rest of the paper: $P(n, k)$ and the theta covers are handled by Theorems 44, 80 and 81, which are direct eliminations and never pass through $\det M(\zeta)$. Theorems 85 and 129 kill the claim for arbitrary and for regular bases respectively; what replaces regularity as the hypothesis is Theorem 91 below.

Remark 90 (What this does and does not reach). Theorem 82 removes the two-vertex restriction, but not by settling the general conjecture: it settles the bases on which the elimination *chains*, namely those whose underlying graph is a path and whose connecting edges are single. The chain does not terminate when the base contains a cycle — the last elimination feeds back into a fibre already eliminated — and it solves for nothing when two vertices are joined by parallel edges, since the relevant row then contains two unknowns of the next fibre; that is the θ case of Theorem 81. Theorem 83 disposes of the first obstruction for *pure* cycle bases, by a degeneration rather than by an elimination. What remains genuinely open is the case that is neither a path nor a cycle nor two vertices: a base carrying both a cycle and extra structure, such as a cycle with loops, or K_4 , where neither argument applies — and, for an arbitrary base, nothing at all, since Theorem 85 shows the statement is false there, and for a regular base likewise, by Theorem 129. The live question is the condition the proofs actually use.

Theorem 91 (Leading-coefficient criterion). *Let B be a base with voltages, and let $c_{\max}$ be the coefficient of the top power of ζ in $\det M(\zeta)$, regarded as a polynomial in the edge weights and the diagonal entries of B . If $c_{\max}$ is a nonzero monomial in the edge weights alone — equivalently, a unique permutation of $V(B)$ attains the maximum total voltage and it uses no diagonal entry (Lemma 93) — then $\text{null } A \leq D$ for every $A \in \mathcal{S}(B^n)$ and every n .*

The condition is what every proof in this paper uses. On the two-loop base the top coefficient is the product of the two loop weights; on a theta base, minus the product of the two extreme-voltage edge weights; on a path with a loop at every vertex, the product of the loop weights; on a cycle base, the product of all edge weights. None of these can vanish, which is why each elimination has a leading term to solve for. On each of the three bases where we know the ceiling fails, the top coefficient contains a diagonal entry: on the star with a loop it is $-w_\ell (d_2 w_{03}^2 + d_3 w_{02}^2 - d_0 d_2 d_3)$ with w_ℓ the loop weight; on $K_{2,3}$ it is $-w_{02} w_{12} (d_3 w_{04} w_{14} + d_4 w_{03} w_{13})$; on the K_4 base of Section 21 it is $-w_{12} w_{23} (w_{01} w_{03} - d_0 w_{13})$. A free diagonal in the top coefficient means it can be made to vanish, the span drops, and the counting argument has no leading term; whether the whole determinant can then be made to vanish is a further question, answered for K_4 by a rank-one fibre. The criterion is checked symbolically on every base in the paper (`verification/k4_rank_one.py`). The proof follows. It is not the chained elimination of Theorem 82, whose order can exceed D (Theorem 98), but a count on a window cut by two gradings of the symbol, and it shows that the kernel of the lifted operator on the infinite strip has dimension exactly D for every choice of nonzero weights. The two-vertex, path and cycle theorems are the cases in which one grading already suffices.

The leading-coefficient criterion, proved

Theorem 91 is proved here. The proof is not the chained elimination: that elimination solves for the unknowns of one fibre at a time and, as Theorem 98 shows, its order can exceed the degree

span. What gives exactly D is a count that grades the unknowns twice — once from the top of the symbol and once from the bottom — and reads the kernel off a window that is cut by one grading on each side.

Notation. Throughout, B has vertex set V , $|V| = m$, and for $i, j \in V$ let $E_{ij} \subset \mathbb{Z}$ be the set of exponents of ζ occurring in the entry $M_{ij}(\zeta)$: the voltages v of the edges $i \rightarrow j$, their negatives for $j \rightarrow i$, both $\pm v$ for a loop at $i = j$, and always $0 \in E_{ii}$ for the diagonal. Put $v^+(i, j) = \max E_{ij}$ (and $-\infty$ if $E_{ij} = \emptyset$). By $M(\zeta)^\top = M(\zeta^{-1})$ we have $E_{ji} = -E_{ij}$. Expanding the determinant, the top degree of $\det M(\zeta)$ is

$$V_{\max} = \max_{\sigma} \sum_{i \in V} v^+(i, \sigma(i)),$$

the maximum over permutations σ of V with every $(i, \sigma(i))$ an entry, the bottom degree is $-V_{\max}$, and $D = 2V_{\max}$. The *strip* B^∞ is the derived graph over $\mathbb{Z}$: vertices (x, t) , an edge $(x, t) - (y, t + v)$ for every edge $x - y$ of voltage v and every t . A *strip matrix* is a real symmetric matrix A indexed by $V \times \mathbb{Z}$ with the pattern of B^∞ , every edge weight nonzero and the diagonal arbitrary; no decay is imposed on anything. Its kernel K is the space of all sequences $x = (x_{j,s})$ with $Ax = 0$, and the row of A at (i, t) reads

$$\sum_{j \in V} \sum_{v \in E_{ij}} a_{ij}^v(t) x_{j, t+v} = 0, \quad (3)$$

with $a_{ij}^v(t) \neq 0$ for $v \neq 0$ or $j \neq i$, and $a_{ii}^0(t)$ the free diagonal.

Lemma 92 (Potentials, and the leading matrix). *There are integers $(p_i)_{i \in V}$, $(q_j)_{j \in V}$ with*

$$p_i + q_j \geq v \quad \text{for all } i, j \text{ and } v \in E_{ij}, \quad \sum_i p_i + \sum_j q_j = V_{\max}.$$

Call (i, j, v) tight if $p_i + q_j = v$, and let Γ be the bipartite graph on two copies of V with an edge ij for each tight (i, j, v) . Let L be the $m \times m$ matrix whose (i, j) entry is the coefficient of $\zeta^{p_i + q_j}$ in $M_{ij}(\zeta)$. Then $\det L = c_{\max}$ as polynomials in the weights and diagonals, and the perfect matchings of Γ are exactly the permutations attaining $V_{\max}$.

Proof. The first statement is the duality theorem of the assignment problem with integer weights $v^+(i, j)$ (Egerváry; see [15, Ch. 17]), applied to the complete bipartite graph with $v^+(i, j) = -\infty$ on missing entries; the constraint for $v < v^+(i, j)$ is implied. For the second, the matrix $N(\zeta) = \text{diag}(\zeta^{-p_i}) M(\zeta) \text{diag}(\zeta^{-q_j})$ has entries of degree ≤ 0 whose degree-0 part is L , and $\det N = \zeta^{-V_{\max}} \det M$, so the constant term of $\det N$, which is $\det L$, is the coefficient of $\zeta^{V_{\max}}$ in $\det M$. For the third: if $(i, \sigma(i), v)$ is tight then $v \leq v^+(i, \sigma(i)) \leq p_i + q_{\sigma(i)} = v$, so $\sum_i v^+(i, \sigma(i)) = \sum p + \sum q = V_{\max}$; conversely a maximising σ has $\sum_i (p_i + q_{\sigma(i)} - v^+(i, \sigma(i))) = 0$ with every term ≥ 0 . $\square$

Lemma 93 (What a monomial top coefficient means). *$c_{\max}$ is a nonzero monomial in the edge weights alone if and only if $V_{\max}$ is attained by exactly one permutation σ^* and, for every i , the top coefficient of $M_{i\sigma^*(i)}$ is a single edge weight. In that case $c_{\max} = \pm \prod_i w(i, \sigma^*(i))$, Γ has σ^* as its unique perfect matching, and for every $S \subseteq V$ the induced subgraph $\Gamma[S, \sigma^*(S)]$ has $\sigma^*|_S$ as its unique perfect matching.*

Proof. “If” is the expansion of the determinant. For “only if”, each maximising σ contributes $\text{sgn}(\sigma) \prod_i (\text{top coefficient of } M_{i\sigma(i)})$ to $c_{\max}$, the top coefficient being a sum of the weights of the edges $i \rightarrow \sigma(i)$ of voltage $v^+(i, \sigma(i))$, or d_i at a fixed point without a loop. Expand each contribution

into monomials. A monomial determines the multiset of edges used, hence the 2-regular multigraph they form on V , hence the cycle structure of σ and therefore its sign; two maximising permutations with a common monomial differ only in the orientation of cycles of length ≥ 3 , which must have voltage 0 for both to be maximal. So no cancellation occurs between distinct permutations, and every monomial of every maximising permutation survives in $c_{\max}$. If $c_{\max}$ is a single monomial, no top coefficient involved is a sum and no d_i occurs; and if two permutations attained $V_{\max}$ they would differ by reversing a zero-voltage cycle C of length ≥ 3 , which can be replaced by 2-cycles along the edges of C (and, if $|C|$ is odd, one fixed point, at a vertex that carries no loop, since a loop there would raise the voltage above $V_{\max}$): a third maximising permutation whose monomial contains a squared edge weight or a d_i , contradicting the single monomial. Uniqueness of the perfect matching of Γ is Lemma 92; a second perfect matching of $\Gamma[S, \sigma^*(S)]$ would extend by σ^* on $V \setminus S$ to a second perfect matching of Γ . $\square$

Two gradings. Fix potentials as in Lemma 92 and put $h_i = p_i + q_i \geq 0$ (the constraint for $0 \in E_{ii}$). Grade rows and unknowns of the strip twice:

$$\text{top: } \tau(i, t) = t + p_i, \quad \lambda(j, s) = s - q_j; \quad \text{bottom: } \tau'(i, t) = t - q_i, \quad \lambda'(j, s) = s + p_j.$$

In the row (3) at (i, t) the unknown $x_{j, t+v}$ has

$$\lambda = \tau - (p_i + q_j - v) \leq \tau, \quad \lambda' = \tau' + (p_j + q_i + v) \geq \tau',$$

the second because $-v \in E_{ji}$. Equality holds exactly for the tight (i, j, v) , respectively the tight $(j, i, -v)$. Thus every row sees unknowns at top-level at most its own and at bottom-level at least its own, and $\lambda'(j, s) = \lambda(j, s) + h_j$, $\tau'(i, t) = \tau(i, t) - h_i$. At a fixed top-level τ there is one row $(i, \tau - p_i)$ and one unknown $(j, \tau + q_j)$ for each $i, j \in V$; the $m \times m$ matrix L_τ of the coefficients of the level- τ unknowns in the level- τ rows has support in Γ and carries, on the edges of σ^* , the weights of the lifted edges realising them: nonzero. Likewise the bottom-level matrix L'_τ has support in the transpose of Γ , whose unique perfect matching is σ^{*-1} . By Lemma 93, for every $S \subseteq V$ the sub-block $L_\tau[S, \sigma^*(S)]$ is, after permuting its columns, a matrix with a unique perfect matching in its support and nonzero entries on it: its determinant is $\pm$ the product of those entries, nonzero. The same holds for L'_τ and σ^{*-1} .

Theorem 94 (The strip kernel has dimension exactly D). *Let $c_{\max}$ be a nonzero monomial in the edge weights alone. Then for every strip matrix A , $\dim \ker A = D$.*

Proof. Let $h = \max_i h_i$ and fix integers $\beta < b$ with $b - \beta \geq h$. Put

$$U = \{(j, s) : \lambda \leq b, \lambda' \geq \beta\}, \quad R = \{(i, t) : \tau \leq b, \tau' \geq \beta\}.$$

Both are finite: $(j, s) \in U$ iff $\beta - h_j \leq s - q_j \leq b$, so $|U| = \sum_j (b - \beta + 1 + h_j)$; $(i, t) \in R$ iff $\beta + h_i \leq t + p_i \leq b$, so $|R| = \sum_i (b - \beta + 1 - h_i)$. Hence

$$|U| - |R| = 2 \sum_i h_i = 2 \left(\sum_i p_i + \sum_i q_i \right) = 2V_{\max} = D.$$

Rows in R live on U . An unknown of a row in R has $\lambda \leq \tau \leq b$ and $\lambda' \geq \tau' \geq \beta$.

Rows in R are independent. Order them by top-level. At top-level τ the rows in R are those i with $h_i \leq \tau - \beta$; call this set S_τ . A nontrivial linear relation among the rows of R has a highest level τ with a nonzero coefficient; restricted to the unknowns at top-level τ , which rows of lower level

do not touch, it is a nontrivial relation among the rows of $L_\tau[S_\tau, \cdot]$, impossible since the sub-block $L_\tau[S_\tau, \sigma^*(S_\tau)]$ is nonsingular.

Restriction to U is a bijection from K onto the solutions of R . Let y be a vector on U satisfying every row in R . Extend it downwards: for $\tau' = \beta - 1, \beta - 2, \dots$ in turn, the m rows at bottom-level τ' have top-level $\tau' + h_i \leq \beta - 1 + h \leq b$, are not in R , and involve unknowns at bottom-level $\geq \tau'$; those at bottom-level $> \tau'$ are already defined (they lie in U , having $\lambda \leq b$ and $\lambda' \geq \beta$, or were defined at an earlier step), and the coefficients of the m unknowns at bottom-level exactly τ' form $L'_{\tau'}$, nonsingular. So the step is forced and possible. After all steps y is defined on every unknown with $\lambda \leq b$ (those with $\lambda' < \beta$ have $\lambda < \beta \leq b$) and satisfies every row with $\tau \leq b$: the rows with $\tau' \geq \beta$ are in R , and every row with $\tau' < \beta$ was used. Then extend upwards: for $\tau = b + 1, b + 2, \dots$ the rows at top-level τ involve unknowns at top-level $\leq \tau$, all defined except the m at level τ , whose coefficient matrix is L_τ , nonsingular. The result is an element of K restricting to y , and the same two recursions with zero right-hand sides show that an element of K vanishing on U vanishes everywhere. Therefore $\dim K = |U| - \text{rank } R = |U| - |R| = D$. $\square$

Proof of Theorem 91. Let $A \in \mathcal{S}(B^n)$. The covering $(x, t) \mapsto (x, t \bmod n)$ maps each edge of B^∞ onto an edge of B^n , or onto a vertex if a loop has $n \mid v$. Give each edge of B^∞ a nonzero weight so that the weights of the edges over one edge of B^n add up to its entry in A (if $r \geq 1$ classes of strip edges lie over it, use $1/r$ of the entry on each), and choose the strip diagonal so that, with the loop lifts that close up, the diagonal of A is reproduced. The result is a strip matrix $\tilde{A}$, and $x \mapsto \tilde{x}$, $\tilde{x}_{j,s} = x_{j, s \bmod n}$, is an injection of $\ker A$ into $\ker \tilde{A}$. By Theorem 94, $\text{null } A \leq D$. $\square$

Remark 95 (The monodromy, for every base). The proof gives more than the bound. With A as above, $\tilde{A}$ has n -periodic weights, so the shift $(j, s) \mapsto (j, s + n)$ acts on K , a space of dimension exactly D ; call the action T . Periodic kernel sequences are the fixed vectors of T , so

$$\text{null } A = \dim \ker(T - I) \leq D, \quad \text{with equality iff } T = I,$$

which is Theorem 44 with $2k + 2$ replaced by D and the two-vertex base by any base satisfying the criterion. For equivariant A the eigenvalues of T are the n th powers of the roots of $\det M(\zeta)$, and this recovers Theorem 65.

Remark 96 (Why one grading is not enough). Grading from the top alone, the unknowns at levels below τ that some row at level $\geq \tau$ still needs form a *forward state* of size $\sum_j m_j$, $m_j = \max\{p_i + q_j - v\}$ over the entries of column j , and the level- τ rows determine the level- τ unknowns from it, so the kernel is determined by its forward state and $\text{null } A \leq \sum_j m_j$ for every optimal potential. One computes $\sum_j m_j \geq D$ always, with equality iff the potential p is also, after a sign change, an optimal column potential; for the two-loop base this holds and gives the $2k + 2$ of Theorem 44, but on the three-vertex base with loops of voltages 2, 1, 3 and edges 0–2, 0–1, 1–2 of voltages 0, 3, 2 the minimum over optimal potentials is 13 while $D = 12$, and the kernel does have dimension 12: the forward state carries one direction that never extends backwards. Theorem 98 is this phenomenon on two vertices: its N is the size of a one-sided window, and $N > D$ exactly when the window over-counts. Since the forward state is a zero forcing set of B^n — a bipartite graph with a unique perfect matching has a vertex of degree one, so each level is forced one unknown at a time — one also has

$$M(B^n) \leq D \leq Z(B^n) \leq \min_{(p,q)} \sum_j m_j,$$

and the two outer bounds can differ.

Remark 97 (What is checked by machine). `verification/leading_coefficient.py` checks, on every base of this paper and on random bases with up to five vertices: the equivalence of Lemma 93 against the symbolic $c_{\max}$; the count $|U| - |R| = D$ together with $\text{rank } R = |R|$ for random non-equivariant integer weights, the rank computed exactly over $\mathbb{F}_p$; and $\dim K = D$ independently, as the number of nonzero eigenvalues of the forward-state transfer matrix over one period, which exceeds D in size on 18 of 30 random bases satisfying the criterion and has exactly $\sum_j m_j - D$ zero eigenvalues on each.

Theorems 80 and 81 are the two extreme shapes of a two-vertex base: loops with one connecting edge, and connecting edges with no loops. Between them lies the general case, where the two eliminations compete for the same variables, and there the answer is not uniform. It can be settled exactly, by counting.

Theorem 98 (Exactly when the reduction reaches the ceiling). *Let B carry loops of voltages $p, q \geq 1$ at u and v and connecting edges of voltages $0 = t_1 < \dots < t_m$, and let N be the order of the kernel recurrence. Then*

$$N = p + q + \max(p, t_m) + \max(q, t_m),$$

while the degree span is $D = \max(2(p + q), 2t_m)$. Consequently

$$N = D \iff t_m \leq \min(p, q),$$

and when the loops are absent $N = 2t_m = D$ always.

Proof. In a window of W consecutive fibres there are $2W$ unknowns. A row is usable only when every index it touches lies inside the window: the row at u_i needs $i \pm p$ and $i + t_r$ inside, leaving $W - p - \max(p, t_m)$ usable rows, and the row at v_j needs $j \pm q$ and $j - t_r$ inside, leaving $W - q - \max(q, t_m)$. The usable rows are generically independent, so the solution space has dimension $2W$ minus their number, which is the stated N , independent of W . The equivalence is arithmetic: if $t_m \leq \min(p, q)$ both maxima are the loop voltages and $N = 2(p + q) = D$; if $t_m > \min(p, q)$ then N exceeds both $2(p + q)$ and $2t_m$. $\square$

Verified against the measured order on sixteen bases with no discrepancy, and the equivalence on fourteen; see `verification/band_order.py`.

Remark 99 (What this does and does not decide). When $t_m \leq \min(p, q)$ the reduction proves null $A \leq D$ for every matrix carrying the pattern, and Theorems 80 and 81 are the cases $t_m = 0$ and “loops absent”. When $t_m > \min(p, q)$ the reduction still applies but yields only null $A \leq N$ with $N > D$. This does *not* exhibit a cover whose nullity exceeds the degree span: it shows the technique stops being tight there, not that the ceiling fails. Theorem 91 settles the question: $c_{\max}$ is $w_p w_q$ when $p + q > t_m$ and $-w_{t_1} w_{t_m}$ when $t_m > p + q$, a monomial either way, so null $A \leq D$ holds throughout, with the single exception of the tie $t_m = p + q$, where $c_{\max} = w_p w_q - w_{t_1} w_{t_m}$ can vanish and the question remains open. The same theorem covers bases with any number of vertices.

Theorem 100 (Certified values off every congruence class). *For every n with $17 \leq n \leq 33$,*

$$Z(P(n, 3)) = M(P(n, 3)) = 8.$$

In particular this holds at the primes $n = 17, 19, 23, 29, 31$, and the range is consecutive: seventeen values with no exceptions.

Proof. $Z(P(17, 3)) \leq 8$ is the rotation bootstrap. For the lower bound, take $n = 17$, $k = 3$, $r = 8$, so $N = 34$ and the system above has $85 + 208 = 293$ unknowns and 272 equations, of which $208 + 36 = 244$ are the structural ones. At a numerically located point the Jacobian of the 244 structural equations has rank exactly 244, so fixing the remaining 49 unknowns at explicit rationals with denominator 10^4 leaves a square regular system. Newton in 60-digit arithmetic drives its residual to 3.2×10^{-61} ; the 28 dependent equations, which are not solved for, fall to 4.1×10^{-61} , confirming the structural reduction. With Y an approximate inverse of the square Jacobian,

$$\|Y\|_\infty = 1.54 \times 10^3, \quad \|I - YJ\|_\infty = 2.18 \times 10^{-12}, \quad \|Yf\|_\infty = 2.85 \times 10^{-60},$$

and with $L = 12$ bounding the Lipschitz constant of J in the ∞ -norm, the Krawczyk test on the box of radius $\delta = 10^{-12}$ gives $\alpha = \|I - YJ\|_\infty + \|Y\|_\infty L\delta = 1.85 \times 10^{-8} < 1$ and $\|Yf\|_\infty + \alpha\delta \leq \delta$. So a true zero (w^*, X^*) exists in that box. Then $A(w^*)K(X^*) = 0$ with rank $K = 8$, so null $A(w^*) \geq 8$, and ≤ 8 by Corollary 45. For that last step $A(w^*)$ must be a matrix of $P(17, 3)$, i.e. every edge weight nonzero, and this has to be known for w^* rather than for the box centre w_0 , which is all a direct computation gives. It follows from the box: $\|w^* - w_0\|_\infty \leq \delta$, so every edge weight of $A(w^*)$ is at least $\min_e |w_0(e)| - \delta$, and here $\min_e |w_0(e)|$ exceeds 0.05 times the matrix scale while $\delta = 10^{-12}$, so the bound survives with room to spare. The ninth singular value exceeds 0.126 times the scale, so the box is far from any degenerate stratum and the pattern is exactly that of $P(17, 3)$.

The remaining n are identical in structure and each passes the same test; for $n = 19$, to give one more set of numbers, there are 335 unknowns and 304 equations of which 276 are structural and 28 dependent, 59 unknowns are fixed at rationals, and $\|Y\|_\infty = 4.37 \times 10^2$, $\|I - YJ\|_\infty = 1.06 \times 10^{-12}$, $\|Yf\|_\infty = 1.20 \times 10^{-60}$, with Krawczyk passing at $\delta = 10^{-13}$ and $\alpha = 5.25 \times 10^{-10}$. See `verification/krawczyk.py` and `verification/certify_pipeline.py`. $\square$

Together with Theorem 58 this gives $Z(P(n, 3)) = 8$ at $n = 10$, every $17 \leq n \leq 33$, and every multiple of 10 — a set that no single divisibility condition describes. The search fails only at $n = 13$ and $n = 15$, where the solution variety has dimension 3 and 9 by (4), against 15 at $n = 17$ where it first succeeds. By Lemma 102 the slice searched is complete, so those are failures of the search and not evidence that no matrix of nullity 8 exists there. That distinction is not academic: $n = 18$ resisted the restart budget we used for its neighbours and certified once the budget was raised, so a failure here carries no information beyond the effort spent.

Remark 101 (Three independent numerical routes, one certificate). The certificate at $n = 17$ was located three separate ways — by minimising $\|T - I\|$ over all $5n$ weights, by an alternating solve of $AK = 0$ with the spokes normalised, and by the fully gauge-fixed search of (4) — and all three lead to a Krawczyk-certified matrix of nullity 8, with the gauge-fixed route giving the best conditioning ($\|Y\|_\infty = 2.1 \times 10^2$, ninth singular value 0.167 of the matrix scale, minimum edge weight 0.146). Since stage 1 is only a search and stages 2–3 are a proof, the numerical route taken does not enter the result; that three unrelated ones agree is a check on the search, not on the certificate.

Since 17 and 19 are prime, Theorem 72(i) shows no period-one certificate — in particular nothing in Sections 17 or 19 — can give these values. Indeed no divisibility class containing either of them is available at all: a class $\{n : L \mid n\}$ with $L \geq 3$ contains a prime only when L is that prime, and Γ_{17} offers only Ψ_1 and Ψ_{17} , the latter of degree $8 \neq k + 1 = 4$. These are the first values in this paper that do not sit on a divisibility class.

The reachable set stabilises

That $n = 17$ works and $n = 13$ does not is explained by a measurement rather than by arithmetic. The dimension of the reachable set of monodromies, that is the rank of $\partial T / \partial w$, grows with n and

then *saturates*:

$k = 2, n$	5	6	7	8	9	10	11	12	≥ 13
rank	19	23	26	28	30	31	31	31	31
$k = 3, n$	7	9	11	13	14	15	16	17	≥ 18
rank	27	34	40	46	49	51	51	51	51

Past $n = 11$ for $k = 2$ and $n = 15$ for $k = 3$, increasing n adds no new reachable monodromies. So whether I is reachable stops depending on n in any essential way once n passes roughly $4k + 3$, and the observed pattern of successes and failures follows it: for $k = 3$ the ceiling is attained at $n = 17, 20$ and not at $n = 13 < 15$. Independently, at $n = 17, k = 3$ the solution variety of $f = 0$ has dimension $293 - 244 = 49 = 5n - \binom{2k+3}{2}$, exactly the codimension count for symmetric matrices on the pattern with nullity $2k + 2$.

There is also a hard *necessary* condition, from spending the gauge, and it pays to be careful about what the gauge can and cannot do.

Lemma 102 (A complete slice). *Every symmetric matrix on the pattern of $P(n, k)$ with all weights nonzero is carried, by a conjugation preserving symmetry, the pattern and the nullity, to one with every spoke equal to $+1$ and every outer weight equal to ± 1 — for even n , with a single outer weight left arbitrary. The signs may be taken to be $+1$ except possibly one, so for odd n there are two such slices and for even n one.*

Proof. Two conjugations are available. Positive diagonal $A \mapsto DAD$ scales $b_i \mapsto d_i d_{i+1} b_i$, $c_i \mapsto d_i d_{n+i} c_i$, and sign diagonal $A \mapsto SAS$ with $S = \text{diag}(\pm 1)$ sends $b_i \mapsto s_i s_{i+1} b_i$, $c_i \mapsto s_i s_{n+i} c_i$. Since $d_i d_{n+i} > 0$, each spoke's *sign* is invariant under the first and must be fixed by the second: taking $s_{n+i} = s_i \text{sign}(c_i)$ makes every spoke positive, and choosing the inner d_{n+i} makes every spoke 1. The outer d_i then satisfy $u_i + u_{i+1} = -\log |b_i|$ with $u_i = \log d_i$, a cyclic system whose matrix is invertible for odd n and has corank 1 for even n ; in the latter case drop one equation, leaving one outer weight unnormalised. Finally $s_i s_{i+1} \text{sign}(b_i) = 1$ is solvable exactly when $\prod_i \text{sign}(b_i) = 1$, giving the two sign classes for odd n , while for even n the unnormalised weight absorbs the invariant. $\square$

We verified Lemma 102 numerically before relying on it: random matrices on the pattern are carried into the slice with the spokes exactly $+1$ and $|b_i| = 1$ to 10^{-15} , and the nullity unchanged, for both parities of n (`verification/gauge_fixed.py`). The point of the lemma is that a search restricted to the slice is not restricted at all, so a failure inside it is a failure of the search and not evidence that no matrix exists.

What remains after the fixing is $2n$ diagonal entries and n inner weights, with no residual gauge, and the solution variety of $f = 0$ has dimension

$$[3n + (2n - r)r] - [(2n - r)r + \binom{r+1}{2}] = 3n - (k + 1)(2k + 3),$$

which is nonnegative only when

$$n \geq \frac{(k + 1)(2k + 3)}{3} = 7, 12, 19, 26, 35, 46, 57 \quad (k = 2, \dots, 8). \quad (4)$$

Conjecture 103. *For every $k \geq 2$ there is $N(k)$ with $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all $n \geq N(k)$.*

For $k = 2$ this is Theorem 53, with $N(2) = 11$, proved. We do not conjecture that $N(k)$ is linear: (4) forces $N(k) \geq \lceil (k + 1)(2k + 3)/3 \rceil$, which is quadratic, and at $k = 6$ that bound is 35 — larger than

the value $4k + 3 = 27$ that the saturation data alone might suggest. The two quantities measure different things, and the dimension count is the binding one. Our data is consistent with $N(k)$ exceeding (4) by a small additive amount: $N(2) = 11$ against a bound of 7, and $14 \leq N(3) \leq 17$ against a bound of 12, since the ceiling is attained at $n = 17$ and 20 and not at $n = 13$.

The content of the conjecture for $k \geq 3$ is a reachability statement about products of transfer matrices, with no arithmetic in it — which is why we expect it to behave uniformly in n , unlike everything in Sections 17 and 19.

Remark 104 (Two corrections to our own evidence). Two caveats. First, $n = 10$, $k = 2$ resists the search even though its reachable dimension is already saturated, and $Z(P(10, 2)) = 6$; we took that resistance as reason to expect a strict gap $M(P(10, 2)) < Z(P(10, 2))$. There is none: $M(P(10, 2)) = 6$, by the integer certificate of Remark 55. The resistance was a fact about Γ_{10} , not about the bound — a failed search measuring effort spent, which is the error this paper warns about and walked into twice. Second, the eigenvalue-minimising search we reported earlier is unreliable and its negative results should not be read as evidence: it gave maximum nullity 7 at $P(20, 3)$, whereas the ceiling 8 is attained there both by Theorem 58 and, in the *symmetric* class, by the method of this section. It was collapsing onto the degenerate strata — all spokes zero, or all edges zero — that the formulations here exclude by construction.

A route to all large n

Conjecture 103 asks for one construction covering every large n . Certifying n one at a time cannot do that. But the condition $T = I$ is multiplicative along the cycle, and that suggests assembling the weight sequence from pieces each of which contributes the identity. The obstacle is that T_i is built from the weights in a window of width about $2k + 2$ around position i , so pieces interact across their boundaries and the product does not factorise. The following removes the obstacle.

Definition 105. Fix a *docking pattern*: values of (b, c, e, a, d) at $k + 1$ consecutive positions. A *tile* of length $\ell \geq 2(k + 1)$ is a weight sequence of length ℓ whose first $k + 1$ and last $k + 1$ positions carry the docking pattern. Its *interior* is the remaining $\ell - 2(k + 1)$ positions.

Lemma 106 (Tile independence). *Let the weight sequence on $\mathbb{Z}_n$ be a concatenation of tiles sharing one docking pattern. Then for each tile the product of the transfer matrices over its positions depends only on that tile's interior weights, and the monodromy is the product of these tile products.*

Proof. By Theorem 44, T_i is determined by the coefficients of the i th reduced equation, which involve b at indices in $[i - k - 1, i + k]$, c and a at $i - k, i, i + k$, e at $i - k$ and i , and d at i . So T_i depends only on weights at indices in $[i - k - 1, i + k]$. If the tile occupies $[s, s + \ell)$ and $i \in [s, s + \ell)$, this window is contained in $[s - k - 1, s + \ell + k - 1]$. The part below s lies in the preceding tile's final $k + 1$ positions and the part at or above $s + \ell$ lies in the following tile's first $k + 1$ positions; both carry the docking pattern, as do $[s, s + k]$ and $[s + \ell - k - 1, s + \ell)$. Hence every weight T_i sees is either docking or in this tile's interior. The second statement is associativity of the product $T = T_{n-1} \cdots T_0$ regrouped along the tiles. $\square$

Theorem 107 (Tiling). *Fix k and $L \geq 2(k + 1)$. Suppose that for every ℓ with $L \leq \ell < 2L$ there are interior weights, all of the required entries nonzero, making the tile product of length ℓ equal to the identity. Then*

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2 \quad \text{for every } n \geq L,$$

with no congruence condition.

Proof. Every $n \geq L$ is a sum of integers from $[L, 2L)$: if $n < 2L$ take the single term n , and if $n \geq 2L$ take a term L and recurse on $n - L \geq L$. Concatenating the corresponding tiles gives a weight sequence on $\mathbb{Z}_n$ whose monodromy is, by Lemma 106, the product of the tile products, hence I . All required weights are nonzero, so the matrix carries the pattern of $P(n, k)$ and Theorem 44 applies; by Corollary 45, null $A = 2k + 2$. With $M \leq Z$ and the rotation bootstrap $Z \leq 2k + 2$, both are equal to $2k + 2$. $\square$

Lemma 106 was checked before use: perturbing one tile's interior leaves the other tiles' products unchanged to the last bit, 0 exactly, for $k = 2, 3, 4$ (`verification/tiles.py`). Two remarks on the shape of Theorem 107. First, the hypothesis over a full interval $[L, 2L)$ is what makes the conclusion gapless; using two tile lengths a, b with $\gcd(a, b) = 1$ instead would give only n of the form $ia + jb$, leaving every $n < (a - 1)(b - 1)$ uncovered, which for the smallest solvable lengths at $k = 3$ is already several hundred. Second, the hypothesis is a *finite* list of independent finite problems: a tile of length ℓ has $5(\ell - 2(k + 1))$ free weights and the requirement is $(2k + 2)^2$ equations, so it is solvable on parameter count once ℓ is a little above $2(k + 1) + (2k + 2)^2/5$. For $k = 3$ that is $\ell \geq 21$, and $L = 21$ gives every $n \geq 21$.

The tiles exist, and are certified

We carried the computation out for $k = 3$. The parameter count is sharp in practice: at $\ell = 19, 20$ the systems are underdetermined (55 and 60 unknowns against 64 equations) and the residual stalls near 4×10^{-1} ; and from $\ell = 21$, the first length with more unknowns than equations, identity tiles are found. A Newton solve produced one for *every* length in the required range, $21 \leq \ell \leq 41$, which is exactly $[L, 2L)$ for $L = 21$; we in fact have tiles for every $21 \leq \ell \leq 43$.

Each of those 23 tiles is then *certified*, which needs one observation to avoid interval arithmetic on a rational map. The condition $T_{\text{tile}} = I$ is a rational equation in the tile weights, so a Krawczyk bound on it directly would require interval matrix products. It is unnecessary: by Corollary 45 the monodromy of the cyclic matrix built from a *single* tile on $P(\ell, k)$ is the tile product, so

$$T_{\text{tile}} = I \iff \text{null } A_{\text{one tile}} = 2k + 2,$$

and the right-hand side is certified by the bilinear machinery of Theorem 100, whose Jacobian is affine. The only extra requirement is that the tiles still compose, which needs them all to carry the same docking pattern; this is arranged by pinning the $5(2k + 2)$ docking weights at their exact integer values ($b = c = e = 1$, $a = d = 0$) rather than letting the solver move them, so they belong to the fixed set and the certified solution carries them exactly.

All 23 certifications at $k = 3$ pass. For $\ell = 22$: 398 unknowns, 352 equations of which 324 are structural and 28 dependent, 40 docking pins and 74 unknowns fixed at rationals in all; high-precision Newton drives the structural residual to 3.1×10^{-61} and the dependent equations to 2.7×10^{-61} ; $\|Y\|_\infty = 5.4 \times 10^2$ and the Krawczyk test passes with $\alpha = 6.7 \times 10^{-11}$. Across all lengths $\alpha \leq 2.2 \times 10^{-10}$, every certified matrix has nullity exactly 8, every docking weight agrees with its prescribed integer to 10^{-14} , and the minimum edge weight is 0.15 of the matrix scale. See `verification/certify_tiles.py`.

Remark 108 (The tile equations are degenerate, and by how much). The system $T_{\text{tile}} = I$ has $(2k + 2)^2$ equations but they are not independent. Measuring the rank of the Jacobian of $w \mapsto T_{\text{tile}}$ at a solution gives exactly

$$\text{rank} = \binom{2k + 3}{2}, \quad \text{deficiency} = \binom{2k + 2}{2},$$

namely 36 and 28 at $k = 3$ and 55 and 45 at $k = 4$, at every tile length tested. Two consequences. First, it explains the behaviour of the solve: the residual has $\binom{2k+2}{2}$ components that vanish automatically near a solution, so a least-squares step is working on a degenerate system, and that is consistent with the erratic convergence reported in the remark below. Second, the count that a tile must satisfy is $5(\ell - 2(k+1)) \geq \binom{2k+3}{2}$ rather than $\geq (2k+2)^2$, which for $k = 3$ would permit $\ell \geq 16$ rather than $\ell \geq 22$ and so a smaller L . Whether short tiles exist is a separate question from whether the count allows them, and the answer is in between. With a larger restart budget $\ell = 21$ *does* yield an identity tile, certified like the rest, so $L = 21$ rather than 22 at $k = 3$ and the required range is $[21, 42]$; this does not change Theorem 109, which already reaches $n \geq 17$ through Theorem 100. Below that the searches fail and the failures look different in kind: $\ell = 19$ and 20 stall at 1.4×10^{-1} and 3.7×10^{-1} , and $\ell = 16, 17, 18$ at 8.3×10^{-1} , 7.8×10^{-1} and 8.2×10^{-1} , whereas every length that we eventually solved first appeared as a residual near 10^{-2} or below. So the count is not the binding constraint and I appears not to be a product of fewer than about twenty transfer matrices. We record that as a computation and not a proof: we have no lower bound on the number of factors needed to write I , and $\ell = 21$ is a reminder that a stalled search at these sizes can mean only that the budget was too small.

We note that $\binom{2k+3}{2} = \dim Sp(2k+2)$, and that the space of skew forms B with $T^\top BT = B$ has dimension $k+1$. It is tempting to conclude that T is symplectic with respect to a weight-dependent form. We do not claim it: our attempts to exhibit a nondegenerate invariant form were numerically unstable, giving residuals of 10^{-12} at $k = 2, 4, 5$ but 10^{-1} at $k = 3$, so the coincidence of dimensions is suggestive and no more.

Theorem 109 (All large n , for $k = 3$ and $k = 4$).

$$Z(P(n, 3)) = M(P(n, 3)) = 8 \quad \text{for every } n \geq 17,$$

$$Z(P(n, 4)) = M(P(n, 4)) = 10 \quad \text{for every } n \geq 29.$$

Proof. For $k = 3$, identity tiles exist for every ℓ with $21 \leq \ell < 42$, and for $k = 4$ for every ℓ with $29 \leq \ell < 58$; each is certified as above. (We in fact have certified tiles for every $21 \leq \ell \leq 43$ at $k = 3$ and every $29 \leq \ell \leq 61$ at $k = 4$, comfortably more than $[L, 2L)$ requires.) A certification exhibits a box containing a true real solution of $AK = 0$ with K of full rank $2k+2$, so the one-tile cyclic matrix has nullity at least $2k+2$ and hence exactly $2k+2$ by Corollary 45, so $T_{\text{tile}} = I$; and its docking weights are the prescribed integers exactly, since they are held fixed throughout, so tiles with different ℓ compose. Theorem 107 with $L = 21$ and $L = 29$ respectively gives $n \geq 21$ and $n \geq 29$, and Theorem 100 covers $17 \leq n \leq 20$ at $k = 3$. $\square$

All $23 + 33 = 56$ certifications pass. To give the numbers at both ends: at $k = 3$, $\ell = 22$ has 398 unknowns and 352 equations of which 324 are structural and $28 = \binom{8}{2}$ dependent, with 40 docking pins, high-precision residual 3.1×10^{-61} , and Krawczyk $\alpha = 6.7 \times 10^{-11}$; at $k = 4$, $\ell = 61$ has 1425 unknowns and 1220 equations of which 1175 are structural and $45 = \binom{10}{2}$ dependent, with 50 docking pins, residual 4.7×10^{-61} , and $\alpha = 3.7 \times 10^{-10}$. Across all 56, $\alpha \leq 1.0 \times 10^{-9}$, every certified matrix has nullity exactly $2k+2$, every docking weight matches its prescribed integer to 10^{-14} , and the minimum edge weight is 0.15 of the matrix scale. See `verification/certify_tiles.py`.

These are the statements Conjecture 103 asks for, and they are proofs rather than computations reported as such: the chain is Theorem 107, Lemma 106 and Corollary 45, all proved, together with 56 certifications whose Krawczyk hypotheses are verified in exact rational arithmetic (Section 20). No step rests on a floating-point nullity. With Theorem 53 for $k = 2$, the three smallest k beyond the trivial case are now settled for all large n , with

$$L(2) = 9, \quad L(3) = 17, \quad L(4) = 29,$$

against the tile lower bound $2(k+1) + (2k+2)^2/5$, which is 13.2, 20.8 and 30.

Theorem 110 (Complete determination for $k = 2$, $k = 3$ and $k = 4$). *For every n with $2k < n$, the value $Z(P(n, k))$ is now known when $k = 2$, $k = 3$ or $k = 4$:*

n	5	6	7	8	9	10	11	≥ 12	n	7	8	9	10	11	12	≥ 13
$Z(P(n, 2))$	5	4	6	5	6	6	6	6	$Z(P(n, 3))$	6	6	6	8	7	7	8

n	9	10	11	12	13	14	15	16	17	≥ 18
$Z(P(n, 4))$	6	6	7	6	8	8	9	8	9	10

No value is left open in any of the three rows.

Proof. The tails are Theorem 53 for $k = 2$, Theorem 111 for $k = 3$, and Theorem 109 for $k = 4$, which gives $Z(P(n, 4)) = 10$ for every $n \geq 29$; the finitely many remaining values, including $18 \leq n \leq 28$ at $k = 4$, are exhaustive searches over vertex subsets. $\square$

For $k = 2$ this completes what [6] began, which gave $Z(P(n, 2)) = 6$ only for $n \geq 10$. For $k = 3$ it is the first determination of any kind: the value claimed in [6] for that case is false, and [7], which found the error, leaves the threshold open. Note that neither row is monotone and neither is eventually constant from the first time the bound is attained — $k = 3$ reaches 8 at $n = 10$, drops to 7 at $n = 11, 12$, and only then settles. Any argument that assumed monotonicity would have got this wrong, and one did.

The same exhaustive computation at $k = 4$ gives

n	9	10	11	12	13	14	15	16	17	18
$Z(P(n, 4))$	6	6	7	6	8	8	9	8	9	10

n	19	20	21	22	23	24	25	26	27	28
$Z(P(n, 4))$	10	10	10	10	10	10	10	10	10	10

which is more erratic still: the value falls at $n = 12$ and again at $n = 16$, and the bound 10 is first attained at $n = 18$. These reach $n = 28$ because the rotation $\rho(i) = i + 1$ lets the search assume $u_0 \in S$, or $v_0 \in S$ if S is entirely inner, which is exhaustive up to symmetry and so still exact; at $n = 28$, ruling out every 9-set takes about 1.2×10^9 candidates. Together with Theorem 109 for $n \geq 29$ this leaves *no* value of $Z(P(n, 4))$ undetermined, which is why $k = 4$ appears in Theorem 110.

The matrix statement is weaker than the forcing statement at exactly one value. $M(P(n, 4)) = 10$ is proved for every $n \geq 29$ by Theorem 109, for $n = 25, 26, 27, 28$ by individual certificates, and for every n divisible by 60, 70 or 90 by Theorem 69. Only at $n = 24$ do we know $Z = 10$ without a certificate, so $M(P(24, 4))$ is not determined; nothing suggests $M < Z$ there.

That one is not a search failure, and it is worth saying why, because the same mechanism governs the two exceptions at $k = 2$. In a period-one matrix the block M_j is singular only if $(d_{\text{out}} + ax_j)(d_{\text{in}} + by_j) = c^2$ with $x_j = 2 \cos(2\pi j/n)$ and $y_j = 2 \cos(2\pi jk/n)$; since $c \neq 0$ neither factor vanishes, so two singular blocks with $y_j = y_{j'}$ force $x_j = x_{j'}$ and hence $j' \equiv \pm j$. The singular blocks therefore have pairwise distinct y values, and

$$\text{null } A \leq \sum_{\text{distinct } y \text{ values}} (\text{largest } \pm \text{class in that value}),$$

a bound on every period-one matrix at once. It gives 8 at $(n, k) = (24, 4)$ against a ceiling of 10, so no rotation-invariant matrix attains the ceiling at $n = 24$ and no restart budget would have

found one; it gives 5 at $(10, 2)$, which is the general form of the computation in Remark 55; and at $k = 2$ the values of n it blocks are exactly 8 and 10, the two exceptions, derived rather than observed (`verification/period1_ceiling.py`). The bound constrains period-one matrices only — the period- d and tiling constructions are not affected, and it is an upper bound, so a value below it need not be attainable.

Theorem 111 ($k = 3$ completely, and the stabilization threshold).

$$Z(P(n, 3)) = 8 \quad \text{for every } n \geq 13,$$

and 13 is optimal: $Z(P(11, 3)) = Z(P(12, 3)) = 7$. Moreover $Z(P(n, 3)) = M(P(n, 3)) = 8$ for every $n \geq 17$ and for every n divisible by 10.

Proof. For $n \geq 17$ this is Theorem 109. For $n = 13, 14, 15, 16$ exhaustive search over vertex subsets shows no forcing set of size 7 exists, so $Z = 8$ by the bound $Z \leq 2k + 2$. The values $Z(P(11, 3)) = Z(P(12, 3)) = 7$ are likewise exhaustive. $\square$

This proves Conjecture 5 of [7], stated there as

“Conjecture 5. $Z(P(n, 3)) = 8$ for every $n \geq 13$,”

and described in that note’s abstract as needing “a lower-bound proof valid for all large n ”. Theorem 111 is that statement, and the missing ingredient is exactly what Section 20 supplies: $M \leq Z$ turns a matrix into a lower bound, and the tiling construction produces such matrices for all large n at once rather than one n at a time. The upper bound $Z \leq 8$ for $n \geq 9$ is already in [7], as are the values for $7 \leq n \leq 20$ by exhaustive search; what is new here is the lower bound for every $n \geq 13$, and with it the stabilization threshold, which the same note records as open alongside a rigorous lower bound for $k \geq 4$. It is worth being exact about what that note does and does not claim, because it is unusually careful on the point. Of Conjecture 5 it writes that “the content is the lower bound $Z(P(n, 3)) \geq 8$ for all $n \geq 13$, which we verified only for $n \leq 20$ ”, that proving it “means excluding 7-vertex forcing sets uniformly in n ”, and that “we will not assert it from finite computation, which would reproduce the gap this note corrects”. So the missing ingredient is named precisely, and extrapolation from finite data is explicitly declined. Uniformity in n is exactly what tiling provides: one finite family of certified tiles yields a matrix for every $n \geq L$ at once. The sequence is genuinely non-monotone below it — $Z(P(n, 3))$ runs 6, 6, 6, 8, 7, 7, 8, 8, ... for $n = 7, \dots, 14$ — which is why the threshold is not simply the first n attaining the bound, and why Conjecture 41 is stated in the form it is.

The certification is exact, not floating-point

A Krawczyk bound is a theorem about real numbers, and evaluating its three norms in double precision is not a proof of its hypotheses. Our first implementation did exactly that: $\|Y\|_\infty$, $\|I - YJ\|_\infty$ and $\|Yf\|_\infty$ were computed in float64 with no directed rounding, and J and f were evaluated at the double-rounded point rather than at the high-precision one. The margins are enormous — α near 10^{-10} against a threshold of 1 — so the conclusion was surely true, but “certified” should not rest on a margin.

It need not. Three observations make the whole test exact at essentially no cost.

1. *The box centre is exactly rational.* A stored certificate is a float64 vector, and every float64 is a dyadic rational, so z_0 is known exactly. We snap it to the grid $2^{-60}\mathbb{Z}$, which moves z_0 by at most 2^{-61} and is harmless because z_0 is only the centre of the box and the test re-measures $f(z_0)$ after the move.

2. *f and J have integer coefficients.* Since $f(w, X) = A(w)K(X)$ is bilinear, J is affine and every entry of $J(z_0)$ is 0, 1, or a single coordinate of z_0 . So $J(z_0)$ and $f(z_0)$ are exactly rational with denominators 2^{60} and 2^{120} , and are built here as integer matrices.
3. *Y is arbitrary.* The Krawczyk test holds for *any* matrix Y ; Y has only to make $\|I - YJ\|_\infty$ small, and need not be $J(z_0)^{-1}$. So we take the float64 inverse and read it as an exact dyadic rational, losing nothing whatsoever.

Hence $\|Y\|_\infty$, $\|I - YJ(z_0)\|_\infty$ and $\|Yf(z_0)\|_\infty$ are exact rationals, and the two comparisons $\alpha < 1$ and $\|Yf\|_\infty + \alpha\delta \leq \delta$ are exact integer comparisons. The one obstacle is speed, since α needs every entry of the $\text{rk} \times \text{rk}$ product $YJ(z_0)$ with rk up to 1175. We multiply by *balanced limb decomposition*: each integer is written in base 2^{20} with digits in $[-2^{19}, 2^{19})$, each pair of digit matrices is multiplied in float64 — those products stay below 2^{53} and are therefore themselves exact integers — and the limb products are recombined with exact integer arithmetic. This gives the exact integer product at the speed of BLAS; the routine refuses to run if the limb bound is not met, so it cannot silently return an inexact answer.

All 56 tile certifications and all 17 per- n certifications of Theorem 100 pass the test evaluated this way, with $\alpha \leq 1.0 \times 10^{-9}$. To give the two that the thresholds rest on: $\ell = 21$ at $k = 3$ has $\text{rk} = 308$, $\|Y\|_\infty = 1.5743 \times 10^3$, $\alpha_0 = 2.5071 \times 10^{-11}$ and $\alpha = 2.139912 \times 10^{-10}$; $\ell = 29$ at $k = 4$ has $\text{rk} = 535$, $\|Y\|_\infty = 1.4416 \times 10^3$, $\alpha_0 = 9.2248 \times 10^{-12}$ and $\alpha = 1.822215 \times 10^{-10}$. These agree with the float64 values to three figures, which is the expected outcome and the reason the original conclusion was not in doubt; the point is that they are now *bounds* rather than estimates. See `verification/exact_krawczyk.py` and `verification/exact_certify.py`.

Two smaller points belong here. First, the Lipschitz constant. Row (i, t) of f involves the ≤ 4 weights whose cells meet row i — one diagonal and three edges, since $P(n, k)$ is cubic — each contributing one entry of K , and the ≤ 4 columns $X[m, \cdot]$ for which $A[i, \ell] \neq 0$, each contributing one weight. Both groups are Lipschitz 1 in the corresponding coordinate, so the row sum of $|J(z_1) - J(z_2)|$ is at most $8\|z_1 - z_2\|_\infty$ and $L = 8$ is valid; it is also attained, in the all-ones direction. We nevertheless run every test with $L = 12$, which is conservative: a larger L makes α larger and the test strictly harder to pass, so using 12 can only understate the margin.

Second, the nonvanishing of the edge weights is deduced over the whole box exactly as in the proof of Theorem 100, and for the tiles the docking weights are not perturbed at all: they sit among the fixed coordinates, so they are exact integers at z_0 and the box moves only the free coordinates. That is what lets tiles of different lengths compose — the certified w^* carries the docking pattern exactly, not approximately.

What a general existence proof would need, and what it reduces to

Theorems 107 and 109 together have an unsatisfying shape: the implication is proved for all k , but its hypothesis is verified by *computing* the tiles, separately for each k . A proof that identity tiles exist for every k and every sufficiently long ℓ would give $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all k and all $n \geq L(k)$ in one statement, and would subsume Sections 17 and 19 entirely. This subsection identifies what that proof is a proof *about*.

Fix k and a docking pattern, and for $\ell \geq 2(k + 1)$ let $\mathcal{S}_\ell \subseteq \text{GL}_{2k+2}(\mathbb{R})$ be the set of tile products of length ℓ realisable with all weights nonzero. Existence of an identity tile of length ℓ is exactly the statement $I \in \mathcal{S}_\ell$. Three facts constrain $\mathcal{S}_\ell$.

Observation 112 (Symplectic structure). For a fixed docking pattern there is a skew-symmetric B , unique up to scale, with

$$P^\top B P = B \quad \text{for every } P \in \mathcal{S}_\ell \text{ and every } \ell,$$

and B is nondegenerate. Hence $\mathcal{S}_\ell \subseteq Sp(B) \cong Sp(2k+2, \mathbb{R})$, a connected simple Lie group of dimension $\binom{2k+3}{2}$.

The mechanism is not special to this problem: the recurrence of Theorem 44 is self-adjoint, because A is symmetric, and the transfer matrices of a self-adjoint difference system carry a Wronskian-type skew form from one step to the next. A tile whose two ends carry the same docking therefore returns that form to itself. Concretely, solving $T_i^\top B_{i+1} T_i = B_i$ along a tile has a $(k+1)$ -dimensional solution space for any one tile, and imposing the same docking across many different interiors cuts it to the single B above.

We label this an observation rather than a theorem: we verified it numerically for $k = 2, \dots, 6$ at two tile lengths each, finding in every case exactly one invariant skew form, nondegenerate, preserved to between 10^{-13} and 10^{-8} , and we have not written out the algebraic proof (`verification/symplectic_structure.py`).

Observation 113 (Full dimension). The rank of the differential of $w \mapsto P_\ell(w)$ equals $\binom{2k+3}{2}$ exactly — the dimension of $Sp(B)$ — for every (k, ℓ) we computed, $2 \leq k \leq 6$. So for ℓ past the threshold of Remark 108, $\mathcal{S}_\ell$ has nonempty interior in $Sp(B)$.

Lemma 114 (Concatenation). $\mathcal{S}_m \cdot \mathcal{S}_\ell \subseteq \mathcal{S}_{\ell+m}$.

Proof. Immediate from Lemma 106: the docking pattern makes the product over each tile independent of its neighbours, so concatenating a tile of length m after one of length ℓ realises the product as a tile of length $\ell + m$. $\square$

Together these say that $\mathcal{S} = \bigcup_\ell \mathcal{S}_\ell$ is a subsemigroup of $Sp(2k+2, \mathbb{R})$ with nonempty interior, and the question “do identity tiles exist?” becomes

is I in the interior-carrying subsemigroup generated by these transfer matrices?

That is a controllability question about a Lie semigroup, and the honest status is that it does not follow from the above alone: $Sp(2n, \mathbb{R})$ does admit proper subsemigroups with nonempty interior — compression semigroups of convex cones are the standard examples — so one must show the generators are not confined to such a wedge. What the observations do establish is that the group is identified, the dimension obstruction is absent, and the remaining question is of a recognised type rather than an unstructured search. We note that I is in fact reachable for every k where we have looked ($k = 3$ at all $21 \leq \ell \leq 43$, $k = 4$ at all $29 \leq \ell \leq 61$, $k = 5$ at 21 lengths in $[54, 75]$), which is evidence for the general statement and not a proof of it.

Existence of identity tiles without finding them

The controllability question above can be answered. The answer rests on two facts about a single, explicitly chosen tile — the tile whose interior *is* the dock — and the rest is an argument about compact subgroups. It yields $Z(P(n, k)) = M(P(n, k)) = 2k+2$ for all sufficiently large n at every k for which the two facts are verified, and they are verified exactly for $k \leq 9$; in particular at $k = 6, \dots, 9$, where no identity tile has ever been found. The price is that the threshold $L(k)$ is not effective.

Throughout, a *dock* is a constant weight pattern $(a_0, b_0, c_0, d_0, e_0)$ with $b_0 c_0 e_0 \neq 0$, normalised to $b_0 = 1$; the *dock symbol* is the rotation-invariant symbol of §17 with these weights,

$$F_{\text{dock}}(s) = (a_0 + s)(d_0 + e_0 t_k(s)) - c_0^2, \quad t_k(s) = 2T_k(s/2),$$

and S_0 is the common step matrix of the constant weight sequence, so that a tile whose interior equals the dock has product S_0^ℓ .

Lemma 115 (The invariant form, exactly). *Fix a dock. For the state $Z_0 = (x_{-k-1}, \dots, x_k)$ let $B = B(\text{dock})$ be the skew form*

$$Z^\top B Z' = \sum_{\substack{\text{edges } pq \\ p \leq -1 < q}} A_{pq} (z_p z'_q - z'_p z_q),$$

the flux of two kernel sequences across the cut between positions -1 and 0 , with inner coordinates eliminated by $y_m = -(x_{m-1} + a_0 x_m + x_{m+1})/c_0$. Then $P^\top B P = B$ for every tile product P , and

$$\det B = e_0^{2k}/c_0^{4k} \neq 0.$$

Hence every tile product lies in $Sp(B) \cong Sp(2k+2, \mathbb{R})$.

Proof. For two kernel sequences z, z' of the strip operator, the flux across the cut between j and $j+1$ changes, when the cut moves one position, by $\sum_p (z_p (Az')_p - z'_p (Az)_p)$ over the vertices that cross, which is 0. So the flux is the same across every cut. Across the cut $(-1, 0)$ it involves the outer edge $u_{-1}u_0$ and the k inner edges $v_m v_{m+k}$ with $-k \leq m \leq -1$, whose inner coordinates are linear in $x_{-k-1}, \dots, x_k$: this is the form B , and it depends only on the weights at positions $-k-1, \dots, k$. Transporting the state from the cut $(-1, 0)$ to the cut $(\ell-1, \ell)$ by the tile product preserves the flux, and at both cuts the weights in range are dock weights (the tile's own dock at one end, the neighbouring tile's at the other), so both forms are B : $P^\top B P = B$. The determinant is computed in exact arithmetic for the docks used below (`verification/tile_wronskian.py`); the closed form was checked on random rational docks for $k \leq 8$ and is stated as what it is. $\square$

Lemma 116 (Elliptic docks exist for every k). *Let $\varepsilon_k = 1$ for k even and $\varepsilon_k = (-1)^{(k-1)/2}$ for k odd. For $c_0 > 0$ small enough the dock $(0, 1, c_0, 0, \varepsilon_k)$ has $F_{\text{dock}}(s) = \varepsilon_k s t_k(s) - c_0^2$ with $k+1$ distinct real roots in $(-2, 2)$; hence S_0 is diagonalisable with all eigenvalues on the unit circle, and $\{S_0^m : m \geq 0\}$ has compact closure, a group.*

Proof. $g(s) = s t_k(s)$ has the k simple zeros $2 \cos \frac{(2j-1)\pi}{2k}$ of t_k in $(-2, 2)$ and the zero $s = 0$. For even k these are $k+1$ distinct simple zeros, $g(\pm 2) = \pm 4 \neq 0$, and simple zeros persist under the small perturbation $-c_0^2/\varepsilon_k$. For odd k , t_k is odd, so $s = 0$ is a double zero of g with $g(s) = t'_k(0)s^2 + O(s^4)$ and $t'_k(0) = k(-1)^{(k-1)/2}$; the equation $\varepsilon_k g(s) = c_0^2$ then has two real roots $\pm c_0/\sqrt{k} + O(c_0^3)$ near 0 because $\varepsilon_k t'_k(0) = k > 0$, and the other $k-1$ zeros are simple and persist. The eigenvalues of S_0 are the z with $F_{\text{dock}}(z + z^{-1}) = 0$; real roots $s \in (-2, 2)$ give $z = e^{\pm i\theta}$ with $s = 2 \cos \theta$, distinct for distinct s . A diagonalisable matrix with unimodular eigenvalues generates a relatively compact cyclic group, whose closure is a compact abelian group. $\square$

Theorem 117 (Identity tiles for every large length). *Fix $k \geq 2$ and a dock satisfying Lemma 116. Suppose that for some length ℓ_0 the differential of the tile map $w \mapsto P_{\ell_0}(w)$ has rank $(k+1)(2k+3) = \dim Sp(2k+2)$ at some point. Then there is L such that an identity tile of every length $\ell \geq L$ exists. Consequently $Z(P(n, k)) = M(P(n, k)) = 2k+2$ for all $n \geq L$.*

Proof. Write $G = Sp(B)$, a connected Lie group by Lemma 115, and fix a left-invariant metric. The set where the differential of P_{ℓ_0} has rank $< \dim G$ is a proper algebraic subset of the weight space (the rank is maximal on an open dense set), so there are full-rank points w^* arbitrarily close to the dock point w_0 (interior = dock), and at such points all edge weights are nonzero. At a full-rank point P_{ℓ_0} is a submersion onto G , hence open: S_{ℓ_0} contains a ball $B_\varepsilon(P^*)$ around $P^* = P_{\ell_0}(w^*)$. Since P^* is close to $S_0^{\ell_0}$ and $S_0^{\ell_0}$ is diagonalisable with distinct unimodular eigenvalues — an open condition in Sp , because unimodular eigenvalues of a symplectic matrix can leave the circle only by colliding — we may take P^* elliptic: the closure C of $\{P^{*m} : m \in \mathbb{Z}\}$ is compact.

Let κ bound the Lipschitz constant of conjugation by elements of C on a neighbourhood of the identity, and put $B' = B_{\varepsilon/\kappa}(I)$. For $v_1, \dots, v_M \in B'$ set $u_i = P^{*M-i}v_iP^{*-(M-i)} \in B_\varepsilon(I)$, so that $P^*u_i \in B_\varepsilon(P^*) \subseteq S_{\ell_0}$ and, telescoping,

$$(P^*u_1)(P^*u_2) \cdots (P^*u_M) = P^{*M}v_1v_2 \cdots v_M.$$

The sets B'^M increase with M and exhaust an open subgroup of G , which is G itself; the set $K = \{P^{*-M}S_0^{-r} : M \geq 0, 2k+2 \leq r < 2k+2+\ell_0\}$ is contained in $C \cdot \{S_0^{-r}\}$, a compact set, so $K \subseteq B'^{M_0}$ for some M_0 . Given $N \geq M_0\ell_0 + 2k+2$ write $N = M\ell_0 + r$ with $M \geq M_0$ and $2k+2 \leq r < 2k+2+\ell_0$, choose $v_i \in B'$ with $v_1 \cdots v_M = P^{*-M}S_0^{-r}$, and concatenate the M tiles P^*u_i with a pure-dock tile of length r (product S_0^r): by Lemma 114 the result is a tile of length N with product $P^{*M}v_1 \cdots v_M S_0^r = I$. Theorem 107 with Theorem 44 gives the nullity $2k+2$ and hence $Z = M = 2k+2$ for every $n \geq L := M_0\ell_0 + 2k+2$. $\square$

Remark 118 (Three directions per position). The hypothesis is checkable, and the length at which it holds is predictable. Rescaling the coordinate at u_i and at v_i by nonzero constants is a diagonal similarity $A \mapsto DAD$, which preserves the pattern, the nullity and the monodromy up to conjugation, and moves two of the five weights at position i . So only three weights per position act on the tile product, and the generic rank of the differential is

$$\text{rank } dP_\ell = 3(\ell - 2k - 2) + (k + 1) = 3\ell - 5k - 5$$

until it saturates at $\dim Sp = (k+1)(2k+3)$, a law we observe exactly at every (k, ℓ) computed. The first submersive length is therefore $\ell_F(k) = \lceil (k+1)(2k+8)/3 \rceil$: 19, 27, 36, 47, ... for $k = 3, 4, 5, 6, \dots$, and the identity tiles of Theorem 109 first appeared at 21 and 29 — two above ℓ_F — which also explains why the count $5(\ell - 2k - 2) \geq (2k+2)^2$ used earlier overestimated the parameters available. At the dock point itself the rank is $\dim Sp - (k-2)$: a constant perturbation of a constant background moves the $k+1$ eigenphases of S_0 only through the three parameters of the symbol, the obstruction of §17 in infinitesimal form.

Proposition 119 (The hypothesis, verified exactly). *For $2 \leq k \leq 9$, with the integer docks of Table 6: the dock symbol has $k+1$ distinct real roots in $(-2, 2)$ (Sturm's theorem over $\mathbb{Q}$); $\det B(\text{dock}) \neq 0$ (exact rational arithmetic); and at the integer point w^* of the table the differential of P_{ℓ_0} , computed by exact forward-mode differentiation over $\mathbb{F}_p$, $p = 33554393$, has rank $(k+1)(2k+3)$ — a lower bound for the rank over $\mathbb{Q}$. Hence Theorem 117 applies, and*

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2 \quad \text{for all sufficiently large } n, \quad 2 \leq k \leq 9.$$

Remark 120 (What is new, and what is not). For $k = 3, 4, 5$ the statement was already known with effective thresholds (Theorems 109, 126) by finding and certifying the tiles; for $k \geq 6$ no tile has been found, and Proposition 119 is the first statement of any kind about $Z(P(n, k))$ for all large n at those k . The bound $L(k)$ is ineffective because M_0 comes from compactness. The rank hypothesis is proved for every k in §20: the $k-2$ directions missing at the dock point are supplied at second order.

The rank hypothesis holds for every k

Theorem 117 is conditional on one rank. We now show the rank condition holds for every k , at every nonresonant elliptic dock, so that the conclusion is unconditional.

Fix an elliptic dock and write $z_1, \dots, z_{2k+2}$ for the eigenvalues of S_0 , distinct and unimodular, $s_j = z_j + z_j^{-1} \in (-2, 2)$ the roots of the dock symbol, and $\phi_j = (z_j^{-k-1}, \dots, z_j^k)^\top$, ψ_j the right and left eigenvectors, normalised by $\psi_j^\top \phi_j = 1$. Call the dock *nonresonant* if no ratio z_j/z_l ($j \neq l$) is a root of unity, the only relations $z_j z_b = z_a z_l$ among the eigenvalues are the trivial ones ($\{j, b\} = \{a, l\}$ or $(a, b) = (\bar{l}, \bar{j})$ with $z_{\bar{j}} = z_j^{-1}$, which identifies the two copies of each root space of $\mathfrak{sp}$), and no z_j equals a ratio z_a/z_b . For the explicit family of Lemma 116 this holds for all but countably many c_0 , since each excluded relation is a nontrivial algebraic condition on c_0 and the eigenvalue angles are nonconstant in c_0 .

Let w_0 be the dock point and $A = dP_\ell(w_0)$, right-translated to $\mathfrak{sp}(B)$. Since every step matrix at w_0 is S_0 , the direction of a weight perturbation at position t is

$$Y_t = \text{Ad}(S_0^{m_t})(X), \quad m_t = \ell - 1 - t,$$

with X independent of t for each weight type, and $X = \sum_i e_r \otimes v_i$ over the rows i in which that weight occurs, where e_r is the last basis vector and v_i the derivative of row i of the recurrence. In the eigenbasis, $\text{Ad}(S_0^m)$ multiplies the (j, l) entry by $(z_j/z_l)^m$; the diagonal entries form the Cartan subalgebra $\mathfrak{h}$, of dimension $k+1$ in $\mathfrak{sp}$ (entries come in opposite pairs), and the off-diagonal positions are the root spaces.

Lemma 121 (First order). *At w_0 , for ℓ large, the image of A is the sum of all root spaces together with a 3-dimensional subspace of $\mathfrak{h}$. In particular $\text{rank } A = \dim \text{Sp}(2k+2) - (k-2)$.*

Proof. Root spaces. Take the weight a at position t . It enters the rows $i = t - k, t, t + k$, with v_i supported at the single offset $k, 0, -k$ and values $-e_0/c_0, -d_0/c_0, -e_0/c_0$ (from $y_t = -(x_{t-1} + a_t x_t + x_{t+1})/c_t$). Since $\psi_j^\top e_r \neq 0$ — if the last entry of a left eigenvector of the companion matrix S_0 vanished, the shift structure would force all its entries to vanish — the (j, l) entry of Y_t is, up to the nonzero factor $\psi_{j,r}/(\text{lead} \cdot z_l)$ and the phase $(z_j/z_l)^{m_t}$,

$$-\frac{1}{c_0} \left(d_0 + e_0 z_j^k + e_0 z_j^{-k} \right) = -\frac{d_0 + e_0 t_k(s_j)}{c_0} = -\frac{c_0}{a_0 + s_j} \neq 0,$$

using $(a_0 + s_j)(d_0 + e_0 t_k(s_j)) = c_0^2$ at a root. It does not depend on l . So every root space is touched, and as m_t ranges over $\ell - 2k - 2$ values the phases $(z_j/z_l)^{m_t}$ are distinct for distinct root spaces (nonresonance), so the Y_t span every root space once ℓ exceeds the number of root spaces plus a constant.

Cartan. The diagonal entry j of Y_t is independent of m_t and equals the first-order shift of the eigenphase θ_j under the perturbation, i.e. the derivative of the dock symbol at s_j with respect to the perturbed weight, divided by $F'(s_j)$. With $u_j = a_0 + s_j$ these derivatives are

$$\partial_a F = \frac{c_0^2}{u_j}, \quad \partial_d F = u_j, \quad \partial_e F = c_0^2, \quad \partial_c F = -2c_0, \quad \partial_b F = \frac{c_0^2 s_j}{u_j} = c_0^2 - \frac{a_0 c_0^2}{u_j},$$

all evaluated at the roots, so the Cartan images lie in $\text{span}\{(1)_j, (u_j)_j, (1/u_j)_j\}$, of dimension exactly 3 for $k+1 \geq 3$ distinct u_j . Hence the Cartan part of the image has dimension 3. $\square$

Lemma 122 (Second order fills the Cartan). *Let δ_1 be the a -perturbation at one interior position. For all sufficiently small $\varepsilon \neq 0$ and ℓ large, $\text{rank } dP_\ell(w_0 + \varepsilon \delta_1) = \dim \text{Sp}(2k+2)$.*

Proof. Write $A(\varepsilon)$ for the right-translated differential at $w_0 + \varepsilon\delta_1$ and $B = \frac{d}{d\varepsilon}A(\varepsilon)|_0$, which lies in $\mathfrak{sp}$. By the block form of $A(\varepsilon)$ in a basis adapted to $\ker A$ and $\operatorname{im} A$, for small $\varepsilon \neq 0$

$$\operatorname{rank} A(\varepsilon) \geq \operatorname{rank} A + \operatorname{rank}(\pi \circ B|_{\ker A}),$$

with π the projection onto $\mathfrak{sp}/\operatorname{im} A$, which by Lemma 121 is a $(k-2)$ -dimensional quotient of $\mathfrak{h}$. For a perturbation δ at position t and δ_1 at position t_1 , differentiating the right-translated differential gives the bracket

$$B[\delta] = [Y_{t_1}, Y_t] \quad (\text{the two orderings give opposite signs}),$$

whose Cartan part at j is, writing $Y_t = \operatorname{Ad}(S_0^{m_t})(\alpha\beta^\top)$ in the eigenbasis (rank one, by the proof of Lemma 121), $\gamma_j = \alpha_j\beta_j$ and $\Delta = m_{t_1} - m_t$,

$$h(\Delta)_j = \gamma_j \left(c_{-\Delta} z_j^\Delta - c_\Delta z_j^{-\Delta} \right), \quad c_\Delta = \sum_l \gamma_l z_l^\Delta.$$

Now take $\delta = \sum_t \lambda_t \delta_t$ over many positions t with λ chosen so that $A\delta = 0$: this imposes one linear condition per root space, $\sum_t \lambda_t (z_j/z_l)^{m_t} = 0$, and three on the Cartan, $\sum_t \lambda_t = 0$ (the Cartan part of Y_t is m_t -independent). Then $\pi B[\delta] = \pi \sum_t \lambda_t h(\Delta_t)$. The map $\lambda \mapsto (\text{constraints}, \sum_t \lambda_t h(\Delta_t))$ is a Vandermonde-type map in the frequencies z_j/z_l , 1, and $z_j^{\pm 1}$; under nonresonance these frequencies are pairwise distinct, so for enough positions the map is onto, and $\pi B|_{\ker A}$ has image $\pi(\operatorname{span}_\Delta h(\Delta))$. Finally $\operatorname{span}_\Delta \{h(\Delta)\} = \mathfrak{h}$: a linear relation $\sum_j \mu_j h(\Delta)_j = 0$ holding for all Δ in an infinite set is a vanishing exponential polynomial in Δ with frequencies $z_j z_l^{\pm 1}$, which by nonresonance are distinct, so (Skolem–Mahler–Lech in its elementary form for nondegenerate sequences) every coefficient $\mu_j \gamma_j \gamma_l$ vanishes, and $\gamma_j \neq 0$ by the first-order computation. Hence $\operatorname{rank}(\pi B|_{\ker A}) = k - 2$. $\square$

Theorem 123 (All k). *For every $k \geq 2$ there are L and identity tiles of every length $\ell \geq L$; hence*

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2 \quad \text{for all sufficiently large } n.$$

Proof. Choose a nonresonant elliptic dock (Lemma 116 and the remark after it). Lemma 122 supplies a point at which the tile differential has full rank, Lemma 115 supplies the group, and Theorem 117 finishes. $\square$

Remark 124 (What is checked by machine, and what is not). Lemma 121 is verified numerically: at every $k \leq 9$ computed the dock-point rank is $\dim Sp - (k - 2)$ at a nonresonant dock, every root space carries a nonzero component, and the Cartan part of the image has dimension exactly 3 (`verification/tile_controllability.py`). Lemma 122 is verified in the strongest form: one a -perturbation at one position lifts the rank to $\dim Sp$ at $k = 3, 4, 6, 7, 8$ with nonresonant docks, and fails to at the symmetric dock $(0, 1, 0, 1)$ for $k = 5$, whose symbol is even in s so that $z \mapsto -z$ pairs eigenvalues — a resonance, exactly as the proof predicts. The exact certificates of Proposition 119 remain the independent check for $k \leq 9$. The nonresonance hypothesis is used only to separate frequencies in the Vandermonde arguments; we expect a weaker condition suffices, and we have not pursued it. The threshold $L(k)$ remains ineffective.

Remark 125 (What this does not give). Theorem 109 leaves $7 \leq n \leq 16$ open at $k = 3$, apart from $n = 10$ which Theorem 58 settles; the construction of Section 20 fails at $n = 13$ and 15, where (4) gives the solution variety dimension only 3 and 9. At $k = 4$ it leaves $9 \leq n \leq 30$ open apart from the multiples of 60, 70 and 90 covered by Theorem 69.

At $k = 5$ the picture is different from what we first reported, and the correction is worth stating plainly. The count $\ell \geq 2(k+1) + (2k+2)^2/5$ gives $\ell \geq 41$ at $k = 5$ and $\ell \geq 54$ at $k = 6$, and we initially searched only $41 \leq \ell \leq 45$ at $k = 5$ and found no identity tile, the residual stalling between 7×10^{-2} and 2×10^{-1} . We drew no conclusion from that, correctly: the count is a *lower bound* on where tiles can exist, not a prediction of where they start, and at $k = 3$ the gap is already large (21 against a count of 20.8, but nothing at all below 21). Searching higher, we now have identity tiles at $k = 5$ for

$$\ell = 54, 55, \dots, 72, 74, 75,$$

21 lengths in all, and *all of them are certified*, by the same pipeline and to the same standard as $k = 3$ and $k = 4$: the exact Krawczyk test of Section 20 passes at every one, with α up to 8.4×10^{-10} , nullity exactly 12, and the docking pattern exact.

This is enough for an all-large- n statement, and by a route worth spelling out because it shows the $[L, 2L)$ hypothesis of Theorem 107 is sufficient rather than necessary. We are missing $\ell = 73$ and $\ell = 76, \dots, 107$, so $[54, 108)$ is *not* covered and Theorem 107 does not apply as stated. But what the construction actually needs is only that n be a *sum* of certified lengths, and the numerical semigroup generated by $\{54, \dots, 72, 74, 75\}$ contains every integer $n \geq 162$. Hence:

Theorem 126 ($k = 5$, all large n).

$$Z(P(n, 5)) = M(P(n, 5)) = 12 \quad \text{for every } n \geq 162,$$

and also for every n in $[54, 162)$ that is a sum of certified tile lengths, which is 64 further values.

Proof. Identical to Theorem 109 except for the last step. Each certified tile gives a one-tile cyclic matrix of nullity exactly $2k + 2 = 12$ by Corollary 45, hence $T_{\text{tile}} = I$, and the docking weights are the prescribed integers exactly, so tiles of different lengths compose (Lemma 106). Concatenating tiles of lengths summing to n therefore gives a matrix on $P(n, 5)$ of nullity $2k + 2$. Instead of invoking $[L, 2L)$ we check the semigroup directly: the largest integer not representable as a sum of $\{54, \dots, 72, 74, 75\}$ is 161, so every $n \geq 162$ is representable. See `verification/verify_all.py`, section 9. $\square$

The threshold 162 is certainly not optimal — it is an artefact of the two missing lengths, and filling $\ell = 73$ and $76 \leq \ell \leq 107$ would bring it down to 54. We report it as it stands rather than waiting, since it is a theorem either way.

Nothing in Lemma 106 or Theorem 107 refers to k , and the certification scaled from 8×8 transfer matrices at $k = 3$ through 10×10 at $k = 4$ to 12×12 at $k = 5$ with no modification at all.

We considered the natural explanation — that the product of ℓ transfer matrices becomes too ill-conditioned for the residual $T - I$ to carry information — and it does not survive contact with the numbers: at $k = 5$, $\ell = 44$ the product has condition number about 2.5×10^6 , whereas at $k = 4$, $\ell = 61$, which *does* yield a certified tile, it is about 1.1×10^{10} . So conditioning is not the discriminating factor and we do not have a diagnosis. The most likely mundane explanation is restart budget: we gave each $k = 5$ length four to six restarts, and at $k = 4$ every length that failed at that budget succeeded when given fifteen to sixty. On the evidence we have, the right statement is that we did not find $k = 5$ tiles, not that they do not exist.

One practical observation, since it governs how much compute the method needs. The Newton solve for a tile is erratic: at the restart budget we first used, roughly half the lengths at $k = 4$ failed, including lengths with *more* unknowns than ones that succeeded. Every such failure that we retried with a larger budget — $\ell = 31, 32, 33, 35, 38$ — then solved to 10^{-14} . So a failed tile search at these sizes carries no information beyond the effort spent, and we were careful not to read

the first pass as evidence about $k = 4$. Extending a neighbouring tile as a starting point does *not* substitute for restarts: the damped Newton stalls near 10^{-2} from such starts at the lengths that needed retrying.

21 A cubic family on which the symbol stays at 6 and the nullity does not

The ceiling, where it holds, says the maximum nullity of a cover is read off its base. This section records a family on which we believed it would hold, and the matrix that shows it does not. The family was proposed, in an earlier version of this paper, as a cubic separation between M and Z with an unbounded gap; it is instead the counterexample to the regular-base conjecture of Remark 89. We keep the certificate and the Z values because they are correct, and the record of how the conjecture failed because it is the point.

Let $B = K_4$ with voltages 0 on $0-1$, 1 on $0-2$, 0 on $0-3$, 2 on $1-2$, 0 on $1-3$ and 1 on $2-3$. Every cover B^n is cubic, simple and connected, on $4n$ vertices, and the generic degree span of $\det M(\zeta)$ is $D = 6$ for every n .

Proposition 127 (An exact certificate at the span). *Give every edge of B^n weight 2 and the four fibres diagonal entries 4, 4, 5, 4. The resulting integer matrix has null $A = 6 = D$, verified by exact rank over $\mathbb{Z}$ at $n = 7$ and $n = 14$.*

Proof. $\det M(\zeta)$ is palindromic, since $M(1/\zeta) = M(\zeta)^\top$ for real weights, so in $x = \zeta + \zeta^{-1}$ it is a cubic. Writing $\zeta^3 \det M(\zeta) = A(\zeta^6 + 1) + B(\zeta^5 + \zeta) + C(\zeta^4 + \zeta^2) + E\zeta^3$ and using $\zeta^3 + \zeta^{-3} = x^3 - 3x$, $\zeta^2 + \zeta^{-2} = x^2 - 2$, the symbol in x is $Ax^3 + Bx^2 + (C - 3A)x + (E - 2B)$. The minimal polynomial of $2 \cos \frac{2\pi}{7}$ is $x^3 + x^2 - 2x - 1$, also a cubic, and matching the two up to scale forces $A = B = C = E$. With all six edge weights equal this reduces to $d_0 = d_1 = d_3 =: t$ and $d_2 = (3t + 4)/((t - 1)(t + 2))$; taking $t = 2$ gives $d_2 = \frac{5}{2}$, and scaling the matrix by 2 clears the denominator. The symbol is then $16(\zeta^7 - 1)/(\zeta - 1)$, which vanishes at each of the six primitive 7th roots of unity, so six blocks are singular. $\square$

Proposition 128 (The zero forcing number grows). *$Z(B^n) = n + 2$ for $n = 4, \dots, 8$, each value computed by a CP-SAT model with optimality proved rather than merely attained.*

n	$ V(B^n) $	D	Z	$Z - D$
4	16	6	6	0
5	20	6	7	+1
6	24	6	8	+2
7	28	6	9	+3
8	32	6	10	+4

Theorem 65 bounds every equivariant matrix with $\det M(\zeta) \neq 0$ by 6, and Proposition 127 attains it. We had concluded that the equivariant maximum nullity of the family is 6 while Z grows without bound, and conjectured $M(B^n) = 6$ — which would have been an infinite family of cubic graphs with bounded maximum nullity and unbounded zero forcing number, the gap explained by a parameter read off the base. Both the conclusion and the conjecture are false, and the reason is in the voltages: the three edges of voltage 0 are $0-1$, $0-3$ and $1-3$, a triangle, so each fibre index i carries a triangle $\{(0, i), (1, i), (3, i)\}$, and the only edges leaving it go to the fibre over vertex 2.

Theorem 129 (The regular-base ceiling fails). *On the cover B^n above, for every $n \geq 4$, there is an equivariant $A \in \mathcal{S}(B^n)$ with $\text{null } A \geq n$. Hence $M(B^n) \geq n$ against $D = 6$, and $Z(B^n) - M(B^n) \leq 2$ wherever $Z(B^n) = n + 2$.*

Proof. Take $u \in \mathbb{R}^3$ with all entries nonzero and let the principal submatrix of A on each triangle $\{(0, i), (1, i), (3, i)\}$ be $uu^\top$: its off-diagonal entries $u_a u_b$ are nonzero, as the pattern requires, and its diagonal entries are free. Give the remaining edges any nonzero weights and the fibre over 2 any diagonal. Let S be the $3n$ triangle vertices and S^c the n vertices over 2. Then $G[S]$ is n disjoint triangles and $A[S]$ has rank n ; the block $A[S, S^c]$ has n columns; so the rows indexed by S have rank at most $2n$, the rows indexed by S^c have rank at most n , and $\text{rank } A \leq 3n$. This is Proposition 88 with $\text{mr}(G[S]) = n$ and $|S^c| = n$. In the equivariant picture $M(\zeta)$ has a rank-one 3×3 block, so its first three columns span at most two dimensions and $\det M(\zeta) \equiv 0$: every block is singular, which is the case Theorem 65 excludes. The ranks were also computed exactly over $\mathbb{Q}$ for $n = 4, \dots, 10$, where the nullity is exactly n (`verification/k4_rank_one.py`). $\square$

What survives of the section is this. The generic span $D = 6$ is attained, by an integer matrix; $Z = n + 2$ by exact search; and $M \geq n$, so the family is not a separation between M and Z , and the degree span is not the parameter Y with $M \leq Y < Z$ that the survey [2] asks for — not on regular bases, where we had proposed it. Whether $M(B^n)$ is n , $n + 1$ or $n + 2$ we have not determined, and one piece of evidence should be recorded with its standard attached. Maximising the nullity over the *full* pattern at $n = 4$ — not the equivariant subclass — reaches $5 = n + 1$, with the five smallest singular values between 10^{-9} and 10^{-11} and the sixth at 0.24 of the matrix scale, while $n + 2 = 6$ is not reached and stalls at 6×10^{-4} , nowhere near zero. If that witness is genuine then the rank-one construction is not optimal and $Z - M \leq 1$ at $n = 4$. We label it *numerical*: unlike $P(10, 2)$ it resisted exact rationalisation, so there is no certificate, and this project has twice been misled by a converged optimum. Its structure is at least not the one the proof above uses — the triangle blocks come out with ranks $(2, 2, 2, 3)$ rather than all 1, so Proposition 88 does not account for it (`verification/k4_nplus1.py`). Certifying or refuting it is the open problem. The documented strict separations remain tree-like — the Barioli–Fallat tree and a graph derived from it — and the published cubic results reach $Z = 3$ and $Z = 4$ [9]; we do not have a regular example. The two attacks of Remark 89 both missed this matrix, for a reason worth recording: the diagonal-zeroing attack varies only diagonals, and the gradient search starts from generic weights, while the construction lives on the locus where the triangle blocks are singular, which neither explores. The matrix was found by reading the voltages, not by searching.

22 Why lower bounds resist standard methods

A set is zero forcing iff it meets every *fort* — a nonempty F such that every vertex outside F has 0 or ≥ 2 neighbours in F — so Z is a minimum hitting set over forts. Forts, and the covering formulation of Z they give, are due to Brimkov, Fast and Hicks [4]. Three standard consequences all fail here, and all degrade as k grows.

- **Disjoint forts.** Pairwise disjoint forts $F_1, \dots, F_t$ force $Z \geq t$. But forts in $P(n, k)$ have 7–13 vertices while the graph has $2n \leq 56$, and a greedy search found at most *two* disjoint forts in every case tested: the certificate cannot exist at these sizes.
- **The fort LP.** Computing the exact LP value by cutting planes gives $\text{LP}/Z \approx 0.43\text{--}0.50$, e.g. $\text{LP} = 3.00$ against $Z = 7$ for $P(12, 3)$, and 3.21 against $Z = 10$ for $P(18, 4)$. The integrality

gap is about 2 and widens with k . This also explains why fort-based exact solvers converge slowly on this family: branch and bound must close a factor-2 gap by branching.

- **Spectral.** The Hoffman-type bound $Z \geq N(1 + 2\lambda_{\min}/(d - \lambda_{\min}))$ yields values in 1.2–2.9 against true Z of 6–12, and also weakens as k grows.

23 What this contributes, and what it does not

It is worth separating the contribution from the framing before listing the claims, because the two are not the same size.

The clearest contribution is Part II. It closes three specific gaps in the literature on a family that is used as a test case. A published theorem — $Z(P(n, 3)) = 8$ for $n \geq 12$, [6, Thm. 3.6] — is false; the error was found by [7], which exhibits $Z(P(12, 3)) = 7$, and what was missing there, a lower bound valid for every large n , is Theorem 111. Conjecture 5 of [7] is thereby proved and the stabilisation threshold that note leaves open is fixed at 13. And the values at $k = 4$ are the first exact ones at that parameter at all, as well as the first holding for an infinite family at any $k \geq 4$. Separately, the ceiling of Corollary 45 is a statement about the method rather than the family: the matrix-certificate approach either settles $Z(P(n, k))$ completely or cannot settle it at all.

Part I is a complete classification of a recognised kind. Droll did this for unitary Cayley graphs and Le–Sander [12] for integral circulants of prime power order; in both cases, as here, only finitely many members qualify. The contribution is exactness and completeness, not surprise: anyone who knows that abelian covers do not expand would have predicted the shape of the answer. What was not available before is *which* members, and that no member exists beyond $k = 45$. For $k \leq 9$ other than 6 and 8 the answer is a theorem rather than a list (Theorem 25): the classification is an initial segment, cut at B_k , and at $B_k/2$ for odd n when k is odd, because the negative set of D_k consists of the end bands alone — a fact about the real roots of one integer polynomial, decided by Sturm’s theorem. Where the list has gaps, it is because an interior band appears, and from $k = 10$ on it always does.

Three things we are careful not to claim. First, the graph Riemann Hypothesis is not the classical one: by the Ihara determinant formula it is equivalent to a finite eigenvalue check, which is exactly why it is decidable. The analogy lives in the zeta function, not in the difficulty. Second, Theorem 30 refines a known principle rather than establishing a new one; what is new is the explicit constant and the exact criteria of Theorems 27 and 28 for the two-vertex bases. Third, the verification apparatus is methodology, not mathematics: it is why we believe the numbers, and it is not a theorem.

24 Status of the claims

Added 4 October (§20). Proved: the invariant form of a tile is the discrete Wronskian and tile products lie in $Sp(B)$ (Lemma 115); elliptic docks exist for every k (Lemma 116); identity tiles of every large length exist whenever the tile differential has full rank at one point (Theorem 117). The rank hypothesis holds at every nonresonant elliptic dock, for every k (Lemmas 121, 122), so $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all sufficiently large n , for every $k \geq 2$ (Theorem 123), with an ineffective threshold; the hypothesis is also certified exactly for $k \leq 9$ (Proposition 119). Open: an effective $L(k)$. Also proved the same day (§20): the leading-coefficient criterion, Theorem 91, formerly a conjecture — on every base whose top coefficient is a monomial in the edge weights, null $A \leq D$ for every $A \in \mathcal{S}(B^n)$ and every n , and the kernel of the lifted operator on the infinite

strip has dimension exactly D for every choice of nonzero weights (Theorem 94); the two-vertex, path and cycle reductions are special cases. Every claim below is labelled by the standard it meets. Four standards appear, in decreasing strength: *proved*, meaning a proof in the text; *proved by an exact certificate*, meaning an explicit matrix whose nullity is computed in $\mathbb{Z}$ or $\mathbb{Q}(\zeta_n)$; *proved by interval certification*, meaning a Krawczyk contraction bound establishing that a true real solution exists in an explicit box; and *numerical*, meaning a floating-point computation only. We also list what we have refuted, including our own earlier claims.

- **Proved.** Theorems 36, 40 and Theorem 44 with Corollary 45; Propositions 46 and 47; Theorem 63; Theorem 64 and Theorem 60, the nullity classifications of the adjacency matrix and of its signed variant; Theorem 49, that $c^2 = 0$ if and only if the symbol factors as $(s+a)(L_k+t)$, with Corollaries 50 and 51 and Proposition 52; Theorem 62, that no rational period-one symbol attains 14 at $k = 6$ in either matrix class; Theorem 72, that a period-one certificate covers exactly one divisibility class, which contains at most one prime, together with the identity $e_2 = -k$ for every $k \geq 3$; Theorem 65 — under the hypothesis $\det M(\zeta) \neq 0$, which an earlier version omitted — with Theorem 66 for cyclic covers; Theorems 80, 81, 82 and 83; the two refutations, Theorems 85 and 129; and Propositions 86 and 88.
- **Proved by an exact certificate.** $Z(P(n, 2)) = 6$ for every $n \geq 9$ with $n \neq 10$, and for every n divisible by 7 (Theorem 53); $Z(P(n, 3)) = 8$ for every n divisible by 10 (Theorem 58); $Z(P(n, 4)) = 10$ for every n divisible by 60, 70 or 90; $Z(P(n, 5)) = 12$ for every n divisible by 24; $Z(P(n, 7)) = 16$ for every n divisible by 120 (Theorem 61). The $k = 4, 5$ values were the first exact values for $k \geq 4$; the $k = 7$ value replaces a floating-point claim we previously made at a single n . In the $k = 2$ ($7 \mid n$), $k = 3$ and $k = 7$ cases the certificate is an integer matrix with entries in $\{0, \pm 1\}$.
- **Proved by exact certification.** $Z(P(n, 3)) = M(P(n, 3)) = 8$ for every $n \geq 17$, and $Z(P(n, 4)) = M(P(n, 4)) = 10$ for every $n \geq 29$ (Theorem 109), and $Z(P(n, 5)) = M(P(n, 5)) = 12$ for every $n \geq 162$ (Theorem 126), from Theorem 107 together with certified identity tiles of every length in $[21, 44)$ at $k = 3$ and $[29, 62)$ at $k = 4$, plus Theorem 100 for $17 \leq n \leq 20$; at $k = 5$ the certified lengths are $[54, 73) \cup \{74, 75\}$, which do not form an interval $[L, 2L)$ but generate every $n \geq 162$ as a sum. No congruence condition appears and the primes are included. All 77 tile certifications — 23 at $k = 3$, 33 at $k = 4$, 21 at $k = 5$ — and all 17 per- n certifications are Krawczyk contraction bounds with $\alpha \leq 1.0 \times 10^{-9}$, each evaluated in *exact rational arithmetic* (Section 20); no step rests on a floating-point nullity. Each rests on a Krawczyk contraction bound for the bilinear system $A(w)K(X) = 0$, with the square subsystem chosen structurally so that it is equivalent to $AK = 0$ rather than merely a maximal-rank selection of it. By Theorem 72 no period-one certificate can reach these n at all, so this is the only route to them that we know.
- **Proved (finite).** Every tabulated value of Z , by exhaustive search over vertex subsets. Combined with the above this gives Theorem 110: $Z(P(n, k))$ is determined for every admissible n when $k = 2$, $k = 3$ and $k = 4$, with no value left open in any of the three rows. In particular Theorem 111, $Z(P(n, 3)) = 8$ for every $n \geq 13$ with 13 optimal: the four values $n = 13, 14, 15, 16$ are exhaustive searches and the rest is Theorem 109, so the stabilization threshold at $k = 3$ — recorded as open in [7] — is exactly 13. This is the one result here whose statement needs no matrix theory to check.
- **Numerical, not exact.** Theorem 76: $Z(P(n, 6)) = 14$ at $n = 60, 72, 84, 90, 96, 108, 120$ and $Z(P(96, 8)) = 18$. The certificates are period-three matrices whose weights come from a

Newton solve, verified on the full $2n \times 2n$ matrix with a wide spectral separation and every required entry a substantial fraction of the matrix scale, but not in exact arithmetic. We do not claim these at the standard of the items above. ($Z(P(96, 7)) = 16$ was in this category and is now exact.)

- **Observation supported by computation, not proved.** The rank law $\text{rank} = \min(2d + 1, k + 1)$, verified for $3 \leq k \leq 13$ and $1 \leq d \leq 7$. Also the saturation of the reachable set of monodromies, at dimension 31 for $k = 2$ and 51 for $k = 3$.
- **Conjectural.** Conjecture 41; and Conjecture 103 for $k \geq 6$. The leading-coefficient criterion, a conjecture in earlier versions, is now Theorem 91; what remains open there is the case of a top coefficient with two or more monomials, such as the tie $t_m = p + q$ of Theorem 98. For $k = 3$ and $k = 4$ the conjecture is a theorem (Theorem 109), and for $k = 5$ it is a theorem with a non-optimal threshold (Theorem 126, $N(5) \leq 162$); what remains open is whether $N(k)$ exists for every k , for which we prove only the necessary condition (4).
- **Refuted (ours).** The regular-base ceiling conjectured in Remark 89, and with it the conjecture of an earlier version that $M(B^n) = 6$ on the K_4 family of Section 21, which would have given $Z - M \rightarrow \infty$ on cubic graphs: Theorem 129 gives $M(B^n) \geq n$. The accompanying statement that the *equivariant* maximum nullity of that family is 6 was also false, because Theorem 65 needs $\det M(\zeta) \neq 0$ and a rank-one fibre makes it vanish identically. The conjecture had been written into the paper, the board and the interview script as the strongest result in the project.
- **Refuted (ours).** That the period-one construction “never exceeds 6 for every $k \geq 2$ ”. Three roots can be prescribed and no more, but the remaining $k - 2$ can land on the grid by Galois conjugacy, and do. The period-two construction we built to get past 6 was unnecessary and its values are superseded.
- **Refuted (ours).** That $k = 3$ and $k = 7$ admit no rational period-one symbol attaining $2k + 2$. That enumeration imposed $c^2 > 0$, the condition for a *symmetric* certificate, where $c^2 \neq 0$ suffices: c^2 is the product of the two spoke weights and only its vanishing is fatal. Six symbols at $k = 3$ and one at $k = 7$ survive, giving Theorems 58 and 61.
- **Refuted (ours).** The unconditional bound $Z(B^n) \geq D$ for cyclic covers, by $Z(P(8, 2)) = 5 < 6$. Only the conditional form $Z(B^n) \geq \nu$ for an attained ν survives.
- **Refuted (ours).** The single-threshold form $n_0(k) = 5k - 2$, by $Z(P(29, 6)) = 12$; and any bound of the form $Z \leq 2(\gcd(n, j) + \gcd(n, k))$, by $Z(P(11, 3)) = 7 > 4$.
- **Refuted (ours).** That the threshold for reaching the ceiling with a general matrix is $n \geq (2k + 2)^2/5$. Spending the gauge gives (4), $n \geq (k + 1)(2k + 3)/3$, which is larger; the earlier figure led us to test $k = 6$ at $n = 27, 29, 31$, all below the true threshold of 35.
- **Refuted (ours).** That $N(k)$ in Conjecture 103 is $O(k)$. The saturation data alone suggests $4k + 3$, but (4) forces $N(k) \geq \lceil (k + 1)(2k + 3)/3 \rceil$, which is quadratic.
- **Refined (ours).** We earlier recorded as refuted the claim that the monodromy is symplectic, on the grounds that the *standard* form is not preserved. That is true and was the wrong conclusion to draw. With the docking pattern fixed there is a unique invariant skew form B , and it is nondegenerate, so tile products are symplectic with respect to B (Observation 112).

The earlier statement — that the space of invariant skew forms for a single product has dimension $k + 1$, so eigenvalues come in reciprocal pairs — remains correct; it is the family, not the individual, that pins B down.

- **Observation, not proved.** Observations 112 and 113: the invariant form and the full-dimensionality of the reachable set, verified for $2 \leq k \leq 6$ but not proved. These are what reduce the existence of identity tiles for all k to a controllability question in $Sp(2k + 2, \mathbb{R})$, stated in Section 20. We do not claim that reduction settles the question: subsemigroups with nonempty interior in $Sp(2n, \mathbb{R})$ need not be the whole group.
- **Inconclusive, not negative (ours).** The per- n search of Section 20 at $k = 6$ returned nothing at $n = 41, 43, 45, 47, 49, 51$, and the tile search at $k = 5$ nothing at $41 \leq \ell \leq 45$ — the latter now explained: the first $k = 5$ tile is at $\ell = 54$, so that search was below the range where tiles exist. Both ran at a restart budget that we have since seen condemn a length that does solve: $\ell = 21$ at $k = 3$ stalled at 5.6×10^{-2} on four restarts and produced a tile at 1.4×10^{-14} on twenty-five. We therefore record these as searches we did not push far enough, not as evidence about $k = 5$ or $k = 6$. The one regularity we have found is that every length eventually solved first appeared at a residual of 10^{-2} or below, and the $k = 5$ and $k = 6$ attempts sit at 10^{-1} .
- **Withdrawn as unreliable (ours).** The eigenvalue-minimising search whose plateaus we previously reported as evidence that the ceiling is unreachable at various (n, k) . It gave maximum nullity 7 at $P(20, 3)$, where 8 is attained — by Theorem 58 and, in the symmetric class, by the method of Section 20. It collapses onto the degenerate strata (all spokes zero, or all edges zero). Its negative results should not be read as evidence, and the conclusion we drew from them, that exactness for $k \geq 6$ was unlikely, was not supported.
- **Open.** Conjecture 103 for $k \geq 6$, where we have no tiles at all and the count gives only $\ell \geq 54$. At $k = 5$ the conjecture now holds for $n \geq 162$ (Theorem 126) but not below it: we have 21 of the 54 lengths $[54, 108)$ would require, and $54 \leq n < 162$ is settled only at the 64 values that are sums of them. Also whether $M = Z$ in general, and the rank law. No (n, k) is proved to have $M(P(n, k)) < 2k + 2$, and the one case where we expected a strict gap, $(n, k) = (10, 2)$, turned out not to be one: $M(P(10, 2)) = Z(P(10, 2)) = 6$ by the integer certificate of Remark 55.
- **Superseded (ours).** The claim that the period- d method “cannot settle prime n at all” and that a different construction would be needed there. That is correct for the period- d method and is now a theorem (Theorem 72), but it is no longer a limitation of the paper: Section 20 settles prime n for $k = 3$.
- **Not determined.** Whether Theorem 40 is attained for $k \geq 3$; whether the degeneracy of Section 19 at $n = 48, 54$ is essential.

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$$Q(x, y) = 4x^2 + 4y^2 - 8\sqrt{2} |xy| + 3 \geq 0$$

one criterion, four fixed lenses, the same for every k

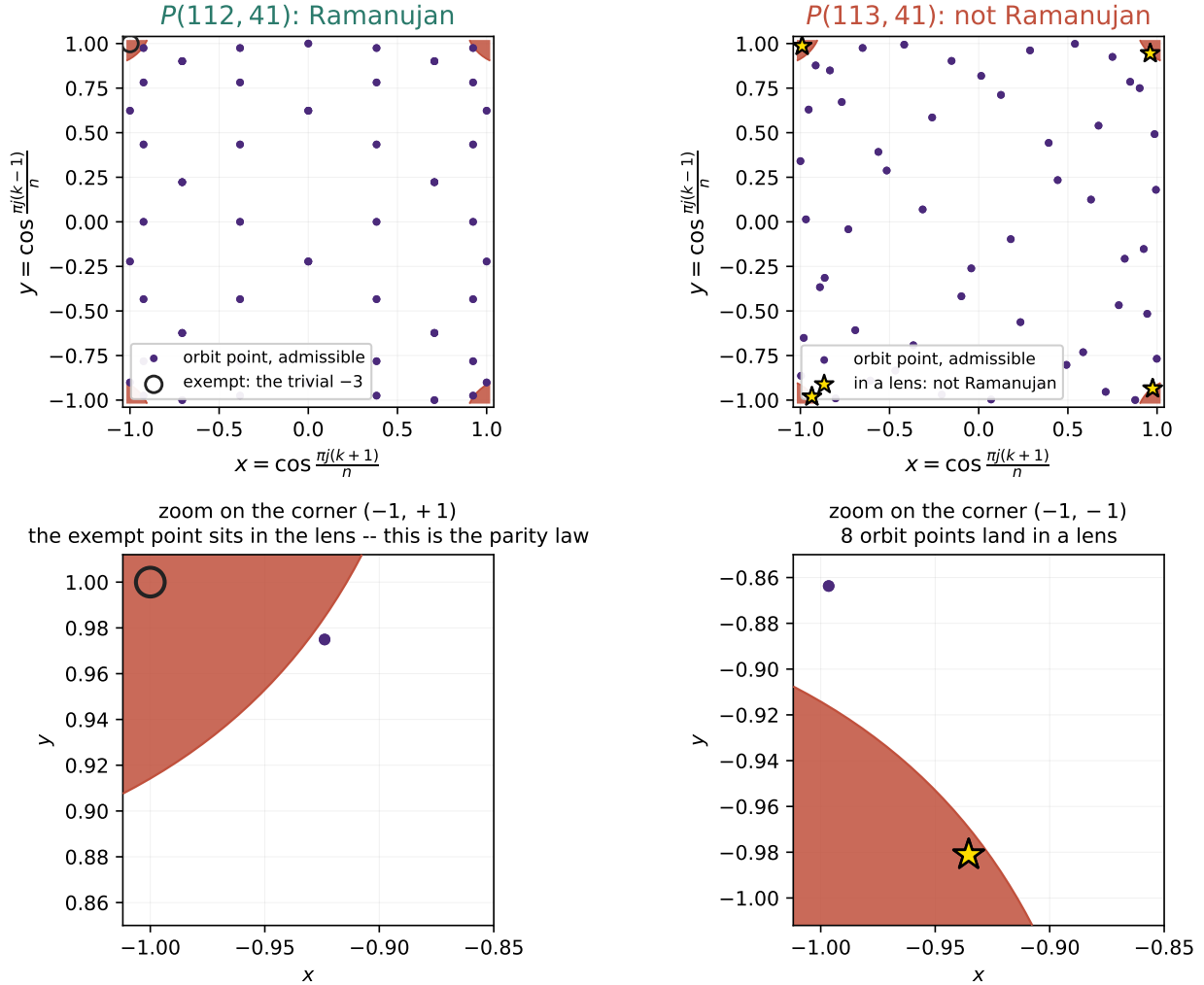


Figure 1: The corner criterion. Left, the largest Ramanujan member of the whole family; right, the cover one step larger, which fails. The four forbidden regions are the same in both pictures and for every k — only the map into the square changes. In the left-hand zoom the single point inside a region is the exempt trivial -3 , which is why bipartiteness helps.

k	B_k	count	the n for which $P(n, k)$ satisfies the RH
1	15	9	3–8, 10, 12, 14
2	23	19	5–23
3	32	18	7–16, 18, 20, 22, 24, 26, 28, 30, 32
4	42	34	9–42
5	51	28	11–26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50
6	61	20	13–16, 18, 20–23, 25, 27–28, 30, 32, 35, 37, 39, 42, 44, 49
7	71	39	15–36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70
8	81	16	17, 19–22, 24, 26, 28–29, 31, 33, 35, 38, 40, 42, 49
9	91	50	19–46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90
10	101	14	21, 23, 25–26, 28, 30, 32, 34, 37, 39, 41, 43, 50, 52
11	111	36	23–26, 28–32, 35–38, 40–42, 46–48, 50, 52–53, 58, 60, 62, 64, 70, 72, 74, 82, 84, 92, 94, 96, 104, 106
12	121	9	27, 29–30, 32, 34, 38, 40, 45, 51
13	131	23	28–30, 32, 34–37, 42–44, 46, 48–49, 56, 58, 60, 70, 72, 74, 84, 86, 98
14	141	7	31, 33, 35–36, 38, 46, 53
15	151	19	33–36, 38, 40–42, 49–50, 52, 54, 56, 66, 68, 70, 82, 84, 98
16	161	6	35, 39–40, 42, 44, 52
17	171	16	37–40, 42, 44, 46–47, 56, 58, 60, 62, 74, 76, 78, 94
18	181	3	45–46, 50
19	191	17	42–44, 46, 48, 50, 52–53, 64, 66, 68, 70, 84, 86, 88, 104, 106
20	201	3	45, 49–50
21	211	11	46, 48, 50, 54, 56, 58, 70, 72, 76, 92, 96
22	221	1	49
23	231	10	52–54, 56, 60, 62, 76, 78, 84, 106
24	241	1	53
25	251	8	56, 58, 60, 66, 68, 84, 90, 92
27	271	8	60, 62, 64, 70, 72, 74, 90, 98
29	291	9	64, 66, 68, 70, 76, 78, 96, 98, 106
31	311	4	70, 74, 82, 84
33	331	5	74, 76, 78, 86, 90
35	351	5	78, 80, 84, 92, 96
37	371	3	82, 84, 88
39	391	3	88, 102, 106
41	412	2	98, 112
43	432	2	96, 98
45	452	2	100, 108

No k outside this table admits any n : the list is finite in both variables.

Table 1: The complete classification, every n and every k . For each k , B_k is the per- k bound of Theorem 12; the rows stop at $k = 45$ because by Theorem 13 no larger k admits any n at all. Generated directly from the certification run.

k	bound $2k + 2$	least n attaining it	least n from which it holds always
2	6	7	9 (it fails at $n = 8$)
3	8	10	13 (it fails at $n = 11, 12$)
4	10	18	18 (Theorem 110)
5	12	23	not determined
6	14	28	not determined; $Z(P(29, 6)) = 12$

Table 2: The bound is attained well before it is attained *always*: at $k = 3$ it first holds at $n = 10$, fails again at $n = 11$ and 12 , and holds for every $n \geq 13$ (Theorem 111). The two columns are different quantities and conflating them is easy; at $k = 6$ the bound is attained at $n = 28$ and lost at $n = 29$, so no stabilization threshold is known there. At $k = 4$ the two columns happen to coincide at 18, which is not a pattern: they differ at $k = 2$ and $k = 3$. All values by exhaustive search.



Figure 2: The deficit $\delta^*(n)$, the best achievable nontrivial spectral radius at each size, measured from the Ramanujan threshold. Green points are sizes at which an optimal member exists; the last is $n = 112$. The curve rises to the asymptote $3 - 2\sqrt{2}$, which by Proposition 33 is the most that can ever be lost. The two branches are the parity law.

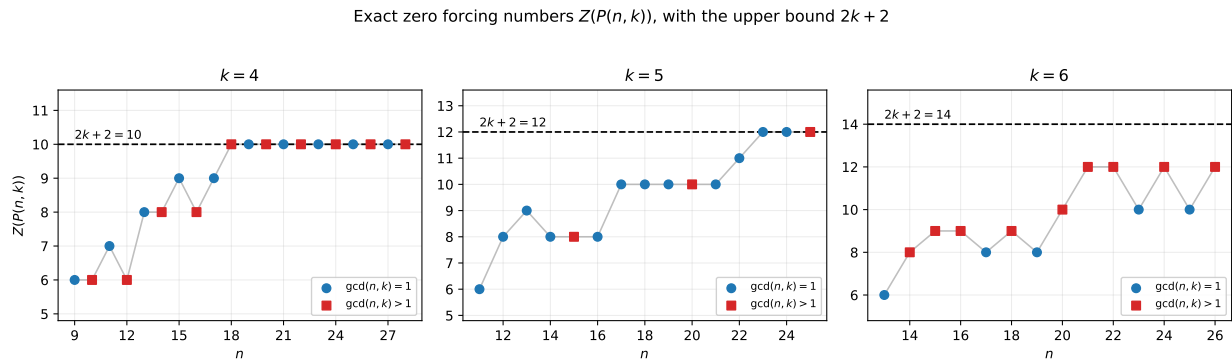


Figure 3: Exact $Z(P(n, k))$ for $k = 4, 5, 6$ against the bound $2k + 2$. Points are separated by whether $\gcd(n, k) = 1$. The sequences are non-monotone: $k = 4$ drops at $n = 12, 16$ and $k = 6$ at $n = 23, 25$.

k	$\mathcal{D}$	$\text{lcm}(\mathcal{D})$	F	a	d	c^2	nullity
2	$\{3, 10\}$	30	$sL_2 - 1$	0	0	1	$6 = 2k + 2$
2	$\{14\}$	14	Ψ_{14}	-1	0	1	$6 = 2k + 2$
3	$\{1, 4, 5\}$	20	$\Psi_1 \Psi_4 \Psi_5$	-1	-1	1	7
4	$\{4, 30\}$	60	$(s+1)L_4 - s - 2$	1	-1	1	$10 = 2k + 2$
4	$\{5, 14\}$	70	$\Psi_5 \Psi_{14}$	0	1	1	$10 = 2k + 2$
4	$\{5, 18\}$	90	$\Psi_5 \Psi_{18}$	1	0	1	$10 = 2k + 2$
5	$\{3, 6, 24\}$	24	$sL_5 - 1$	0	0	1	$12 = 2k + 2$

Table 3: Exact period-one certificates. The $k = 2$, $k = 4$ and $k = 5$ rows give $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for every n divisible by the stated modulus.

k	$2k + 2$	best period-one nullity	attained at
2	6	6	every n with $9 \leq n \leq 130$ except $n = 7, 8, 10$
3	8	8	$n = 40, 60, 80, 120$
4	10	10	$n = 60, 70, 90, 120$
5	12	12	$n = 24, 48, 72, 96, 120$
6	14	8	$n = 35, 37, 70, 74, 105, 111$
7	16	10	$n = 60, 120$
8	18	8	$n = 21, 26, 27, 39, \dots$

Table 4: The period-one construction reaches the ceiling $2k + 2$ for $k \leq 5$ and, in the computed range, falls well short for $k \geq 6$.

k	$2k + 2$	period d	$\lceil k/2 \rceil$	n	smallest required entry
6	14	3	3	60, 72	0.188, 0.162
7	16	4	4	96	0.107
8	18	4	4	96	0.260

Table 5: Period- d certificates attaining the ceiling for $k \geq 6$. The period used is in every case the one the rank law prescribes.

k	dock $(a_0, c_0, d_0, e_0), b_0 = 1$	ℓ_0	rank = $\dim Sp$	$\det B$
2	(0, 1, 0, 1)	15	21 = 21	1
3	(1, 1, 1, 1)	21	36 = 36	1
4	(0, 1, 1, 1)	29	55 = 55	1
5	(0, 1, 0, 1)	39	78 = 78	1
6	(0, 1, -1, 2)	49	105 = 105	4096
7	(0, 1, 0, -1)	61	136 = 136	1
8	(-1, 1, 0, 2)	75	171 = 171	65536
9	(0, 1, 0, 2)	89	210 = 210	262144

Table 6: The docks and lengths at which the hypothesis of Theorem 117 is certified. Generated by `verification/verify_all.py`.